

# The Supply Side of Social Media\*

Shakked Noy<sup>†</sup>

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Aakaash Rao<sup>‡</sup>

Northwestern Kellogg

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### Abstract

We study an overlooked feature of social media platforms: their distinctive model of content production. Unlike legacy media outlets, platforms do not professionally produce content ex-ante optimized to appeal to viewers; instead, they rely on large numbers of amateur users creating content in pursuit of social status, expressive satisfaction, or creative fulfillment. The resulting scale of their content libraries allows them to ex-post cherry-pick the pieces most appealing to viewers. We develop an equilibrium model of this process. We show that because the scale powering platforms' cherry-picking concentrates where creators are eager to produce, creators' supply-side preferences can influence what viewers consume. Collecting new data on Instagram, TikTok, and X/Twitter, we show that most of the content in viewers' feeds is created by non-professionals who tend to produce envy-provoking "lifestyle" and partisan political content. To separate supply- and demand-side preferences, we experimentally manipulate viewers' feeds and pay creators to post different types of content. Piece-by-piece, lifestyle and partisan political content is less likely to highly engage viewers than more benign entertainment. But the satisfaction creators derive from creating such content means it is produced at such a large scale that platforms can cherry-pick the minority of pieces viewers find more interesting or entertaining than the marginal piece of entertainment. Estimating our model using the experimental variation, we calculate that supply-side preferences account for one- to two-thirds of the prevalence of lifestyle and partisan content in viewers' feeds.

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<sup>†</sup>[snoy@mit.edu](mailto:snoy@mit.edu)

<sup>‡</sup>[aakaash.rao@kellogg.northwestern.edu](mailto:aakaash.rao@kellogg.northwestern.edu)

# 1 Introduction

Over the past twenty years, successive generations of social media platforms have gradually displaced legacy media outlets to become many users’ primary source of news and entertainment. Critics allege that by serving users content provoking personal insecurities and political grievances, and fragmenting their attention spans with short-form feeds, these platforms drive recent declines in political comity, institutional trust, mental wellbeing, and academic achievement.<sup>1</sup>

The common view in both popular commentary and academic research is that these problems originate on the demand side.<sup>2</sup> As the story goes, social media platforms excel at predicting what will engage viewers, and envy, outrage, and other negative emotions are highly engaging. And by designing short-form feeds that, like slot machines, offer a stream of variable and unpredictable rewards, platforms keep viewers compulsively scrolling. On this view, what differentiates social from legacy media is a superior ability to identify and exploit latent demand.

This paper develops a new and complementary way of thinking about what makes social media different from legacy media. We focus on what is arguably social media’s most distinctive feature: its unusual *production* side. Unlike legacy media outlets, social media platforms do not professionally produce content ex-ante optimized to appeal to viewers. Instead, they harvest content created by amateur users in pursuit of social status, expressive satisfaction, or creative fulfillment. They then compensate for the low average appeal of such content by using the enormous scale at which it is produced, combined with their ability to experimentally test pieces on subsets of viewers, to facilitate granular ex-post *discovery* of the pieces most appealing to each viewer.

This business model has two side effects. First, because the production scale fueling platforms’ cherry-picking will concentrate selectively in regions of the content space where creators are most eager to produce, creators’ supply-side preferences may end up shaping what viewers consume.

For example, suppose that politically extreme users derive more expressive satisfaction from arguing about politics than moderate users, and therefore produce much more political content. Then even a viewer who prefers moderation *ceteris paribus* may find her feed full of extreme political posts which, thanks to the size of the pool from which they are carefully picked, land much closer to what interests or animates her than the nearest available moderate posts.

Or suppose attractive users pursue social status by posting self-centered “lifestyle” content, so the supply of such content outpaces the supply of more effortful and less ego-gratifying comedy sketches. Then even if a viewer finds lifestyle content mostly less interesting and more prone to triggering unfavorable social comparisons than comedy, she may receive a feed of cherry-picked lifestyle content so amusing she prefers it to the best available comedy despite the envy it provokes.

Second, social media’s business model favors content types disaggregated enough to allow maximally granular experimentation and cherry-picking—for example, short-form content.

Overall, social media may hence offer viewers content they find more entertaining and interesting than legacy media, but which is also more likely to trigger political or personal discontent—due to

<sup>1</sup>See Allcott et al. (2020); Levy (2021); Braghieri et al. (2022); Haidt (2024); Bursztyn et al. (2025a).

<sup>2</sup>See Allcott et al. (2022); Beknazar-Yuzbashev et al. (2024); Bao (2026); Solheim and Wang (2026); Bolte et al. (2026).

reliance on scale generated through conflict- and status-motivated production—and fragment their attention spans—due to reliance on cherry-picking from disaggregated pools of short content.

In this paper, we collect observational data on, and conduct experiments with, social media users and creators. First, we develop a model formalizing the aforementioned characterization of social media’s business model, and compile a tapestry of evidence supporting it and demonstrating the importance of non-professional content supply. Second, we use experimental variation to estimate our model and decompose the equilibrium prevalence of two types of content—envy-provoking “lifestyle” content and partisan political content—into supply- and demand-driven components.

**Observational data.** We begin by collecting new data on the consumption and production of social media content. We pay 393 young Americans to download their personal data on Instagram, TikTok, and YouTube and install a Chrome extension recording their X feeds, yielding second-by-second histories of the posts they have seen on these platforms. We scrape the posts appearing in these histories, use LLMs to classify their contents, and scrape information about their creators.

We link these data to three supplementary data sources. First, we use TikTok’s Research API to construct a representative sample of the videos *created* on TikTok, allowing us to trace the composition of content from initial production through to final consumption. Second, we contact and survey 740 of the creators appearing in our participants’ histories, asking about their production process and motivations, and measure selection into content creation by surveying 3,490 platform users. Third, we pay the participants providing their viewership histories to first spend a week using an iPhone app we built that detects when they are on social media and pings them periodically to ask how they are feeling. Combined, these data provide a rich descriptive picture of content production, consumption, and experienced emotions on social media.

**Framework.** To structure our investigation, we develop a model of the equilibrium production and consumption of social media content. Pieces of content inhabit a high-dimensional characteristic space over which viewers have heterogeneous preferences. Creators choose what to create, weighing the fixed costs or expressive benefits of producing different content types against pecuniary and non-pecuniary rewards for attracting viewership. Content creation is a noisy process: when produced, a piece draws a partly stochastic vertical quality and horizontal features, making its appeal to viewers ex-ante unpredictable. A platform collects the produced pieces, tests each on a sample of viewers to discover its final characteristics, and allocates pieces to viewers to maximize engagement.

Equilibrium consumption of a given content type depends on viewers’ taste for the features distinguishing it *and* on creators’ taste for producing it. Even if viewers dislike a content type’s distinguishing features, the more pieces of that type are produced, the more opportunities for a piece to land so close to viewers’ preferences on *other* dimensions that they consume it.

The influence of creators’ supply-side tastes on consumption is mediated by two factors. First, supply-side tastes matter more when viewers have high-dimensional and unpredictable preferences. Such preferences prevent professional creators from consistently optimizing for viewer tastes and create local scarcity of content close to viewers’ ideal points. Consumption of a given content

type becomes determined not by viewers’ tastes for the few dimensions distinguishing it, but by how many pieces of the type are produced and therefore how many shots it has at landing close to viewers’ ideal points on the many other dimensions they care about. Second, the less creators care about attracting viewership on the margin, the weaker the feedback from viewers’ demand for content types to production of those types. These comparative-static predictions guide our analysis.

**Supporting evidence.** Our framework predicted that large-scale amateur content supply may pass through to consumption thanks to scarcity created by complicated and unpredictable viewer preferences. We confirm this prediction using our TikTok data. Content production is dominated by small creators who, when surveyed, report posting their preferred content in pursuit of creative fulfillment, expressive satisfaction, and social status. TikTok experimentally inserts videos from across the distribution of produced content into viewers’ feeds to test them for wider dissemination; we identify such “test” videos and show that test viewers’ engagement drives their subsequent reach.

Highly appealing videos emerge in an unpredictable, sparse, and fairly even way from across the distribution of production, from small creators to large influencers. This prevents consolidation of viewership around a few professional creators optimizing for viewer preferences: most of the videos in viewers’ feeds are one-off hits from small creators who never make money from content. The sparse and unpredictable origins of successful videos suggest highly specific and difficult to target viewer preferences. As our model predicts, such preferences create local scarcity of appealing content: although viewers engage less with “stale” content posted longer ago, after 30 minutes of scrolling, half their feed is stale, indicating a severe paucity of engaging fresh material.

**Social versus legacy media.** Having characterized social media’s business model of relying on broad-based amateur production to enable ex-post cherry-picking of unpredictably appealing pieces, we pause to compare it to that of legacy media. Legacy media’s professional production model was adapted to pre-Internet bottlenecks on content distribution (Waldfoegel, 2019). The Internet enabled rapid experimentation with, and subsequent personalized distribution of, media. This raised the returns to production *scale*, favoring media cheap to procure in large quantities—for example, low-effort, intrinsically-motivated social posts—and media disaggregated enough to facilitate granular cherry-picking—for example, short-form media. Hence the rise of short-form social media.

**Supply and demand for partisan and lifestyle content.** We then move on to study whether certain social media content types might be prevalent for supply- rather than demand-driven reasons. To motivate our focus, we show that the creators dominating viewers’ feeds are an unrepresentative minority selected on attractiveness and political extremity, which predict production of self-centered lifestyle and partisan political content. These content types, mainly driven by lifestyle, make up a third of the typical TikTok feed and about half of the content created on TikTok.

Our model clarifies the identification challenges facing any attempt to separate demand- from supply-side preferences for these content types. The videos in viewers’ feeds are selected on the realization of unpredictable appeal shocks, making it impossible to infer viewer tastes from observational variation in engagement. Inability to observe the latent viewership of videos not produced in

equilibrium similarly confounds inferences about viewer tastes from aggregate viewership patterns. And certain content types may be produced at high rates because creators enjoy producing them or because creators chase viewer demand for those types.

**Viewership experiment.** To identify demand-side preferences, we build a TikTok clone and populate it by randomly sampling from the pool of videos created on TikTok and adding videos whose production was induced by the randomized offers from our creator experiment. We recruit 586 TikTok users to scroll through it for 20 minutes and measure their engagement behavior.

Our results show that lifestyle content faces a clear demand penalty relative to other content types. For example, viewers are about 4 percentage points (SE 0.4), or 26%, less likely to watch a random lifestyle video for at least ten seconds compared to an entertainment video. However, consistent with viewers caring about many dimensions of appeal other than those captured by our coarse typology, viewers still engage with a substantial minority of lifestyle videos. This creates room for supply to matter. For example, if lifestyle videos are 26% less likely to highly engage a viewer than entertainment videos, they only need a 36% supply advantage over entertainment in order to appear in viewer feeds at a higher rate. In fact, they enjoy a 71% supply advantage.

Observational evidence corroborates the finding that lifestyle content faces a demand penalty. TikTok’s Creator Rewards program pays eligible creators for each view they receive on videos at least a minute long. Financially motivated creators pursuing viewership should therefore bunch above 60 seconds. Our data show that crossing from 59 to 60 seconds discontinuously reduces the prevalence of lifestyle content—which has the lowest estimated demand in our experiment—and increases the prevalence of entertainment—which has the highest estimated demand. This bunching is specific to accounts eligible for Creator Rewards and emerges immediately after its rollout.

Meanwhile, partisan political content enjoys a small demand advantage over non-partisan content in our experiment. In observational data the partisanship of political content is smooth across the 60-second threshold, indicating no demand premium, perhaps because (as we show) the general population of TikTok users is more moderate than the participants in our experiment.

To distinguish genuine preference dispersion from noisy behavior, we ask participants at the end of the feed to choose between pairs of videos they saw, and use the concordance of their choices with their prior engagement to estimate and purge behavioral noise; this barely affects results.

Because we are interested in the probability that a video of a given type will fall far enough into the upper tail of a viewer’s preferences to make it into their feed, we estimate a parametric model of viewer utility that allows us to interpret and extrapolate from the experimental engagement data. Our experimentally-derived estimates pass a range of overidentifying tests in the observational data: they reproduce the distribution of consumption on TikTok given the distribution of production, and quantitatively match several patterns relating to the shape of the upper tail of viewer preferences.

**Supply experiment.** Our viewership experiment shows no substantial demand premium for lifestyle or partisan political content, suggesting their considerable supply advantage may reflect intrinsic creator preferences rather than attempts to chase demand. To confirm this, we conduct an experiment

attempting to disentangle the reasons creators choose to produce certain content types but not others.

We recruit 979 TikTok creators. For each of several content types, we elicit their incentivized willingness to make a post of this type; their incentivized expectations about viewership; and their incentivized marginal valuation of viewership. We calculate the last object by eliciting their willingness to pay for an intervention designed to increase their post’s viewership and dividing it by their incentivized expectation about the intervention’s effects. We use their marginal valuation of viewership and their indicated indifference points between different increases in viewership at varying starting points to calibrate a simple parametric model of their utility from viewership. The randomized offers we extend at the end of the survey induce the creation of 111 TikTok posts.

We find that there is substantial variation in creators’ willingness to post different content types but almost none of this variation is explained by expectations about demand. Creators expect only slightly lower viewership for content types they do not post in equilibrium, and in fact receive slightly *higher* viewership for these types when paid to post them. Moreover, although creators highly value attention, their *marginal* valuation of viewership is low, reflecting strongly diminishing marginal utility from attention, consistent with prior work (Srinivasan, 2024).

Instead, creators appear willing to supply certain types of content primarily because they find them easy or satisfying to create, and unwilling to supply other types because it would be difficult or uncomfortable to do so. When asked, 82% of creators give such supply-side explanations for their answers, and more than half of creators unwilling to post a content type in equilibrium report that doubling their expected views for that type would not increase their likelihood of posting it.

Our results are robust to a much simpler design varying whether we ask creators about their binary willingness to *post* a certain content type versus their willingness to *privately create* a piece to be shown to a few other experimental participants. Responses to these questions are highly correlated: for example, more attractive participants are more willing to post lifestyle content *and* more willing to privately create a lifestyle video they are prohibited from posting; and more partisan participants are more willing to post *and* to privately create political content. This suggests enviable and partisan users derive more expressive satisfaction from creating content about themselves and their views, and are more comfortable having a few other people see it; and that this, rather than expectations of higher on-platform demand, drives their greater willingness to post.

Overall, our results suggest that the prevalence in production of lifestyle and partisan political content reflects supply-side preferences rather than attempted catering to demand.

**Quantification.** We combine the estimates from our viewer and creator experiments to estimate the parameters of our model and perform decompositions. We have two sources of demand-side data—our viewership experiment, and viewership expectations from our creator experiment—and results are robust to using either source. We find that equalizing creators’ intrinsic preferences for producing different content types would reduce equilibrium consumption of both lifestyle and partisan political content by one-third to one-half.

**Other platforms.** For data quality reasons, our experiments and descriptive results focus on TikTok. However, we show that our descriptive findings replicate on Instagram and X/Twitter. YouTube looks closer to television than to the other social media platforms: it is a highly professionalized environment where viewership is much more consistent and predictable. Participants in our surveys express much more positive views of YouTube than other social media platforms.

**Implications.** Our framework predicts social media viewers may be matched to content they find aversive on certain creator-driven dimensions, but this should be compensated by greater appeal on other dimensions. To suggestively test this, we link the viewership histories of our participants to their contemporaneous emotions collected via our iPhone app. Envious lifestyle and political content drive the bulk of negative emotions experienced on social media. However, participants rarely report regretting having watched such videos, and report finding them entertaining, interesting, and otherwise appealing. In a followup survey where we show participants creators from their feeds and ask if they would like to block them, participants do not want to block envious or political creators at a higher rate than other creators. These results are consistent with our framework’s predictions, but the outcomes we collect are too short-run to support confident welfare conclusions.

**Related literature.** We contribute to two primary branches of literature. The first studies the political economy of media; the canonical question in this literature, reaching back to Mullainathan and Shleifer (2005) and Gentzkow and Shapiro (2006, 2010), is whether features of equilibrium media content (such as slant) reflect supply or demand.<sup>3</sup> Because most papers in this literature study legacy media, the supply-side distortions they look for—including editorial pressure from wealthy owners or repressive governments—are examples of *top-down* capture of media by enfranchised elites (Herman and Chomsky, 1988). We are the first to study a different kind of supply-side distortion: *bottom-up* capture of media by the masses of people willing to spend much of their time posting online, who may be *negatively* selected on factors like institutional enfranchisement and satisfaction with the status quo. The Internet-precipitated transition from media susceptible to top-down pressures to those susceptible to bottom-up pressures offers one lens through which to interpret political trends over the last 20 years in liberal democracies, including the rise of populist parties and fringe ideologies and declining popularity of mainstream parties and institutions.

We also contribute to the large literature on social media (Aridor et al., 2024), the vast majority of which studies social media’s *consumption* side.<sup>4</sup> We make two contributions. First, we are the first to provide systematic quantitative evidence on social media’s *production* side, alongside a new theory of social media centering the features and implications of its unique supply side.<sup>5</sup> Second, by focusing on the supply side, we connect the strands of research on social media’s *political*

<sup>3</sup>See also Besley and Prat (2006); Durante and Knight (2012); Gentzkow et al. (2015); Martin and Yurukoglu (2017); Qin et al. (2018); Martin and McCrain (2019); Beattie et al. (2021); Szeidl and Szucs (2021); Boxell and Conway (2022); Cagé et al. (2025); Noy and Rao (2025).

<sup>4</sup>See Allcott et al. (2020, 2022, 2025b); Levy (2021); Braghieri et al. (2022, 2024, 2026); Acemoglu et al. (2024a,b); Aridor et al. (2024, 2025); Bursztyn et al. (2025b, 2026, 2019); Solheim and Wang (2026); Bao (2026); Beknazar-Yuzbashev et al. (2024, 2025b,a); Kalra (2025); Fujiwara et al. (2024).

<sup>5</sup>See also Srinivasan (2024); Filippas et al. (2025); Balbuzanov et al. (2025); Donati and Song (2026).

and *mental health* effects by noting that a common supply-side mechanism could underlie both: self-selection into content production by creators prone to causing negative reactions in viewers.

Finally, as we discuss later, our comparison of social and legacy media extends the insights of a small literature studying the effects of the Internet on legacy entertainment industries (Brynjolfsson et al., 2003; Waldfogel, 2016, 2017, 2019; Aguiar and Waldfogel, 2018).

**Roadmap.** Section 2 introduces our data. Section 3 theoretically and empirically describes social media’s business model. Section 4 motivates our focus on lifestyle and partisan political content and uses experiments to decompose their equilibrium prevalence into demand- and supply-driven components. Section 5 explores welfare implications. Section 6 concludes.

## 2 Observational Data

We provide an overview of the observational data we collect, with further details in Appendix C.

**Recruitment.** We recruit social media *viewers* from the survey platforms Prolific and MobLab, as is standard in the literature (e.g., Allcott et al., 2025a), and to a small extent via TikTok ads. We recruit *creators* via Prolific and through a large-scale emailing and messaging campaign targeting creators who appear in the viewership histories of our respondents (see below).

Our recruitment procedures involve one innovation relative to preceding literature: across most data collections, we require respondents to begin by logging in with their social media accounts using an authentication flow we developed drawing on the TikTok Login Kit, the X/Twitter `users.read` API, and Google OAuth. The tool collects and authenticates respondents’ social media profiles and checks their numbers of posts and followers before letting them into our surveys, allowing us to verify that they genuinely use these platforms, to automatically screen out respondents whose accounts fall below certain size thresholds for our creator experiments, and to scrape respondents’ accounts to recover the contents of their recent posts.

**Viewership histories and emotion-sampling.** Our data on the contents of users’ social media feeds and their contemporaneous emotions come from a panel of 393 Americans aged 18-30 recruited from Prolific.<sup>6</sup> We pay participants to download our iPhone app, which pings them randomly throughout the day—with particularly high frequency when it detects they are using social media—and asks about their emotions. After a week of using the app, participants download and send us their viewership histories on Instagram, TikTok, and YouTube. Their final payment depends on their submission of those histories and response rate to our emotion sampling notifications. Attrition is 50%, but because we have no randomized treatment, selective attrition is not a concern.

Instagram, TikTok, and YouTube comply with data ownership regulations by giving users the option to download their data, which is where our viewership histories come from. These histories contain second-by-second records of each post the user has viewed and each search they have made, reaching back one week, six months, and all time, respectively, for the three platforms. We link

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<sup>6</sup>Our hypotheses and empirical strategy for estimating the effects of social media content on emotions were pre-registered on the Open Science Foundation at <https://osf.io/8ctq7>.



these histories by timestamp to the emotion responses collected via our app, allowing us to observe exactly which post a respondent was viewing when they were pinged by our app.

X/Twitter’s “request my data” feature does not yield comparable data, so we follow up with participants and pay them to briefly use a Chrome extension that records the top 300 posts on their X/Twitter feed; we obtain these data, which are not linked to emotions, for 57 respondents.

**Random sampling of produced posts.** In addition to viewership histories for a panel of users, TikTok offers a Research API that allows us to randomly sample from the pool of posts produced on the platform in the US in any specified date range. This allows us to compare a representative sample of the content *produced* on TikTok to the TikToks appearing in viewers’ feeds. For this reason, as well as other data-quality reasons described in Appendix C, we focus most of our results on TikTok, showing results for other platforms in Appendix B (to the extent our data allow).

**Post metadata and classification.** We take the post URLs appearing in our viewership histories and API-based random sampling and scrape comprehensive characteristics of those posts, including captions, transcripts, metadata (including like and view counts and flags for commercial sponsorships or ad spending), and thumbnail images. We assign these posts to categories using GPT-5-mini and a classification schema described in Appendix C, developed on the basis of open-ended data gathered from social media users and validated by human coders.

**Creator survey.** We conduct a large-scale descriptive survey of the creators who produce content on the platforms we study. Our viewership history data, which allow us to see the characteristics of creators appearing in viewers’ feeds, allows us to re-weight our survey responses to create a sample representative (in terms of observables) of the creators appearing in viewers’ feeds. We use randomized incentives to test for non-representativeness on unobservables (Dutz et al., 2026).

We contact creators in two ways. First, we scrape the profiles of creators who appear in our respondents’ viewership histories and extract email addresses where available. Second, we recruit creators on Prolific, using our automated log-in flow to automatically screen out creators with fewer than 200 followers or 10 posts (TikTok) or 500 followers (X/Twitter).

**User survey.** To measure selection into content creation, we survey 3,490 users of these platforms recruited through Prolific. We ask respondents to log in with their accounts, allowing us to automatically grab their numbers of followers and posts and to scrape the contents of recent posts from their profiles.<sup>7</sup> We then elicit survey outcomes and correlate those outcomes with features of their posting behavior, as described below.

### 3 The Social Media Business Model

Section 3.1 develops a framework for thinking about the business model of social media platforms. Section 3.2 documents a sequence of empirical facts supporting key features of the framework. Section 3.3 compares the social media business model to that of legacy media outlets, explains

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<sup>7</sup>Since Instagram does not expose anything allowing us to build a login tool, we ask users directly for their handles.

which technological advances enabled it, and shows why it gives more weight to creators’ supply-side preferences and favors short-form content.

### 3.1 Framework

We begin by developing a model of social media that defines our demand- and supply-side objects of interest, explains how they interact to produce equilibrium content consumption, and generates predictions about the conditions under which supply-side preferences exert greater influence on consumption.

**Microfoundations.** Underlying our model is the assumption that pieces of content are drawn from a high-dimensional characteristic space  $\mathcal{X}$ . Viewers have common preferences over some (“vertical”) characteristics in this space and conflicting preferences over other (“horizontal”) characteristics. Viewer  $i$ ’s utility from consuming piece  $p$  with characteristics  $x_p$  is  $u_i(x_p)$ .

When we refer to “types” of content indexed by  $k$ , we mean a coarse partition of  $\mathcal{X}$  into subsets  $X_1, \dots, X_K$ . For example, given a binary characteristic  $x^1 \in \{1, 2\}$ , one such partition is  $X_1 = \{x \in \mathcal{X} : x^1 = 1\}$ ,  $X_2 = \{x \in \mathcal{X} : x^1 = 2\}$ . In this example, the content types  $k = 1, 2$  differ along the focal dimension  $x^1$ ; *within* each type, different pieces still vary on all other dimensions, generating residual variation in their appeal to viewers. If  $k = 1$  represents partisan political content and  $k = 2$  represents moderate content, residual variation in characteristics could include variation in topic, style, phrasing, visuals, and so on, all of which may matter to viewers. When we refer to “high-dimensional preferences,” we mean there are many residual dimensions viewers care about beyond the focal dimensions captured by our content types.

Critical to our framework is the assumption that content production is an unpredictable process: a creator producing a piece may target a general region of  $\mathcal{X}$ , but the final characteristics of the piece—and hence its appeal to potential viewers—emerge with some randomness.

Our assumption that pieces of media emerge stochastically from a high-dimensional characteristic space is motivated by two stylized facts from the literature on media economics (Caves, 2003; Waldfogel, 2019). First, it is extremely difficult to predict ex-ante which pieces of media will resonate with audiences and which will not. This is sometimes called “Goldman’s Law,” after the screenwriter William Goldman’s insistence that “the single most important fact, perhaps, of the entire movie industry [is that] *nobody knows anything*” about which projects might succeed (Goldman, 1983).<sup>8</sup> Quantitative studies of legacy entertainment industries bear out this assertion (De Vany and Walls, 1999; Waldfogel, 2019). The Law also seems to be in force on social media, where creators routinely express profound puzzlement about what draws audiences to their content and why seemingly-identical videos end up differing wildly in performance (Andrews,

<sup>8</sup>Such quotes abound in entertainment industries. The legendary director Sidney Lumet confessed that “I don’t know what makes a hit; I don’t think anyone does” (Lumet, 1995). The president of CBS told the *New York Times* that “all [TV] hits are flukes” (Carter, 1990); the former head of Paramount told a reporter that “if I had turned down every picture I greenlit, and greenlit every picture I turned down, I’d have the same number of hits and flops” (Finke, 2013). At the aptly named Penguin Random House, an editor insisted that “publishing is really gambling” (Euchner, 2019).

2020; Sweet, 2023; Tait, 2023; Cruz, 2024; Ronson, 2026; O’Meara, 2026).<sup>9</sup>

Second, viewer preferences exhibit extreme heterogeneity: for example, different evaluators’ assessments of media correlate only very weakly for movies (Wallisch and Whritner, 2017), images (Vessel and Rubin, 2010), jokes (Rosenbusch et al., 2022), and poems (Belfi et al., 2018).

These facts both arguably stem from the impossibility of tracing the experiences evoked by media to a small set of identifiable dimensions of content (Sibley, 1959). Pieces of media succeed when viewers find them funny, moving, suspenseful, or transporting, reactions produced by unpredictable collisions between subtle characteristics of the media and consumers’ personal cognitive processes and memories. This makes it hard to optimize for consumer preferences and means that coarse classifications of content capture only a small fraction of the dimensions mattering to viewers.

**Setup.** Because the inscrutable and high-dimensional nature of viewer preferences prevents us from comprehensively identifying and measuring the utility-relevant characteristics of the space  $\mathcal{X}$ , we write the model in terms of the reduced-form objects we will directly estimate.

The model involves viewers  $i \in I$ , potential creators  $j \in J$ , content types  $k = 1, \dots, K$ , and a platform intermediating between creators and viewers.

**Viewers.** Viewer  $i$ ’s utility from post  $p$  of type  $k$  is:

$$u_{ip} = m_k + \tilde{m}_{ik} + q_p + e_{ip} \quad (1)$$

Here,  $m_k$  represents viewers’ average taste for pieces of type  $k$ ;  $\tilde{m}_{ik}$  is individual  $i$ ’s personal taste for type  $k$  relative to other viewers;  $q_p$  is the piece’s vertical quality, drawn upon its creation; and  $e_{ip}$  is its idiosyncratic appeal to viewer  $i$ , also drawn on its creation. Residual variation in utility from a piece conditional on its content type—in  $\tilde{m}_{ik}, q_p, e_{ip}$ —hence captures a combination of viewer preference heterogeneity and residual variation, both predictable and unpredictable, in a piece’s characteristics given its content type. For example, if viewers generally dislike partisan content  $k$ ,  $m_k$  may be low; but some pieces of partisan content may have high  $q_p, e_{ip}$ , representing other characteristics viewers find appealing, such as well-placed jokes or a resonant style. Viewer preferences can be written this way without loss of generality relative to our microfoundations.

A viewer’s utility from post  $p$  in a sequence is  $u_{ip} - dh_{k(p)}$ , where  $d$  is a parameter capturing diminishing marginal utility from a content type, and  $h_{k(p)}$  is the share of posts in the sequence of  $p$ ’s type. This restricts diminishing marginal utility to operate homogeneously within coarse type.

**Creators.** Each period, each creator  $j$  draws a separate cost shock  $\kappa_{jk}$  associated with creating each type of content  $k$ , drawn from a common type-I extreme value distribution with scale  $1/\psi$ . The creator then picks up to one type and creates a unit of that type. Within type  $k$ , each creator belongs to one of a finite number of internally homogeneous groups indexed by  $g$ . The quality  $q_p$  and vector of shocks  $\{e_{ip}\}_{i \in I}$  of a piece of type  $k$  created by a member of group  $g$  are drawn from a type- and group-specific joint distribution  $F_{kg}$ .

<sup>9</sup>The unpredictability of success may also result from audience herd behavior or strategic complementarities in consumption, a possibility we discuss in Section 3.2.

If creator  $j$  produces a unit of content of type  $k$  which receives  $v$  views, their payoff is:

$$\pi_{jk} = \mu_k + \tilde{\mu}_{kg} + \beta v^\rho + wv - \kappa_{jk} \quad (2)$$

Here,  $\mu_k$  represents creators' average taste for producing pieces of type  $k$ , and  $\tilde{\mu}_{kg}$  is a group-specific deviation from this average taste.  $\beta v^\rho$ , where  $0 < \rho < 1$ , represents non-pecuniary satisfaction from attracting views.  $wv$ , where  $w \geq 0$  is the wage-per-view paid by the platform, represents the pecuniary rewards to viewership.  $\kappa_{jk}$  is the stochastic cost shock.

Creators' supply-side preferences are captured by the type-specific  $\mu_k$ , which allow the fixed cost, expressive satisfaction, or status utility associated with producing a piece of content to vary by content type. We could also allow  $\beta$  to vary by type, letting viewership bring more non-pecuniary satisfaction for some content types than others. We impose a homogeneous  $\beta$  for simplicity and because this restriction turns out to be inconsequential; empirically, we will show that allowing type-specific  $\beta$  results in very similar  $\beta$  estimates across types and does not change our main results.

**Platform.** So far, the model could describe either social or legacy media. Our assumptions now about the platform's *distribution technology* single out social media, as we discuss in Section 3.3.

We first assume that the platform sets a wage-per-view  $w \geq 0$ , and, given this wage, potential creators decide whether and what to produce, resulting in a pool of produced content.

Our model is intended to describe platforms such as Instagram, TikTok, and X/Twitter, where viewers who open the app scroll one by one through a vertical feed of algorithmically curated posts. We therefore assume that viewers enter the platform with a reservation value of time  $\underline{u}_i$  and sequentially consume the posts they are served until their utility from the marginal post falls below  $\underline{u}_i$ . We assume the platform extracts viewer surplus through some form of price discrimination, for example personalized ad loads (Beknazar-Yuzbashev et al., 2025a), and hence maximizes viewer surplus net of wage costs by serving viewer  $i$  all produced posts  $p$  satisfying  $u_{ip} - \underline{u}_i > w$ .

Two empirical premises underlie this assumption. First, we assume the platform is able to identify which pieces will appeal enough to a viewer to include in her feed. We assume this is accomplished through a preliminary experimentation step where the platform serves a new piece  $p$  to a small subset of viewers and infers its quality  $q_p$  through their average engagement and the distribution of its  $e_{ip}$  by combining their engagement with an estimate of the covariance matrix of viewer preferences based on past behavior ("viewers like you liked..."). We do not model this experimentation step, but we trace it out empirically in Section 3.2. We could allow a component of  $e_{ip}$  to be purely idiosyncratic at the individual level and therefore opaque to the platform when assigning content; we show later that this does not affect our empirical results.

Second, we assume that conditional on a video's immediate surplus to a viewer, the platform does not discriminate based on content type. In reality, the platform may wish to apply an algorithmic "markup"  $\delta_k$  to certain types of content, serving them to viewers whenever  $u_{ip} - \underline{u}_i > w - \delta_k$ , if consumption of those types raises its profits beyond immediate effects on viewer surplus. For example, this would be the case if consuming those types has positive spillover effects on future

consumption or engagement with advertisements (Becker and Murphy, 1988, 1993), or if creators of that type particularly value viewership and increasing the reach of their content encourages them to produce. We show in Appendix D that such algorithmic markups can be detected observationally using a method following Kleinberg et al. (2018) and Rambachan (2024); the idea is that if a content type enjoys a markup, viewers' engagement with marginal posts of that type in their feeds should be predictably lower than for marginal posts of other types. We find that advertisements and sponsored videos are algorithmically marked up according to this measure, as expected, but find no evidence of algorithmic markups favoring any of our content types of interest.

**Timing and equilibrium concept.** The game proceeds as follows. (1) The platform sets a wage  $w \geq 0$ . (2) Each creator draws  $\kappa$  for each content type and decides whether and what to produce. (3) Created pieces draw  $q_p$  and  $\{e_{ip}\}_{i \in I}$ . (4) Viewers  $i$  watch all pieces  $p$  for which  $u_{ip} \geq \underline{u}_i + w$ .

An equilibrium consists of a wage  $w^*$ , and, for each  $w \geq 0$ , creator strategies mapping cost shock realizations  $\{\kappa_{jk}\}_k$  into actions, such that creators best-respond to each other given each wage, and  $w^*$  maximizes expected viewer surplus minus wage payments given creator strategies.

**Equilibrium characterization.** Our first result characterizes the equilibrium of this game. Throughout our results, we assume that the platform's profit is single-peaked in its wage.

**Proposition 1.** *An equilibrium exists. If there is no diminishing marginal utility within content types ( $d = 0$ ), the equilibrium is unique and can be characterized recursively as follows.*

1. *A creator from group  $g$  producing content type  $k$  faces a cumulative distribution  $D_{kg}$  of expected views that depends only on demand-side parameters:*

$$D_{kg}(v) = \mathbb{P}_{(\{e_i\}_{i \in I}, q) \sim F_{kg}} \left[ \sum_{i \in I} \mathbb{1}\{m_k + \tilde{m}_{ik} + q + e_i - \underline{u}_i > w\} \leq v \right] \quad (3)$$

2. *The supply of pieces of type  $k$  from creator group  $g$  can be written as a function  $S_{kg}(D)$  of its viewership distribution  $D$ , where  $S_{kg}(\cdot)$  depends only on supply-side parameters:*

$$S_{kg}(D_{kg}) = M_{kg} \frac{\exp(\psi(\mu_k + \tilde{\mu}_{kg} + \mathbb{E}_{D_{kg}}[\beta V^p + wV]))}{1 + \sum_{\ell} \exp(\psi(\mu_{\ell} + \tilde{\mu}_{\ell g} + \mathbb{E}_{D_{\ell g}}[\beta V^p + wV]))} \quad (4)$$

where  $M_{kg}$  is the mass of creators in group  $g$  for type  $k$ , and  $1/\psi$  is the scale of the TIEV creator cost shocks.

3. *Equilibrium consumption of type  $k$  is given by  $C_k = \sum_g S_{kg}(D_{kg}) \mathbb{E}_{D_{kg}}(v)$ .*

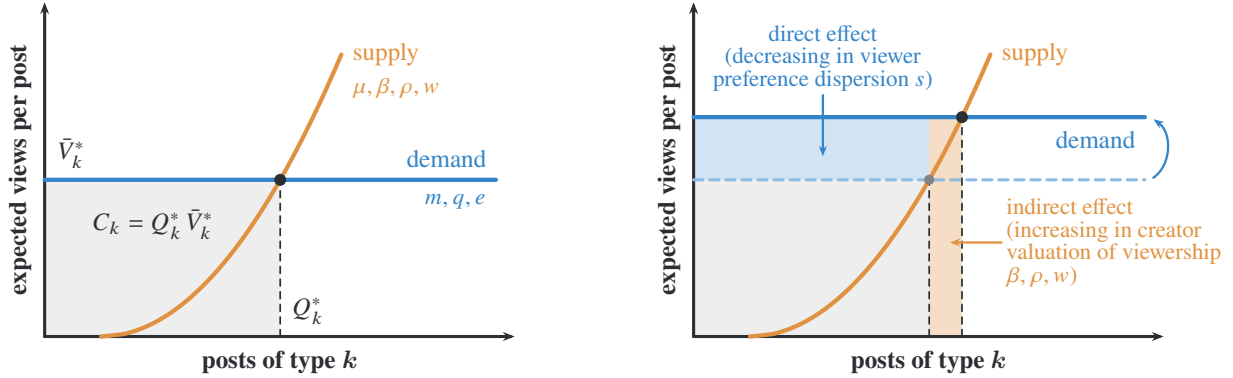
Proposition 1 says that three sets of parameters determine equilibrium consumption of content type  $k$ . The interaction of these parameters is visualized in Figure 1 Panel (a), which plots the quantity of type- $k$  posts produced on the  $x$ -axis and the average viewership of type- $k$  posts on the  $y$ -axis.

First are the demand-side parameters capturing viewers' taste for type  $k$ : the more a viewer likes the distinctive characteristics of type  $k$ , the higher the fraction of type- $k$  posts she will want to watch. In Figure 1 Panel (a), these parameters determine the height of the blue demand curve—the average viewership a post of type  $k$  receives.

**Figure 1:** Supply, demand, and equilibrium content consumption

(a) Equilibrium consumption  $C_k$  of content type  $k$

(b) Comparative statics: parameters mediating the effect of increase in average appeal  $m_k$



Notes: This figure plots the “demand” and “supply” curves determining equilibrium consumption of content type  $k$  under the simplifying assumption that creators make entry decisions according to the mean of expected viewership rather than its full distribution.

Second are the supply-side parameters  $\mu_k$  affecting creators’ taste for producing content type  $k$ . The more that creators enjoy producing type  $k$ , the more pieces they will create. Even if viewers dislike the distinctive characteristics of type  $k$ , meaning  $m_k$  is low, more production entails more draws of  $q$  and  $e$ —more possibilities that a type- $k$  piece will have *other* characteristics appealing enough to make it worth watching. For example, even if a viewer dislikes partisan political posts, greater production of such posts creates more opportunities for a particular piece’s topic, personality, or visuals to interest or entertain this viewer enough to overcome her distaste for partisanship.

In Figure 1, higher values of  $\mu_k$  shift the orange supply curve horizontally outwards by making creators more willing to supply content at any given level of expected viewership.

Third are the supply parameters  $\beta, \rho, w$  governing creators’ utility from attracting viewership. These parameters mediate the feedback from demand to supply: the more that viewers like watching type  $k$ , the more viewership a creator will expect for producing type  $k$ , and therefore the more that creators will choose to produce type  $k$  in pursuit of wages and attention.

Because there is no cross-type congestion in our model—viewers consume all posts above their constant opportunity cost of time—equilibrium consumption of each type is determined separately. Allowing congestion by giving viewers a finite consumption budget or an increasing opportunity cost of time would not affect any of our core insights about relative consumption rates across content types because it would simply alter the way the cutoff for entering a feed is determined. Similarly, allowing diminishing marginal utility within content type,  $d > 0$ , introduces within-type congestion, which simply attenuates differences in consumption across types without changing any directional conclusions. In both cases, allowing for these factors introduces the possibility of multiple equilibria through dynamics that are empirically implausible given the number of actors involved and the constant churn on platforms.

**Comparative statics.** Two factors govern the influence of supply- relative to demand-side preferences on equilibrium consumption. Proposition 2 explores these factors by analyzing the effect of increasing content type  $k$ 's mean appeal to viewers  $m_k$  on its share of consumption.

**Proposition 2.** *Suppose  $d = 0$  and each content type has only one creator group, so expected consumption of type  $k$  can be written as  $C_k = Q_k \bar{V}_k$ , where  $Q_k$  is the expected number of creators choosing to produce type  $k$  and  $\bar{V}_k$  is expected viewership per type- $k$  post.*

*Compare content types  $k = 1, 2$  and consider the effect of increasing the average appeal  $m_1$  of type 1 on their relative consumption. This can be written as the sum of a direct effect holding fixed relative production, and an indirect effect allowing relative production to adjust:*

$$\frac{d \log(C_1/C_2)}{dm_1} = \underbrace{\frac{\partial \log \bar{V}_1}{\partial m_1}}_{\text{direct}} + \underbrace{\frac{\partial \log(Q_1/Q_2)}{\partial m_1}}_{\text{indirect}}$$

*Moreover, the direct and indirect terms obey the following comparative statics:*

- (i) *Without loss of generality, write  $q_p + e_{ip} = s\xi_{ip}$ , where  $s > 0$  controls the variance of the stochastic appeal shocks  $q$  and  $e$ . The direct effect is decreasing in  $s$  if  $\xi_{ip}$  has an increasing generalized failure rate, the distribution of viewers' reservation values of time is log-concave, and each reservation value of time exceeds the average appeal of type-1 posts.*
- (ii) *The indirect effect is increasing in  $\beta$ ,  $\rho$ , and  $w$ .*

**Preference dispersion.** The first factor is the residual *dispersion* of viewer preferences conditional on a piece's type, meaning the variation of the vertical quality  $q$  and horizontal preference  $e$  shocks. Higher dispersion attenuates the effects of demand-side preferences  $m_k$  and amplifies the effects of supply-side preferences  $\mu_k$ , for the following reason.

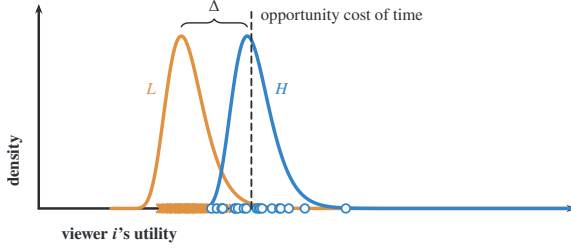
Consider two types of content,  $L$  and  $H$ . In Figure 2 we plot the distributions of utility that pieces of  $L$  and  $H$  content offer a focal viewer  $i$ . Their average appeal determines the means of the distributions, while the dispersion of the  $q$  and  $e$  shocks affects the distributions' variance. We fix the gap between the average appeal of  $L$  and  $H$  at  $m_H - m_L = \Delta$ .

Suppose the variance of the  $q$  and  $e$  shocks is low, as in Panel (a). This means viewer  $i$  has low-dimensional preferences: there are few residual characteristics of content they care about conditional on a piece being  $L$  or  $H$ . Then no matter how many  $L$  pieces are produced, none stand a shot at appealing to  $i$  enough on residual dimensions to overcome her average distaste for  $L$  and end up in her feed. Even a large supply-side force favoring  $L$  will not pass through to consumption.

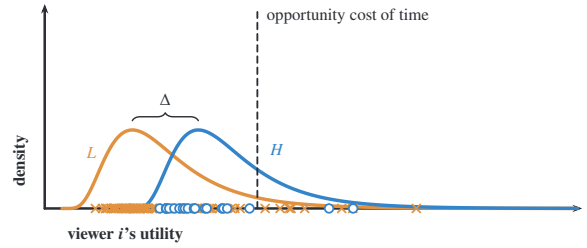
By contrast, suppose viewer  $i$  cares about many dimensions of content other than those captured by the  $L$  vs.  $H$  distinction, so the variance of  $q$  and  $e$  is high, as in Panel (b). Then the first-order determinant of type  $k$ 's representation in consumption becomes not its ex-ante average appeal to viewer  $m_k$  but *how many pieces of type  $k$  are produced* and therefore how many  $q$  and  $e$  draws type  $k$  generates. The mathematical property underlying this is the following: given a distribution

**Figure 2: Preference dispersion and pass-through of production to consumption**

(a) Low dispersion, low pass-through



(b) High dispersion, high pass-through



*Notes:* This figure shows the effects of increasing viewer preference dispersion—the variance of the  $q$  and  $e$  shocks—holding fixed  $m_H - m_L$  and the expected amount of content falling above the focal viewer’s opportunity cost of time.

$X$ , scalars  $s, \Delta > 0$ , and a threshold  $t > \mathbb{E}(X)$ , the ratio  $\frac{\mathbb{P}(sX \geq t)}{\mathbb{P}(sX - \Delta \geq t)}$  is decreasing in  $s$  and attenuates towards 1 as  $s \rightarrow \infty$  as long as  $X$  has an increasing generalized failure rate. The latter is a mild condition on tail shape satisfied by all commonly used distributions in economics.

**Preference dispersion and scarcity.** A more intuitive way to think about the importance of viewer preference dispersion is through the lens of scarcity. Suppose there are a fixed number of slots in viewer  $i$ ’s feed, and  $i$ , all else equal, prefers moderate to partisan political content. Our hypothesized mechanism is that if partisan content enjoys a supply-side advantage, the platform can find pieces of partisan content that viewer  $i$  prefers to the marginal available moderate pieces because the partisan pieces better match her preferences on *other* dimensions (such as style, topic, or personality).

This presupposes local scarcity of content closely matching viewer  $i$ ’s preferences. If the platform’s content library were infinite, any given partisan post could be replaced in the viewer’s feed with a moderate post equally appealing to the viewer on all residual dimensions. The platform would hence serve viewer  $i$  partisan content only if  $i$  had a taste for partisanship. Given the enormity of modern social media platforms, with hundreds of millions of TikToks produced each day (Steel et al., 2025), we might expect their content libraries to be effectively infinite.

Preference dispersion endogenously creates local scarcity of the content most appealing to a viewer. As dispersion grows, the content in viewer  $i$ ’s feed comes from the increasingly thin upper tail of  $q$  and  $e$  draws, where content is scarce and sparsely distributed and pieces have no close substitutes. The underlying mathematical property is that for any distribution  $X$  with a finite mean and scalar  $s > 0$  the average gap between the  $n$ th and  $n + 1$ th highest of  $N$  draws is increasing in  $s$ .

The curse of dimensionality offers one way to make this kind of upper-tail scarcity quantitatively intuitive. If content is characterized by fifty binary dimensions over which viewers have preferences, the probability that a random piece is in a viewer’s top preference bucket is  $\frac{1}{2^{50}}$ , or about one in a thousand trillion. If a platform has a billion pieces of content, a viewer’s favorite hundred will on average be mismatched from her on seven out of fifty dimensions, and from each other on twelve.

**Creator sensitivity to viewership.** Proposition 2 also notes that the parameters  $\beta, \rho$ , and  $w$  mediate the indirect feedback effect from demand to supply. If creators face a high wage-per-view or derive



substantial nonpecuniary satisfaction from attracting more views, they will be pulled towards producing the content types demanded by viewers, amplifying the importance of demand-side tastes  $m_k$  and diminishing the importance of supply-side tastes  $\mu_k$ . By contrast, if wages are low and creators derive little satisfaction from viewership *on the margin*—either because  $\beta$  is low or because  $\rho$  is low, reflecting scope-insensitive or strongly concave nonpecuniary satisfaction from viewership—this feedback effect weakens.

Equilibrium wages on platforms such as Instagram, TikTok, and X/Twitter are very low. Appendix G shows this can occur for multiple reasons. It can occur if nonpecuniary motivations suffice to induce substantial production, meaning higher wages largely transfer money to inframarginal creators. It can also occur due to a market failure *exacerbated* by inter-platform competition (Jeon and Rey, 2024). Suppose platforms cannot enforce exclusivity contracts on creators. (They in fact do not, likely due to practical difficulties discussed in the Appendix.) Then a platform could raise its wages to induce new content creation, but could not appropriate the resulting viewer surplus because this content would also be cross-posted on rival platforms—a practice our survey of creators shows is commonplace. We remain agnostic between these explanations for low prevailing wages.

**Summary.** The model just outlined shapes our subsequent empirical investigation in two ways.

First, it helps us define the objectives of our empirical analysis. We have shown how to write equilibrium content consumption in terms of separate objects depending only on demand- and on supply-side preferences; in Section 4 we estimate these objects and analyze counterfactuals in which we equalize either the supply- or demand-side preferences for different content types in order to decompose observed consumption patterns into supply- versus demand-driven components.

Second, it generates two predictions about the conditions under which supply-side preferences will influence equilibrium consumption. In Sections 3.2 and 4.3, we show that viewer preferences exhibit high dispersion and that this dispersion creates local scarcity of content matching viewer preferences and attenuates demand differences across content types, as our model predicts. In Section 4.4, we show that creators’ marginal valuation of viewership is low due to low wages and strongly concave non-pecuniary rewards to attention.

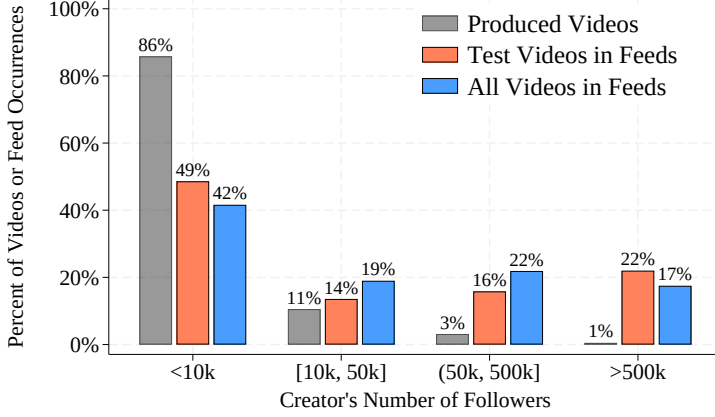
## 3.2 Evidence: amateur production, experimentation, and scarcity

The previous section outlined a model in which social media platforms solicit a pool of content from potential creators, experimentally test pieces from this pool on subsets of viewers to discover their partly stochastic vertical and horizontal characteristics, and act on this information by separately cherry-picking the pieces ex-post most appealing to each viewer.

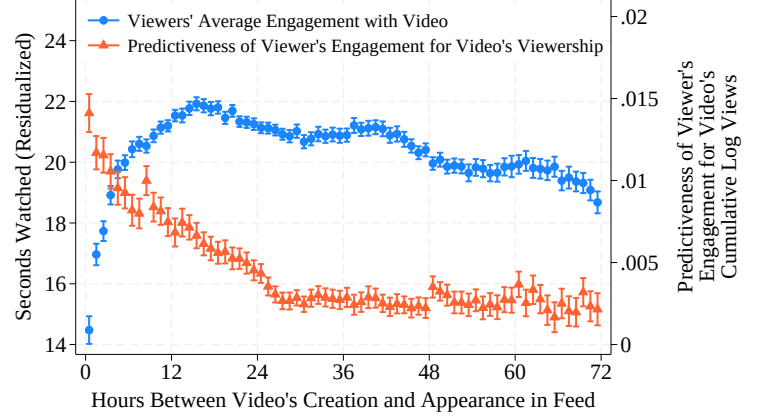
We now draw on our social media data to descriptively document each step of this process. We show that social media platforms benefit from a broad base of production mostly driven by non-pecuniary motivations; that they experimentally uncover appealing pieces of content, which emerge unpredictably from across the whole base of production; and that, as a consequence, most of the content in viewers’ feeds comes from small, non-professional creators. We also show that

**Figure 3: Production, experimentation, and consumption on TikTok**

(a) Distribution of produced, tested, and consumed videos across account sizes



(b) Identifying tested videos



*Notes:* Panel (a) plots the percentage of videos falling into each creator-size bucket in three different samples: (1) a random sample of the videos produced on TikTok, from its Research API; (3) all videos appearing in the histories of participants who sent us their personal TikTok data; and (2) the subset of history videos that appeared in a participant’s feed less than 5 hours after being posted. Panel (b) uses videos in participants’ feeds and plots the number of hours elapsing between the video being posted and appearing in the focal viewer’s feed against (i) the average number of seconds watched by the participant (controlling for viewer by video duration fixed effects and time since the beginning of the session) and (ii) the slope between the focal viewer’s engagement and the video’s total log viewership scraped several months afterwards.

viewer preferences exhibit extreme unpredictable dispersion, resulting in local scarcity of appealing content—a feature Proposition 2 predicted would amplify the influence of the supply side.

Because our richest linked data on production and consumption come from our TikTok viewer-ship histories plus TikTok’s Research API, results in this section focus on TikTok.

**Distribution of production.** The grey bars in Figure 3 Panel (a) use a representative sample of all videos produced on TikTok between November 2025 and February 2026. Panel (a) plots the distribution of the follower counts of these videos’ creators, a measure of those creators’ size and prominence. The panel shows that nearly 90% of production on TikTok comes from very small creators with fewer than 10,000 followers; the median follower count in this bucket is 723.

If a viewer’s TikTok feed were mainly filled with recent posts from accounts they follow, these low-follower creators would account for a large share of production but only a small share of consumption—their share of followership is 7.5%. However, all social media platforms have mechanisms allowing a creator’s post to propagate past their own followers. This mechanism could be a simple “share” button, as with pre-algorithmic versions of Facebook and Twitter. In TikTok’s case, this mechanism is the TikTok algorithm, which fills viewers’ feeds mainly with videos from accounts they do *not* follow but that TikTok predicts they will engage with (Zannettou et al., 2024).

TikTok’s algorithmic recommendations are informed by a preliminary experimentation stage. When a video is posted, TikTok observes basic information about it, such as its creator and hashtags, and acquires more information by applying machine-learning techniques to its transcript and imagery (Boeker and Urman, 2022). However, this featurization step still leaves TikTok with

incomplete information; it therefore serves the video to a pool of “test” viewers and uses their engagement to infer the video’s quality and distribution of potential appeal.

**Identifying test videos.** How deep into the creator size distribution does TikTok reach to draw its test videos? Does it assume that non-followers will never be interested in a video from a small account, or does it believe that widely-appealing gems regularly emerge from such accounts? Answering this requires us to identify instances where TikTok serves a video to a viewer as a test. Panel (b) of Figure 3 does this. We use our participants’ TikTok histories and plot, on the  $x$ -axis, the temporal gap between a video’s appearance in a viewer’s feed and its creation. Low  $x$ -axis values represent videos that appear in a viewer’s feed shortly after being posted.

The blue coefficients in Panel (b) show that viewers’ engagement with videos appearing in their feeds is roughly flat on average for videos appearing between 6 and 72 hours after their creation—as we would expect from an engagement-maximizing algorithm that equates viewers’ marginal propensity to engage with different types of videos. But viewers predictably dislike videos that appear in their feeds less than 6 hours after being posted. This suggests these videos are being served not because the algorithm predicts they will engage the viewer but for an ulterior reason. Indeed, the orange coefficients show that a viewer’s engagement with such a video is highly predictive of the total viewership later accumulated by the video. This predictiveness declines rapidly for videos appearing in a viewer’s feed a longer interval after being posted, consistent with TikTok having already learned these videos’ quality and gradually serving them out to marginal viewers.

“Appearing in a viewer’s feed less than 6 hours after being posted” therefore appears to be a reasonable proxy for being a test video. These videos make up 7% of the typical feed. Half of them are from creators the viewer has never seen before, and in these cases they tend to come from very small creators, with a median follower count of 1,697.

**Distribution of testing and consumption.** The orange bars in Panel (a) of Figure 3 show that although TikTok does statistically discriminate in favor of large creators when deciding which videos to test, it draws test videos from across the distribution of creators. Even though creators in our smallest bin make up less than a tenth of followership, they produce half of all tested videos.

The blue bars plot the distribution of follower counts in participants’ full histories—a measure of final consumption. The blue distribution looks similar to the orange, suggesting that, conditional on being tested, videos from large and small accounts are similarly likely to be broadly disseminated. Overall, nearly half of the videos in viewers’ feeds are from very small accounts.

**Amateurs vs. professionals.** The fact that nearly half of the videos in viewers’ feeds are from very small accounts suggests that, contrary to what we might expect, it is not true that viewers mostly see content produced by professional “influencers” catering to viewers in pursuit of financial rewards.

Indeed, Panel (a) of Figure 4 uses scraped platform data to estimate the percentage of videos in viewers’ feeds that are monetized. Green bars represent income from Creator Rewards, TikTok’s program paying creators based on how many views they receive. Eligibility for the program requires an account with at least 10,000 followers, at least 100,000 views in the preceding 30 days, and a

video that is at least a minute long. Only 9% of the videos in people's feeds satisfy these criteria and therefore at most 9% earn income from Creator Rewards.

Purple bars represent income from third-party sponsors who pay a creator in exchange for featuring their product. These sponsorships must legally be reported, meaning sponsored videos should have tags our scraper can detect. The figure shows that only 8% of the videos seen by viewers receive any sponsorship income. (Our sample includes ads, meaning videos someone has paid to boost into viewers' feeds; 7 out of 8 of these percentage points are ads.)

Appendix Figure A13 instead asks: how many of the videos people see come from creators who *ever* earn money, as suggested by their other videos? The results show that, depending on the conservativeness of our assumptions, at most a third of the videos viewers see on TikTok come from accounts earning money on the platform.

At least two-thirds of the videos TikTok viewers see are therefore produced by non-professionals producing for non-pecuniary reasons. What are these reasons? Panel (b) of Figure 4 plots responses from our creator survey, weighted to be representative of the creators in viewers' feeds.

We first ask respondents how they choose what content to create: do they produce the types of videos that will get the most views? The types that will attract the highest financial payouts? Or the types they personally find satisfying to produce? Responses to such a question may be influenced by social desirability. We therefore randomize the wordings of the response options between two options intended to flip their perceived social desirability, and confirm in a supplementary survey detailed in Appendix C that our manipulations succeed. The bars in the figure plot average responses across both conditions, while the + and – symbols show average responses in each condition.

The figure shows that respondents allocate between half and two-thirds of their points to the first answer: the most important consideration determining their production choices is what they personally find satisfying to produce.

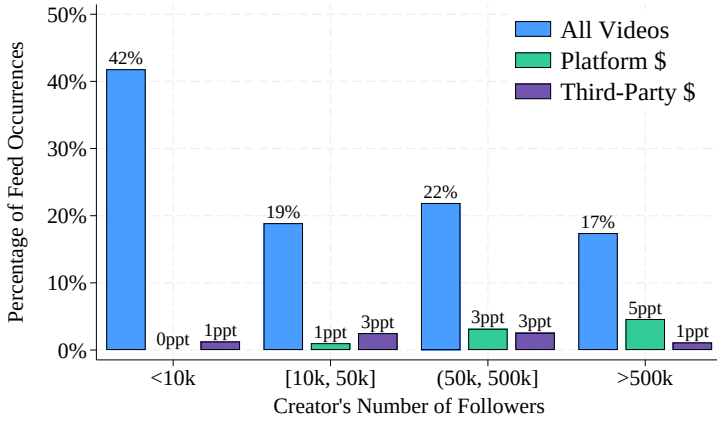
We also ask creators to indicate which kinds of non-pecuniary satisfaction they derive from content production. Panel (b) shows that creative fulfillment and expressive satisfaction are among the most popular answers, while a third of creators report gaining social status through their content.

**Dispersion and scarcity.** The videos that receive positive engagement signals when tested, and end up widely disseminated to viewers, are unpredictable outliers relative to their creator's typical output. The 10th-percentile tested video in terms of ultimate viewership goes on to receive 445 views, while the median receives 11,400; but the 90th-percentile tested video receives 1.9 million.

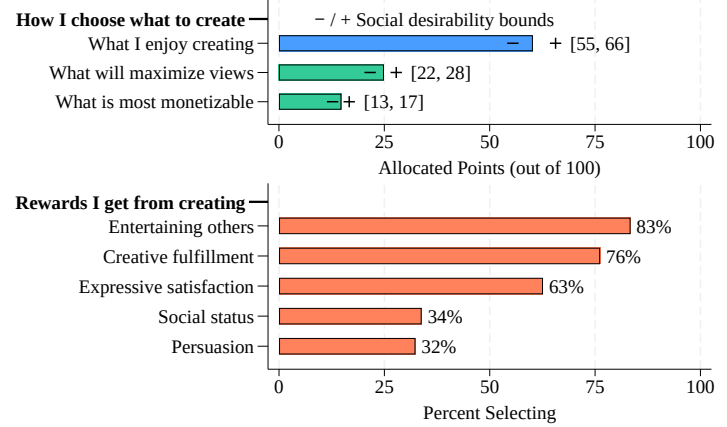
Panel (c) of Figure 4 shows that the median video in a viewer's feed receives 37 times more views than its creator's median viewership on their other videos. For videos posted by creators with fewer than 10,000 followers, this ratio is 212. Panel (d) shows that nearly half of the videos in viewers' feeds are from creators they see once and never again (in the six-month window covered by their TikTok history). These creators have a median of 6,716 followers; half of viewers' feeds are hence filled by one-hit wonders from small creators whose content does not typically perform well, as evidenced by their low follower count.

**Figure 4: Non-professional production, viewer preference dispersion, and local content scarcity**

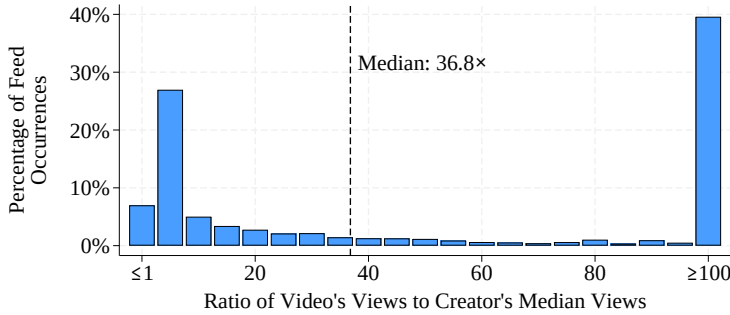
**(a) Non-pecuniary motivations: What fraction of posts in viewers' feeds earn money for their creators? (Admin data)**



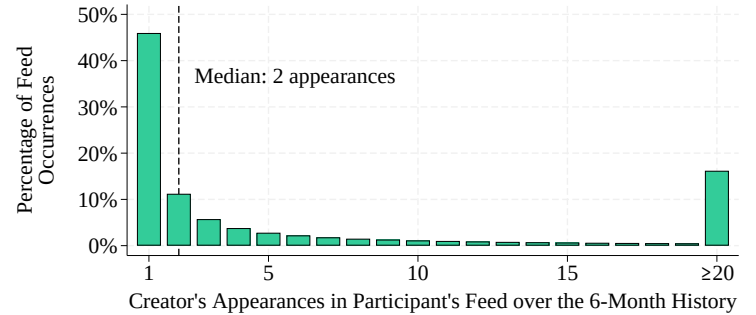
**(b) Non-pecuniary motivations: Which considerations influence the content creators choose to make? (Creator survey)**



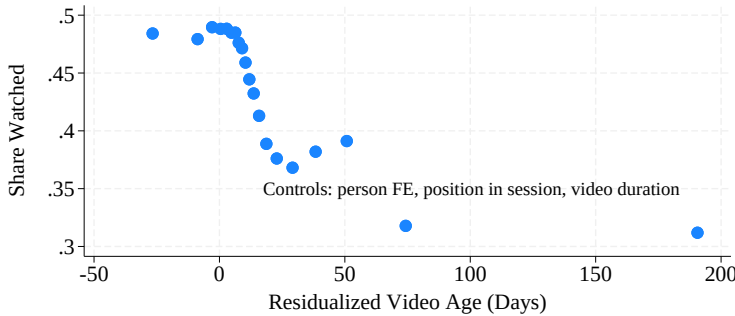
**(c) Dispersion: The videos viewers see are in the far upper tail of their creators' views**



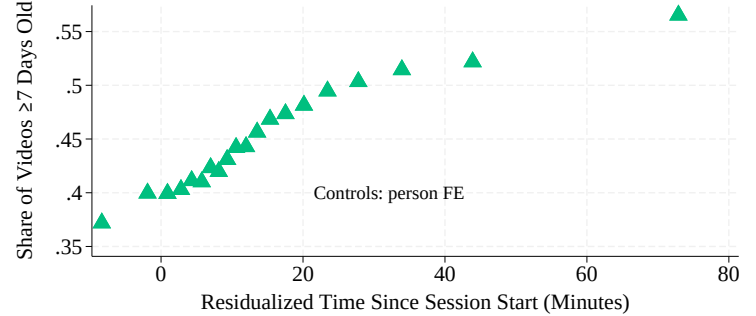
**(d) Dispersion: Nearly half the videos viewers see are from creators they see once and never again**



**(e) Scarcity: Stale videos receive less engagement, but...**



**(f) Scarcity: ... nearly half a typical viewer's feed is stale**



**Notes:** Panel (a) plots videos from the histories of participants who sent us their TikTok data. An observation is a video occurring in a participant's feed. The panel plots in blue bars the percent of observations in each creator size bucket. The green and purple bars plot how many observations in that bucket (i) are eligible for Creator Rewards (creator has at least 10k followers and 100k views in past 30 days and video is at least 60 seconds long and not sponsored) or (ii) are tagged as sponsored. Bars are in percentage points, meaning, for example, that 1 percentage point out of 42 percentage points in the first bucket are sponsored. Panel (b) plots responses from our creator survey, with responses weighted by follower count and posting type to be representative of the creators viewers see in their feeds. The first sub-panel asks respondents to allocate 100 points between three possible considerations determining which types of content they choose to post. Answer options are 50/50 randomized between two sets intended to flip rankings of perceived social desirability as detailed in Appendix C. The second asks them to "tick all that apply" out of a list of non-pecuniary rewards they derive from producing content. Panels (c) and (d) use TikTok histories and the same units of observation as Panel (a). Panel (c) plots the ratio of a video's view count to the median view count of other videos posted by the same creator over the same period spanned by our histories. Panel (d) plots the number of times this observation's creator appears in the viewer's history. Panel (e) uses TikTok histories and plots the share of video a viewer watches against the number of days since the video was posted (controlling for viewer by video duration fixed effects and time since the beginning of the session). Panel (f) plots the share of videos in a viewer's feed that were posted at least 7 days ago as a function of time since the viewer started this scrolling session, controlling for viewer fixed effects.

The ex-ante unpredictable origins of the videos in viewers’ feeds suggest that viewer preferences are hard to target in advance. This explains the failure of professional creators to consistently dominate viewership and suggests viewer preferences cannot be reduced to a small number of targetable dimensions such as video style, production value, or slant.

High-dimensional preferences, combined with a compensatory or non-lexicographic utility function in which a viewer derives utility from each additional dimension of a piece matching their preferences (Tversky, 1969; Fishburn, 1976), in turn produce the kind of extreme residual dispersion in appeal depicted in Figure 2. The fact that the videos in viewers’ feeds are outliers in terms of final viewership is suggestive of this kind of dispersion. However, fat-tailed viewership could also reflect small differences in appeal being amplified into enormous differences in dissemination à la Rosen (1981). In our model, this would manifest as a large density of videos whose qualities place them close to viewers’ opportunity cost of time  $\underline{u}$ . In this case, videos in viewers’ feeds could be outliers in terms of viewership but still have many close substitutes in terms of quality.

To distinguish these possibilities, we test directly whether the videos in viewers’ feeds lack close substitutes in terms of utility. Panel (e) of Figure 4 shows that viewers engage predictably and substantially less with “stale” videos—videos posted more than a week before they are served to a viewer. If, each day, TikTok receives an abundant supply of fresh content closely matching a viewer’s preferences, it should never serve this viewer stale videos. But Panel (f) shows that from the moment the typical viewer begins scrolling, more than a third of the videos in her feed are stale; by the time she has scrolled for 20 minutes, this has risen to more than half. This indicates substantial scarcity of highly appealing content, and the magnitude of this relationship—stale videos receive at least a 30% engagement penalty yet occupy up to half of the typical feed—suggests that the marginal appeal of available content falls very rapidly.

Overall, these patterns indicate substantial dispersion of the  $q$  and  $e$  shocks in our model, which Proposition 2 predicted would amplify the influence of supply- relative to demand-side preferences.

The final alternative interpretation is that videos in viewers’ feeds do lack close substitutes in terms of utility, but this is endogenous to the fact that they have already gone viral—in other words, there is a utility premium to engaging with popular content. This interpretation is perfectly compatible with our story if “potential virality” is a property distributed in the same sparse and unpredictable way as intrinsic appeal. We also do not find evidence of a utility premium for engaging with popular content in our viewership experiment, where we randomize across participants whether videos’ “like” counts are visible; Appendix Figure A3 shows that making like counts visible does not increase engagement with highly-liked videos.

### 3.3 Social versus legacy media

We have characterized social media’s business model as follows: cheaply accumulate an enormous content library by harvesting the output of a broad base of small creators pursuing non-pecuniary rewards, and then ex-post cherry-pick the pieces most appealing to each viewer.

We now explain how and why this business model differs from that of legacy media outlets. Pre-Internet legacy media outlets faced the same unpredictable and heterogeneous consumer preferences (Caves, 2003; Waldfogel, 2017). But they were constrained by technological bottlenecks on distribution (Waldfogel, 2019). The broadcast spectrum only supported three TV channels; cable supported only up to a few hundred, due to cables’ limited bandwidth. Movie theaters had only a few screens. Bookstores could only stock so many books (Brynjolfsson et al., 2003).

This meant legacy outlets could distribute only a limited amount of content, preventing them from fully catering to preference heterogeneity. High distribution costs also prevented outlets from testing new pieces of content on large samples of consumers to learn their vertical quality before deciding whether to widely distribute them (Goldman, 1983; Lumet, 1995). They therefore professionally produced small portfolios of content ex-ante optimized, to the extent possible, to appeal to viewers, then distributed them widely, gambling on opening weekend (Waldfogel, 2019).

The Internet eliminated physical constraints on both personalized distribution and cheap at-scale collection of feedback. Unlike a TV network, a website with a large content library could serve a distinct feed to each visitor at zero marginal cost, and quickly collect feedback in the form of views and shares. This enabled sites with large libraries to test new pieces of content on subsets of consumers and use the resulting feedback to inform selective further distribution to others.<sup>10</sup>

One possible response to the Internet was to deploy these capabilities to streamline distribution of movies, TV shows, and books, increasing the productivity of legacy media. Indeed, this happened to an extent (Waldfogel, 2019), as exemplified by platforms like Netflix and Amazon.

But as with many new technologies (David, 1990), fully harnessing the Internet’s potential for media required a more fundamental redesign of the underlying product. By giving platforms the tools to sift through large libraries of content and locate the best pieces to distribute to each viewer, the Internet raised the returns to content library *scale* and thereby rewarded content that is *cheap to procure at scale*—for example, low-effort content that is satisfying to produce—and *disaggregated enough to facilitate granular cherry-picking and personalization*—for example, short-form content.

A toy version of our model helps illustrate. Suppose there are two types of content,  $L$  and  $H$ , with mean utilities to viewers  $m_L < m_H$  and per-unit procurement costs  $c_L < c_H$ . That is,  $L$  content is less appealing but is also cheaper to procure. When produced, a piece  $j$  of  $L$  or  $H$  content draws vertical  $q_j$  and horizontal  $\{e_{ij}\}_i$  appeal shocks from a distribution common to both types.

Suppose we represent the technological constraints of legacy media as an inability to observe and act on the  $q_j, \{e_{ij}\}_i$  shocks before distributing content to viewers. In this case, an outlet optimally procures a single unit of  $H$  content and distributes it to everyone. As soon as we introduce the

<sup>10</sup>This strategy was initially pioneered by Facebook and Twitter, neither of which initially used recommendation algorithms. Instead, they relied on users to surface and spread high-quality content by deciding what to share or repost, and to personalize their own feeds by deciding who to follow or connect with. They thereby reproduced the decentralized “testing” and dissemination of products through repeated marketplace interactions and social learning, but at vastly greater speed than physical markets. Fueled in part by data from these networks, advances in machine learning then led to algorithms that greased the wheels of this process. Many of the concerns about social media’s pathological effects on mental health or politics partly predate algorithms (see, e.g., Braghieri et al., 2022; Bursztyrn et al., 2019; O’Keeffe et al., 2011; Conover et al., 2011) but do not predate their reliance on amateur content.

Internet—allowing the outlet to experimentally uncover the  $q$  and  $e$  shocks and act on them via personalized distribution—the outlet can do better by procuring several pieces of  $H$  content. This strategy represents Amazon and Netflix. But it may be able to do even *better* by spending the same budget procuring a larger number of  $L$  pieces, compensating for their lower average appeal with a larger number of draws, enabling superior cream-skimming of the best  $q$  and  $e$  draws. This represents social media. The shift from prioritizing content types with high average appeal  $m$  to those with low procurement costs  $c$ —which ultimately derive from the supply-side preferences determining creators’ reservation wages—is the sense in which the social media business model increases the equilibrium influence of supply-side relative to demand-side preferences.

Moreover, suppose the  $q$  and  $e$  draws are separately realized across different subunits of longer pieces of content: for example, some of the jokes in a comedian’s stand-up routine are better than others, or some scenes in a movie appeal more to  $i$  and others to  $i'$ . Then an outlet able to observe and condition distribution on  $q$  and  $e$  benefits from splitting content into the smallest subunits over which these shocks are realized to enable maximally granular ex-post discrimination. Appendix F provides evidence that such discrimination is extensive on social media and summarizes two other mechanisms: (i) short-form content allows more rapid learning about viewer preferences by linking higher-frequency user interaction data to more disaggregated variation in exposure to content characteristics; and (ii) reducing the minimum required length of content lowers barriers to content production. The nascent “microdrama” industry operates according to this logic: produce large volumes of short, cheap entertainment, rely on algorithms to engage in granular experimentation and ex-post matching to viewers, and use variation in performance to learn which specific story beats and set pieces do well with viewers (Kolitz, 2026); see also Bao (2026). Finally, the appendix explains why transitioning to infinite-scroll feeds also follows naturally from this business model.<sup>11</sup>

## 4 Disentangling Supply and Demand on Social Media

We now move on to study one implication of the social media business model’s practice of ex-post cherry-picking from a pool of supplied content: the possibility that certain content types will be prevalent not because consumers prefer them on average, but because creators like producing them, resulting in a deeper produced pool facilitating additional cherry-picking.

Section 4.1 motivates the content types we focus on: status-oriented lifestyle and partisan

<sup>11</sup>Conventional explanations often claim social media content is short-form because the format hijacks our brain’s susceptibility to being tempted by uncertain, variable, and immediate rewards, like those offered by slot machines. But this fails to explain what makes social media different from TV, where content could also be structured to exploit these vulnerabilities, and plausibly *is*. Indeed, 20th-century critics accused TV channels of fragmenting viewers’ attention spans by serving them loosely connected sequences of short clips, often in eerily prescient language: Postman (1985) writes that “what [TV viewers] watch, and like to watch, are moving pictures: millions of them, of short duration and dynamic variety,” and Pynchon (1990) describes a satirical TV-detoxification facility as serving “a retinal diet of scientifically calculated short video clips of what in full dosage would, according to theory, have destroyed [the subject’s] sanity.” Conventional explanations can be rescued only by appending an additional claim about how the tactile sensation of swiping or scrolling somehow activates or exacerbates these psychological vulnerabilities.



political content. Section 4.2 reintroduces our equilibrium model and defines the objects we wish to estimate. Sections 4.3 and 4.4 describe the identification challenges we face and how we overcome these challenges through demand- and supply-side experiments.

## 4.1 Motivating evidence on lifestyle and partisan content

**Production and consumption of content types.** We begin by describing our content types of interest. We take the pool of *consumed* posts from TikTok users’ viewership histories and the pool of *produced* posts sampled from TikTok’s Research API, scrape the posts’ text and imagery, and use LLMs to classify them according to a detailed schema of content types we developed, described in Appendix C. We aggregate the detailed schema into four broad categories depicted in Appendix Figure A11: politics (posts about political events or issues); lifestyle (posts visually emphasizing features of the creator’s life, such as their appearance, living space, or routine); stories/advice (posts where a creator talks to the camera, relating a story, some gossip, or advice); and entertainment (a broad category spanning comedy sketches, hobby demonstrations, edited montages, and so on).

Panel (d) of Figure 5 plots the prevalence of these content types on TikTok, separately as a share of produced videos and as a share of the videos appearing in viewers’ feeds. The plot shows that lifestyle content makes up just under 30% of viewers’ feeds. But it makes up nearly *half* of videos produced on TikTok. Conversely, nearly 40% of viewers’ feeds are entertainment—substantial overrepresentation relative to a production rate of 27%. The first object we will focus on empirically is the relative consumption rates of lifestyle and entertainment content.

Political content makes up just under 5% of viewers’ feeds. Panel (e) plots the partisanship of political content: an indicator for whether a video expresses a clear partisan perspective. The second object we will focus on is the intensive-margin partisanship of political content.

About 60% of produced political content is partisan, slightly higher than the partisan share of consumption. The drop in partisanship from production to consumption is driven by videos posted by news organizations, which make up 27% of the political videos in viewers’ feeds but less than 1% of produced political videos. Less than 20% of news organizations’ political videos are partisan, compared to over 65% of user-generated political content.

This is not mechanical: our data contain partisan news outlets, such as Fox News, which has a 69% partisanship rate for its videos—slightly higher than that of user-generated content. Instead, Appendix Figure A2 Panel (a) shows it is because viewers seem to prefer nonpartisan news outlets. The top outlet for American political news among both Democrats and Republicans in our sample is the Daily Mail, a British tabloid; the next few slots are occupied by CBS, NBC, and CNN. Panel (b) compares the partisanship of the clips posted by news organizations on their TikTok accounts to the partisanship of random political news clips selected from their broadcasts, using the TV data from Noy and Rao (2025). The clips organizations choose to post are *less* partisan than their typical broadcasting and, on TikTok, their more partisan clips get fewer views.

**Selection into content creation.** Section 3.2 showed that most of the content viewers see on social media is produced by small, non-professional creators pursuing personal satisfaction. Figure 5 Panels (a)-(c) show that these creators are heavily selected, relative to the broader user population, on *enviability* and *ideological extremity*, motivating our focus on lifestyle and partisan content.

The figure draws on our survey of 3,490 users of Instagram, TikTok, X/Twitter, and YouTube. We ask users to log in with their accounts, allowing us to capture their total number of posts and scrape their recent posts. We link this information to two survey-elicited characteristics. The first, “enviability,” is the average of a user’s response to a question about how attractive they are and a question about how desirable other people would find their lifestyle. The second, “extremity,” is their distance from the midpoint of a seven-point scale from “very liberal” to “very conservative.”

Panel (a) shows that, conditional on being a user of a platform, both characteristics predict posting volume on the platform. On Instagram, users who post heavily are more enviable and ideologically extreme than non-posting users. On TikTok, heavy posters are more enviable but not more extreme. On X/Twitter—which is a text- rather than image- or video-based platform—heavy posters are substantially more ideologically extreme, but (perhaps humorously) *less* enviable. On YouTube, a highly professionalized platform that pays creators substantially more than Instagram, TikTok, or X/Twitter, creators are not selected on either characteristic.

Panel (b) shows that enviable users are more likely to produce *lifestyle* content specifically, while ideologically extreme users are more likely to produce political content.

The selection of enviable and extreme users into posting lifestyle and political content admits a supply-driven explanation according to which enviable or partisan users derive more status or satisfaction from posting lifestyle or political content holding viewership fixed; it also admits a demand-side interpretation according to which all creators simply pursue viewership, and enviable or extreme users expect their lifestyle or political content will attract more views.

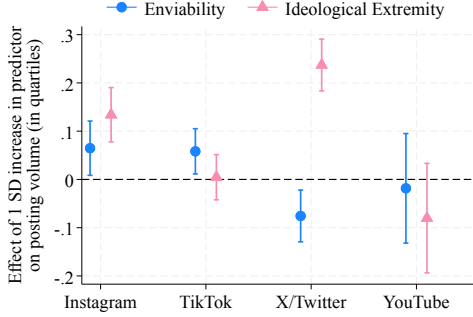
Distinguishing between these possibilities is the central objective of our creator experiment in Section 4.4, but our user survey provides a preliminary piece of evidence. For each of our video types, we ask 50% of respondents if they would be willing to *post* such a video on their account for \$3, and the other 50% if for \$3 they would be willing to privately *create* such a video and send it to us to be shown to a few other participants, but not post it. Willingness to perform the first task should be affected by expectations about the viewership such a post would receive, but willingness to perform the second task should only be influenced by the cost or expressive satisfaction of producing such a video. Panel (c) of Figure 5 shows that enviability and ideological extremity predict willingness to perform these two tasks with almost exactly equal strengths.

## 4.2 Objectives

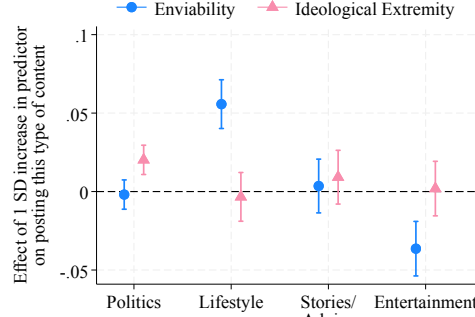
Proposition 1 in Section 3.1 established that equilibrium consumption of a given content type can be written as a function of two objects: the distribution of expected viewership of potential pieces of this type, which depends only on demand-side preference parameters; and the distribution of

**Figure 5: Selection into content production, and content production patterns, on social media**

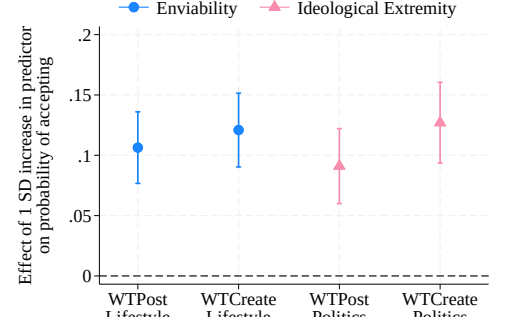
**(a) Enviability and extremity predict total posting volume**



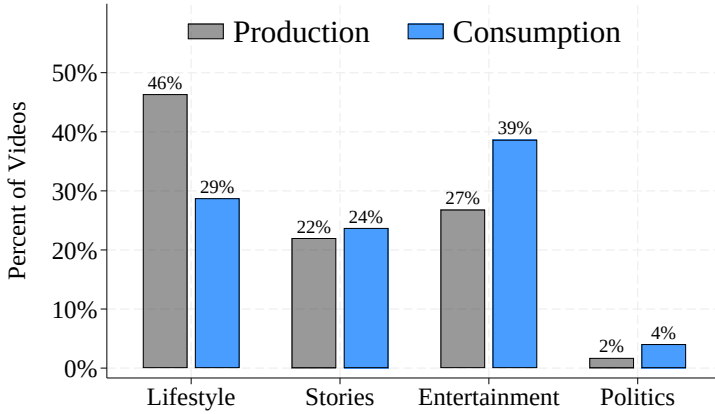
**(b) Enviability predicts lifestyle posting, extremity predicts political**



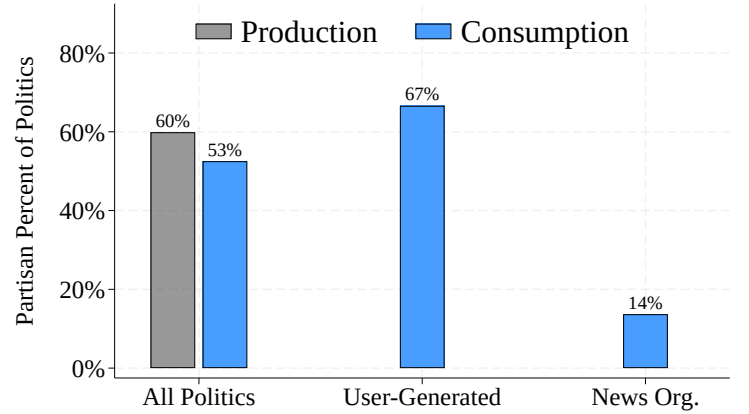
**(c) Enviability and extremity equally predict willingness to post and to create**



**(d) Production and consumption of content on TikTok**



**(e) Partisanship of political videos on TikTok**



**Notes:** Panels (a)-(c) use data from our survey of users of our four platforms of interest. Enviability is measured as the average response to two questions about self-rated attractiveness and lifestyle desirability. Ideological extremity is the distance from the midpoint of a 7-point self-reported ideology scale. Posting volume and the contents of recent posts are measured by requiring users to log in with their accounts, collecting their usernames, and scraping their accounts. Panels (a) and (b) plot the predictiveness of enviability and ideological extremity for total posting volume on a platform (conditional on being a user) and indicators for having recently posted specific content types. Panel (c) plots responses to questions offering respondents \$3 to either post (for 50% of respondents) or privately create (other 50%) different types of content. It correlates willingness to post/create lifestyle content with enviability and political content with extremity, controlling for average willingness to produce other types. Panel (d) uses a random sample of videos created on TikTok from the Research API (production) and TikTok viewer histories (consumption) and plots the percent of videos falling into each of our four aggregated content categories, described in Appendix C. Appendix Figure A1 reproduces Panel (d), measuring consumption by weighting produced videos by their cumulative view counts instead of by using participants' histories. Panel (e) plots the percent of videos classified as being about politics that are classified as partisan, separately for created videos, history videos, and videos in histories that are versus are not posted by news organizations.

creators' willingness to produce pieces of this type as a function of their expected viewership, which depends only on supply-side preferences.

Our goal in this section is to separately estimate the demand- and supply-side parameters underlying each of these two objects. Doing so will enable us to conduct Oaxaca decompositions where we break equilibrium gaps in consumption between content types into components driven by differences in their demand- versus supply-side parameters, and to calculate counterfactuals where we equalize supply-side parameters across content types and simulate effects on consumption.

### 4.3 Demand-side preferences

**Parametrization.** Recall that viewer  $i$ 's utility from piece  $p$  of type  $k$  is given by  $u_{ip} = m_k + \tilde{m}_{ik} + q_p + e_{ip} - dh_k$ , and that each type has associated "groups"  $g$  of potential creators, capturing heterogeneity in creators' taste and facility for creating type  $k$ .

We are interested in five content types: lifestyle, stories/advice, entertainment, partisan politics, and nonpartisan politics. We parametrize the components of  $u_{ip}$  as:

$$u_{ip} = m_k + \tilde{m}_{ik} + \underbrace{\bar{q}_{kg} + \xi_p}_{q_p} + \underbrace{e_{G(i)p}^G + e_{P(i)p}^P + e_{ip}^o}_{e_{ip}} - d h_k \quad (5)$$

where  $G(i)$  and  $P(i)$  are viewer  $i$ 's gender and political leaning; we allow for commonality in the idiosyncratic preferences of same-gender and same-leaning viewers. We assume that  $\tilde{m}_{ik}$  and the three parts of  $e_{ip}$  are normally distributed with mean zero and type-specific variances. The stochastic component of quality  $\xi_p$  follows a lognormal distribution shifted to have mean zero. We allow it to have a type-specific variance but keep its shape parameter common across types.<sup>12</sup>

Within each type, we assume there are ten creator groups  $g$  determined by a creator's follower count (in five bins) crossed with whether the creator produces this type of content in equilibrium. Posts not made in equilibrium, and the share of potential creators belonging to each group, are sourced from our creator experiment. Because of our limited number of posts from creators who do not produce in equilibrium, we assume that groups not producing in equilibrium face a constant type-specific penalty  $\delta_k$  to their quality. In our viewership experiment, sample sizes force us to further pool this parameter for lifestyle, entertainment, and politics; but in our alternative demand estimates sourced from creators' expectations in the creator experiment, we do not impose this restriction, and Figure 10 shows that the two produce almost identical results.<sup>13</sup>

**Identification challenges.** Barriers to identification stem from the fact that the pieces we observe of a given content type are selected. Videos in viewers' histories are selected: since the platform

<sup>12</sup>This is because we calibrate the shape to the observed shape of viewership on TikTok and wish to avoid type-specific observational calibration because our model's ability to reproduce the observed composition of viewership across types is a key overidentification test. Appendix Figure A6 shows the four content types have very similar shapes.

<sup>13</sup>Because of variation in creators' willingness to post content types, our creator experiment generates 23 stories posts that creators were willing to produce absent payment versus 19 unwilling; but for lifestyle, entertainment, and politics, these are 16/4, 21/3, and 6/14, respectively, limiting the power of willing/unwilling comparisons.

should equalize a viewer’s engagement with the marginal video of each type in their feed, videos from lower- $m$  content types should be more positively selected on the realization of  $q$  and  $e$  shocks conditional on appearing in a viewer’s feed. Sampling from the full pool of videos produced on TikTok ameliorates this problem. But the videos of type  $k$  produced on TikTok are still selected on  $\bar{q}_{kg}$  in a way that depends on creators’ supply-side preferences for producing type  $k$ . If creators like producing type  $k$ , they will be willing to produce it even when expecting lower viewership.

### 4.3.1 Viewership experiment

The design and reduced-form analysis of this experiment were pre-registered on the AEA RCT Registry at AEARCTR-0019565 and are described in more detail in Appendix H.

**Building a TikTok clone.** We build a phone-browser-based clone of TikTok. We download real TikTok videos, host them on a server, and pipe them and their metadata into users’ feeds. Appendix Figure A14 shows a screenshot of the clone. It reproduces all of TikTok’s main features: viewers swipe at their leisure through an infinite vertical feed; they can “like” a video, tap to open the top comments or expand the caption, or copy the video’s URL to share it with a friend. We record every aspect of users’ behavior, including how long they spend on a video, whether they “like” it, whether they open its comments or expand its caption, and so on.

**Sampling videos.** We randomly sample 260,000 videos posted in the days preceding the experiment using TikTok’s Research API, classify them, and select a stratified subset covering our four content types, yielding 1,900 politics videos and 5,000 videos each for lifestyle, entertainment, and stories/advice. We also scrape 400 videos recently posted by the hundred news organizations appearing most commonly in the feeds of participants who donated their TikTok viewership histories. Finally, we use 105 videos, including at least 20 of each type, whose creation we induced by paying a participant in our creator experiment \$10 to make a post of this type (described next section).

Each viewer receives a feed of videos randomly drawn from this pool, stratified by content type. The shares of lifestyle and political content in a viewer’s feed are drawn independently from  $\{0, 0.1, 0.2, 0.3\}$  with probabilities  $\{0.2, 0.2, 0.4, 0.2\}$ . Residual videos are split equally between entertainment and stories, with 5% reserved for non-political videos posted by news organizations. Political videos are stratified to be 80% user-generated and 20% posted by news organizations.

**Recruiting viewers.** We recruit Americans aged 18-30 from Prolific, verifying their status as TikTok users by requiring them to log in with their TikTok account to enter the experiment. After logging in, 624 participants start the experiment and 586 complete it, including 94 participants who had earlier donated their viewership histories. For these 94, we include in their feeds several dozen videos recently posted by creators who appeared regularly in their histories, and use their engagement with these videos for an overidentification test reported below. Appendix Figure A15 shows that attrition is not predicted by the content early in a person’s randomized feed and including incomplete sessions does not noticeably affect our results.

**Experimental flow.** Once a participant enters the experiment, we explain its procedures. We tell them they will scroll through TikTok videos for 20 minutes, and should behave as they ordinarily would on TikTok, swiping past videos they dislike and watching ones they like. We incentivize genuine engagement by telling them that they may be invited back to a future scrolling session, and that we will use their engagement behavior in the first session to design a personalized feed for the session just like the TikTok algorithm would operate. To ensure respondents actively watch and scroll, we randomly flash several attention checks throughout the 20-minute window that respondents must tap within a few seconds of their appearance in order to receive their payment.

After 20 minutes of scrolling, we present the respondent with 5 pairs of videos they saw and ask them to indicate which they liked more. Comparing their choices to their prior engagement behavior lets us measure behavioral noise. Finally, participants answer a few demographic questions.

**Identification of parameters.** Identifying utility parameters from viewers’ engagement behavior requires a model linking engagement to utility and accounting for the fact that decisions to swipe past a video are endogenous to the viewer’s outside option of watching the next video in the feed.

We assume that when viewer  $i$  swipes into video  $p$ , her flow utility of watching it starts at  $u_{ip} - dh_k$ , where  $h_k$  is the share of already watched videos that are of the same type and  $d$  captures diminishing marginal utility within type, and her utility declines linearly at rate  $a_k$  as she continues watching. Opportunities to consider swiping to the next video arrive as a Poisson process with rate  $r$ . When one arrives after  $\tau$  seconds, she compares  $u_{ip} - a_k\tau + v_{ip}$ , where  $v_{ip}$  is a behavioral noise term, to her value of starting the next video  $V_i$  minus a swiping cost  $c_i$ .

A more parsimonious engagement model could be written but would struggle to fit to observed engagement behavior, which exhibits several patterns: (i) viewers vary systematically in their average watch duration; (ii) about half of videos are swiped past within the first two seconds; (iii) conditional on watching two seconds, viewers exit gradually with a rate that varies across types.

Figure 6 Panel (a) summarizes how each parameter in Equation (5) is identified from experimental engagement behavior. Intuitively, viewers’ average taste for a content type is identified from their average watch time conditional on swiping into a video of this type; variances of vertical quality and horizontal fit are identified through the covariance matrix across viewers of engagement with videos of each type; and diminishing returns within type are identified from our extensive-margin randomization of the share of a viewer’s feed drawn from each type. The engagement model parameters are identified from the shapes of watch time distributions.

**Reduced-form results.** Figure 6 Panel (b) shows results for one of our pre-registered engagement measures: the probability that a viewer watches at least 10 seconds of a video. Our other pre-registered measures, which produce consistent patterns, are plotted in Appendix Figure A16.

Three patterns are evident. First, lifestyle content suffers a substantial demand penalty: it is the least engaging type. Partisan political content, particularly content concordant with a viewer’s ideology, is more engaging than non-partisan content, but this difference is quantitatively modest.

Second, differences in engagement rates across types are modest, with a substantial minority

of videos of each type engaging viewers. For example, 15% of entertainment videos, the most engaging type, are watched for at least 10 seconds, while the figure is 11% for the least engaging type, lifestyle. This is consistent with high-dimensional viewer preferences attenuating differences in the probability that videos of different types will highly engage viewers: if our coarse content categorizations capture variation along only a few of the dimensions viewers care about, all types will have some videos that appeal to viewers on enough dimensions to appear in their feeds.<sup>14</sup> Such attenuation amplifies the importance of the supply side: if lifestyle videos are  $\approx 30\%$  less likely to highly engage a viewer than entertainment videos, a 43% supply advantage suffices to make lifestyle videos appear in viewers' feeds more commonly than entertainment.

Such attenuation could be produced by genuine preference dispersion or by noisy behavior. We separate the two with the end-of-experiment module where we ask participants to retrospectively indicate which video they preferred out of five pairs they saw. Respondents affirm their earlier revealed preference in 82% of cases (85% in the condition where the module tells them which they engaged with more, and 80% when it does not). Assuming the 18% disagreement rate reflects noise in engagement behavior, we conduct an empirical Bayes procedure to disattenuate engagement differences across content types. Appendix Figure A17 shows this barely changes results.

Third, note that if content production primarily caters to demand, videos produced in equilibrium should engage viewers substantially more than videos not produced in equilibrium. By contrast, if there is important variation in supply-side preferences, some videos may not be created in equilibrium despite facing high latent demand because they would be difficult or unsatisfying to create. The green bars in Figure 6 Panel (b) show that when we pay a participant in our experiment \$10 to produce a video of a type they were *unwilling* to produce for free, these videos receive just as much engagement on average as videos participants *were* willing to create for free, and just as much as actual scraped videos. This suggests variation in production decisions is primarily driven by supply- rather than demand-side preferences, a conclusion our creator experiment affirms.<sup>15</sup>

**Parameter estimates.** Panel (c) of Figure 6 shows estimates of our key utility parameters. We normalize the mean utility of entertainment to zero and the total variance of utility to 1. Results line up qualitatively with Panel (b), with a few deviations since the parameter estimates exploit richer variation in engagement than whether videos are watched for at least 10 seconds.

Two conclusions are again evident. First, entertainment is the highest-utility type, lifestyle is the lowest, and partisan political content enjoys a modest advantage over nonpartisan content. Second, residual preference dispersion is large enough to substantially attenuate the effects of variation in

<sup>14</sup>Since viewers watch 14% of experimental videos for at least ten seconds, we are capturing attenuation in the probability that a video enters their top 15%, rather than their top 0.0001%, which is what is relevant for feed assignment. But this means our results should understate the degree of relevant attenuation as long as the taste shock distribution has a decreasing hazard rate. Appendix Table A1 compares distributions fitted to the upper tail of video viewership. Distributions with a decreasing hazard (lognormal, Pareto, and Weibull) fit well, while those with a constant hazard (exponential) do not.

<sup>15</sup>Marginal created and uncreated videos within a type should experience equal viewership, but the blue bars in the figure capture average inframarginal engagement with created videos, and the first green bar captures average engagement with uncreated videos down to \$10 below the creation threshold.

average appeal. For example, the normalized utility gap between entertainment and lifestyle is 0.29, but the standard deviations of both the vertical quality  $q$  and idiosyncratic appeal shocks among posts estimated to exceed a viewer’s utility threshold are just above 1.

**Simulating consumption.** Using our parameter estimates to simulate consumption requires one more step: estimating the utility threshold above which videos appear in viewers’ feeds. We calibrate this to the probability that a video we randomly sample from TikTok’s Research API appears in a given participant’s viewership history (about 1 in 44,000).

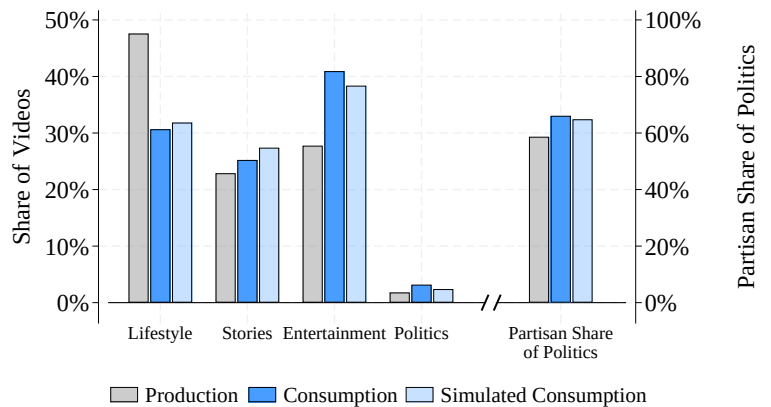
This allows us to take the observed distribution of production across content types (and account sizes within content types) from Figures 3 and 5 and use our demand estimates to simulate the resulting distribution of consumption. Figure 7 shows results. Despite using only experimental variation and one piece of observational information (the 99.9th percentile of view counts among randomly sampled TikTok videos), our demand estimates closely reproduce actual consumption patterns across content types. This is our first and most important overidentification test.

**Other fit and overidentification tests.** Appendix H describes and shows results of our other fit and overidentification tests. Our model closely fits experimental watch-time distributions and variation in watch times across content types. Moreover, its predictions pass a series of overidentification tests in our observational data. The most salient concern about our experimentally-derived estimates is that they will not extrapolate to the thin upper tail of utility where the videos assigned to feeds come from. We therefore focus on whether our model quantitatively reproduces relationships between engagement and feed assignment in the observational data. This includes the relationship between test video engagement and subsequent viewership (Figure 3); the predictiveness of experimental engagement for actual viewership of our experiment’s videos on TikTok; participants’ experimental engagement with videos from creators who appear regularly in their actual TikTok histories; and the relationship between videos’ age and their prevalence in viewer feeds (Figure 4).

**Observational evidence on demand for lifestyle and partisan content.** Figure 8 provides observational evidence directionally corroborating our experimental demand estimates.

The figure shows systematic bunching in the production of different types of content at different video lengths. It exploits two facts. First, TikTok’s interface for filming videos offers default

**Figure 7:** Production, consumption, and simulated consumption of content types



*Notes:* This figure plots, for each of our content types, the type’s observed share of *production* (measured by randomly sampling from produced videos using TikTok’s Research API), its observed share of *consumption* (measured from the viewership histories of our participants), and its *simulated* share of consumption, obtained by taking the observed pool of produced content and applying our model-derived demand estimates to it.

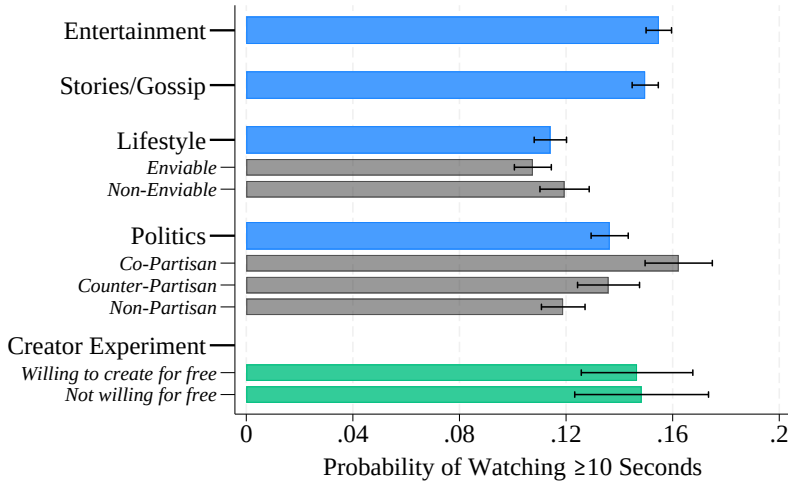


**Figure 6: Viewership experiment: identification, engagement, and viewer utility parameters**

*(a) Identification of viewer utility parameters*

| Parameter                                        | What pins it down                                                                                                                                                                                                                     | Data source                |
|--------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| Mean appeal $m_k$                                | Average engagement with videos of this type.                                                                                                                                                                                          | Experiment                 |
| Variance of $\tilde{m}_{ik}$                     | Cross-viewer variation in average engagement with this type.                                                                                                                                                                          | Experiment                 |
| Group-specific component of $q_p$                | Average viewer engagement with videos of this type created by the relevant group, using videos induced by our creator experiment from creators who do and do not regularly produce this type.                                         | Experiment                 |
| Variance and shape of stochastic quality $\xi_p$ | Variance: cross-video engagement variation that is correlated across viewers who share neither gender nor political leaning, estimated by type. Shape: the upper tail of observational view count distributions, common across types. | Experiment & observational |
| Variance of $e_{ip}$                             | Shared components: within-video within-demographic-group correlation in engagement. Residual component: remainder after partialling out other components of utility and noise.                                                        | Experiment                 |
| Boredom $d$                                      | How engagement with a type declines with its randomized share of the viewer's feed.                                                                                                                                                   | Experiment                 |
| <i>Engagement model (experiment only)</i>        |                                                                                                                                                                                                                                       |                            |
| Swiping costs $c_i$                              | Variation in viewers' average viewing times.                                                                                                                                                                                          | Experiment                 |
| Within-video boredom $a_k$                       | Distribution of within-video view durations, by type. Distinguished from average taste for type by how long participants watch a video conditional on having watched it past a likely reconsideration opportunity.                    | Experiment                 |
| Reconsideration rate $r$                         | The earliest times that different viewers swipe past videos.                                                                                                                                                                          | Experiment                 |
| Variance of noise $v_{ip}$                       | How often viewers re-pick the video they engaged with more in the re-pick module.                                                                                                                                                     | Experiment                 |

*(b) Probability of watching  $\geq 10$  seconds by content type*



*(c) Viewer utility parameters*

| Parameter                                                      | Estimate | SE     |
|----------------------------------------------------------------|----------|--------|
| <i>Mean appeal <math>m_k</math>, relative to entertainment</i> |          |        |
| Lifestyle                                                      | -0.29    | (0.04) |
| Stories                                                        | -0.26    | (0.09) |
| Co-partisan politics                                           | -0.03    | (0.05) |
| Counter-partisan politics                                      | -0.19    | (0.05) |
| Nonpartisan politics                                           | -0.10    | (0.05) |
| <i>Standard deviation among posts in feeds</i>                 |          |        |
| Quality $q_p$                                                  | 1.11     |        |
| Idiosyncratic appeal $e_{ip}$                                  | 1.02     |        |

*Notes:* Panel (a) explains how each viewer utility parameter is identified. Panel (b) plots the average probability that a viewer watches a video of each type for at least 10 seconds, calculated from regressions controlling for viewer, sampling source fixed effects, and video duration decile fixed effects. Panel (c) plots estimates of the parameters of the viewer utility function from our experiment.

duration options of 15 seconds and one minute as well as a normal “record” (indefinitely) button. After selection of one of the default options, the camera will start filming and stop automatically after 15 or 59 seconds unless manually stopped earlier. If creators using these default options tend to be *unsophisticated* producers whose content is not planned out, we should observe bunching of unsophisticated creators at these thresholds, and therefore discontinuous increases in sophistication when crossing them. Second, as mentioned in Section 3.2, TikTok’s Creator Rewards program—which pays eligible accounts with over 10,000 followers for views—pays out only to videos at least 60 seconds long.<sup>16</sup> So creators who are trying to maximize viewership in order to earn Creator Rewards payments should bunch disproportionately at 60 seconds and above.

Summing up, if we sort videos by length and study their underlying creators, we should see a discontinuous increase in creator sophistication as we cross 15 seconds and a discontinuous increase in sophistication *and* financially-motivated attempts to cater to demand as we cross 59 seconds.

Figure 8 Panels (a) and (b) show that crossing these thresholds brings large, sharp decreases in the prevalence of lifestyle content and corresponding increases in the prevalence of entertainment. This validates our earlier result that entertainment enjoys a demand premium over lifestyle.

Panel (c) shows that the prevalence of partisan compared to non-partisan political content is smooth across these thresholds, suggesting partisanship does not enjoy a demand premium. Our experiment found it had a small demand premium. This discrepancy is partly explained by the fact that our experimental participants are more politically partisan than the general population of TikTok users: Appendix H shows that re-weighting for this shrinks the experimental partisanship premium, though it does not eliminate it. Regardless, this suggests our experiment if anything *overestimates* the demand advantage of partisan content.

Two further results corroborate identification. First, only accounts with at least 10,000 followers are eligible for Creator Rewards. Figure 8 shows that the sharp discontinuity at 60 seconds appears only for such accounts, and not for accounts with fewer than 10,000 followers. Second, Panel (d) shows that both bunching of video durations and of content types just above 60 seconds appear rapidly in the months following rollout of Creator Rewards in 2023.

## 4.4 Supply-side preferences

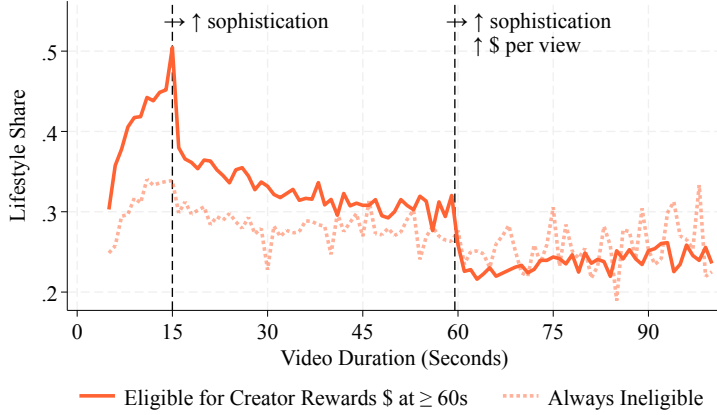
**Objective.** Recall from Proposition 2 in Section 3.1 that a key factor mediating the relative influence of supply-side preferences on equilibrium consumption is creators’ marginal valuation of viewership. If creators primarily care about attracting attention, production will chase demand: certain content types will be produced at a higher rate than others because creators expect those types to receive higher viewership. By contrast, if creators care little about viewership on the margin, supply-side preferences will be a more important driver of production.

Our objective in this section is to assess whether creators’ production decisions are driven by expectations about demand or by supply-side considerations like the difficulty of, or satisfaction

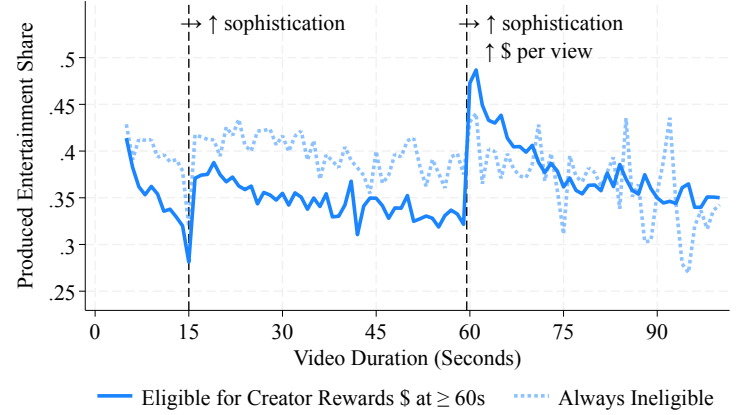
<sup>16</sup>This is presumably why the 60-second default recording option automatically stops at 59, rather than 60, seconds.

**Figure 8:** Lifestyle videos are posted by unsophisticated accounts with weak incentives to attract viewership; the opposite is true for entertainment

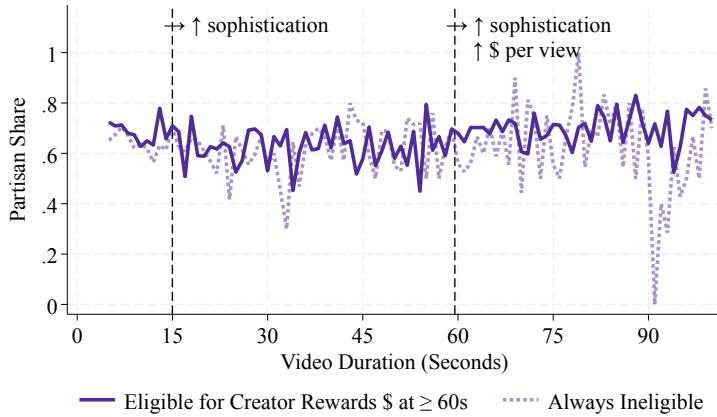
(a) Lifestyle videos are posted by unsophisticated accounts with weak incentives to attract viewership



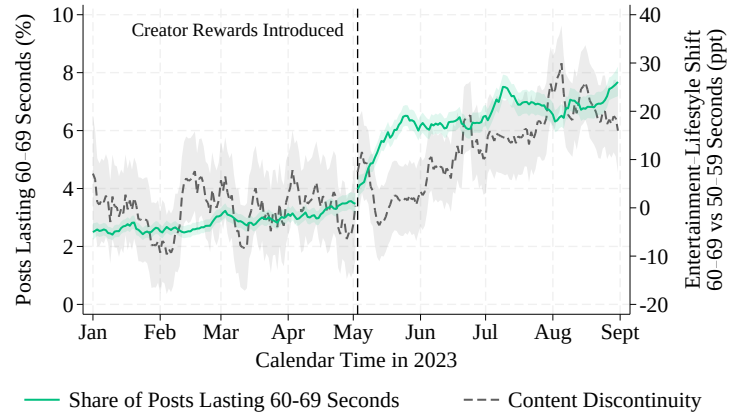
(b) Entertainment videos are posted by sophisticated accounts with strong incentives to attract viewership



(c) Partisanship of political content does not vary with creator sophistication and financial incentives



(d) Bunching of video durations and content composition emerges after introduction of Creator Rewards



**Notes:** This figure plots data from videos produced on TikTok sampled via TikTok's Research API. The x-axis for the first three panels is the duration of the video in seconds. The y-axis shows the share of videos of this duration classified as lifestyle (Panel (a)) or entertainment (Panel (b)), or the share of political videos classified as partisan. Solid lines plot videos produced by accounts with at least 10,000 followers and shaded lines videos from accounts with fewer than 10,000 followers. Panel (d) uses data on videos produced during 2023. It plots, over time, the share of videos between 60 and 69 seconds long (green line, left axis) and the entertainment-minus-lifestyle share gap when comparing videos 60-69 seconds long to videos 50-59 seconds long (black line, right axis). We use a 7-day rolling average on either side of the rollout date. Shaded regions show 95% confidence intervals.

derived from, creating different content types. We conduct an experiment that elicits creators' willingness to accept to post different types of content and decomposes it into a component driven by supply-side preferences and a component driven by expectations about, and valuation of, demand.

**Recruitment.** We recruit 979 Prolific users aged 18-30 who have posted at least once before on TikTok, as verified by logging in with their TikTok account. The median participant has 160 followers and 16 historical posts, while the average participant has 1,755 followers and 88 posts. Appendix I shows that our lifestyle results are robust to restricting to the 19% of participants with at least 1,000 followers, while our politics results become very noisy in the smaller sample because political posting is rare. We also show that our results are qualitatively consistent with descriptive results from our creator survey (Figure 4 Panel (b)), which has a sample representative of the distribution of creators that viewers see on TikTok.

**Experimental design.** The experiment, weighting scheme, and reduced-form analyses were pre-registered on the Open Science Framework at <https://osf.io/24tcz>. The flow of the experiment is depicted in Appendix Figure A25.

Participants complete a survey about content creation on TikTok. We begin by explaining that it is incentive-compatible to answer all questions truthfully, link participants to the full implementation procedure, and tell them they are guaranteed to receive a paid posting offer at the end of the survey. We then define the four content types we will ask about, providing definitions, examples, and links to TikToks in each category, randomly sampled from our observational data.

The first section elicits participants' baseline willingness to accept to post different content types, and their expectations about viewership. We start with a binary question: for each content type, would you be willing to make a post of this type on your account in exchange for \$0.25? We then elicit incentivized expectations about how many views a post of this type would receive.

To enable supplementary analyses, we also elicit participants' binary willingness for \$0.25 to privately *create*, but not post, a video of this type and send it to us to be shown to other experimental participants. And we ask two qualitative questions probing the reasons behind participants' answers.

We then elicit respondents' continuous willingness to accept to post each type. For respondents *unwilling* to post for \$0.25, this is the minimum payment they would require to make a post of this type. For respondents *willing* to post who indicate they were planning to make a post of this type anyways, we elicit the minimum payment they would require to *abstain* from making posts of this type on their account for the next week. (Since we have their account details, we can monitor and enforce this.) We divide their willingness to accept by their average number of posts of this type in weeks where they make at least one, to get a per-post willingness to accept to avoid posting.<sup>17</sup> Our offer generation procedure ensures responding truthfully to each question in this sequence is incentive-compatible; essentially, the procedure is a BDM mechanism and we use a participant's sequence of responses to construct a willingness to accept that can either be positive or negative.

<sup>17</sup>This may underestimate their true valuation if they would simply postpone these posts until after the week ended. Appendix Table A2 shows that results are robust to restricting to creators who report that they would not make these posts at all rather than postponing, and to inflating valuations in these arms to account for potential underestimation.

The second section measures valuation of viewership. For each content type, we tell participants that if we offer them a payment high enough to induce a post of this type, we may offer them an intervention intended to boost their viewership for that post. We randomly offer one of two interventions: a payment from us to boost their post algorithmically (an option TikTok offers), or tips from a professional content creator. Insofar as the latter also boosts their viewership on future posts, we should overestimate participants' marginal valuation of viewership. We elicit participants' willingness to pay for this intervention and their incentivized expectation about its effects on viewership.<sup>18</sup> Their willingness to pay for the intervention divided by their expectation about its effects captures their marginal valuation of viewership. To measure their full viewership valuation function  $\beta v^\rho$ , we estimate  $\rho$  by asking participants to indicate indifference points between various increments of views starting at different viewership levels. For example, we ask: "starting at 1,000 views, how many additional views would a creator need to get so that their satisfaction from this increase in viewership matches the satisfaction of going from 0 to 100 views?" This last object is the only one that is not incentivized.

**Parametrization and identification.** Recall that creator  $j$ 's payoff from creating a piece of type  $k$  which ends up receiving  $v$  views is  $\pi_{jk} = \mu_k + \tilde{\mu}_{kg} + \beta v^\rho + wv - \kappa_{jk}$ . We assume that  $\kappa_{jk}$  is drawn from a Type-1 Extreme Value distribution with the same scale for each content type. To map the experiment into our static model, we assume that a period in the model corresponds to a 24-hour period, so the willingness answers covering the next 24 hours represent one-period utilities.

As pre-registered, we re-weight creators so that the distribution of production implied by their responses matches the observed distribution of production on TikTok. Since our objective is to *decompose* observed production into components driven by different rewards rather than to reproduce observed production, this re-weighting is innocuous. It implicitly imposes a conditional independence assumption: that conditional on follower count and observed posting, creators in our experiment are representative in terms of supply parameters of actual creators on TikTok.

Identification of parameters is simple and summarized in Figure 9 Panel (a). A creator's overall willingness to accept to create a post captures their overall utility from that post. Their expected viewership plugged into their valuation of viewership  $\beta v^\rho$ , estimated from their marginal valuation of viewership and the indifference responses identifying  $\rho$ , yields the component of utility driven by rewards from viewership. Subtracting this out of their overall willingness to accept yields  $\mu_k + \tilde{\mu}_j$ .

The experimental elicitation procedure is obviously complex. And our approach to estimating creator preferences is essentially a residualization: we ask how much of observed variation in willingness to post is explained by the product of variation in expected viewership with implied valuations of viewership. Classical measurement error in the latter will not bias our results because it enters additively in our procedure. But a natural worry is that some form of non-classical response

<sup>18</sup>For 50% of respondents, this is an unincentivized question about its expected effects on their own viewership. (This cannot be incentivized because we would not observe both potential outcomes.) For the other 50%, this is an incentivized question about its expected effects on the viewership of other participants, where aggregating across multiple participants lets us construct this estimate. Appendix Table A3 shows results are robust to restricting to either group.

noise on the right-hand side may bias our results. Our approach is to corroborate our parameter estimates with simple, transparent supplementary quantitative and qualitative results.

**Results.** Parameter estimates are displayed in Figure 9 Panel (b). There is substantial variation in willingness to post across content types: the median gap in willingness to post between types a creator is versus is not willing to post for \$0.25 is \$23.33. But the median gap in implied utility from viewership is only \$0.64, suggesting that almost all of the variation in willingness to post is driven by variation in supply-side parameters rather than expectations about viewership.

Two facts drive this. First, the gap in average expected viewership between content types respondents are versus are not willing to post is small: 813 versus 666 views, or a 22% gap. Second, participants’ implied marginal valuation of viewership—median of \$0.23 per 100 views—is small. This is not because creators do not care about viewership *in general*. Panel (b) shows that for the median post creators are willing to make for free, their implied payoff from viewership is \$9.50 while their overall utility is \$5.00, indicating a roughly \$4.50 cost of producing and posting the video. Rather, it is because returns to viewership are strongly concave, with an estimated  $\rho \approx 0.3$ , meaning that *marginal* variation in viewership is relatively inconsequential for utility.

Such concavity is consistent with scope-insensitive satisfaction from attention: the first few dozen views or likes are very exciting, but after that, a creator’s ability to psychologically differentiate between the attention from 200 views and the attention from 500 views is limited. This is consistent with a large body of work in psychology on diminishing returns to social rewards (Baumeister and Leary, 1995; Ren et al., 2022). It is also consistent with prior work on social media: Srinivasan (2024) studies TikTok creators whose posts go unexpectedly viral and finds a subsequent increase in production that is almost completely insensitive to the size of the virality shock—the effects of receiving 100 likes and receiving 3,500 are indistinguishable. He interprets this as evidence of extremely concave returns to viewership. Outside of economics, Lindström et al. (2021) and da Silva Pinho et al. (2024) find similar results using the same strategy and Scissors et al. (2016) find similar results when surveying Facebook users about satisfaction from receiving “likes.”

Three additional facts corroborate the result that expectations about viewership play little role in explaining variation in willingness to supply different types of content. First, as Figure 9 Panels (c) and (d) show, willingness to *post* a content type is strongly correlated with willingness to privately *create* it, replicating Figure 4 Panel (c)’s implementation of the same exercise in our user survey. The panels show that enviable creators are substantially more willing to post lifestyle content than non-enviable ones, but also substantially more willing to privately create lifestyle content. The same is true for ideologically extreme creators and political content. Respondents’ binary willingness to post and willingness to create in exchange for \$0.25 agree in 73% of cases.<sup>19</sup>

Second, we ask respondents a qualitative question about why they were willing to post certain

<sup>19</sup>Participants are in general more willing to privately create these content types for \$0.25 than to post these content types. One way to make sense of this is that creating a video yields expressive satisfaction that outweighs creation costs, while publicly posting a video brings utility from viewership *and* a fixed cost of exposure to embarrassment.

types for \$0.25 but not others. Figure 9 Panel (e) shows that 82% of respondents say this is because “creating posts of [certain types] would be easier or more enjoyable, or [I] would be more comfortable posting them,” rather than because they expect those posts would attract more likes and views. We randomize the wording of the latter option to flip its perceived social desirability as detailed in Appendix I, which changes answer rates by only one percentage point.

Third, for each content type, we ask respondents to imagine that their expected viewership for this content type doubled. We ask whether this would make them more likely to post content of this type. Figure 9 Panel (f) shows that for content types respondents were already willing to post, 71% say yes. For content types respondents were *not* willing to post, only 40% say yes, and 10% say this would make them *less* likely—concentrated among ideological moderates asked about posting political content, who are evidently uncomfortable having others see their political content.

The main result of the experiment—that variation in willingness to supply different types of content is mostly driven by variation in the costs or expressive benefits of production, rather than expectations about demand—is therefore corroborated by several supplementary pieces of evidence. It is also corroborated by our representative survey of TikTok creators, which showed (Figure 4 Panel (b)) that supply-side considerations are rated as being much more important than expectations about viewership in determining production choices, and that this is robust to a social-desirability wording manipulation.

## 4.5 Quantification

We now combine our demand- and supply-side parameter estimates and conduct decompositions and counterfactuals. We have two sources of demand-side information. First is our viewership experiment, which yields estimates of utility parameters for each content type. Second is our creator experiment, which collects creators’ expectations about viewership for each type of content they could potentially produce. We cannot decompose these expectations into the constituent parameters of viewers’ utility functions but this turns out to be unnecessary for the exercises in this section. We can therefore show robustness of our results to using both sources of demand-side information.

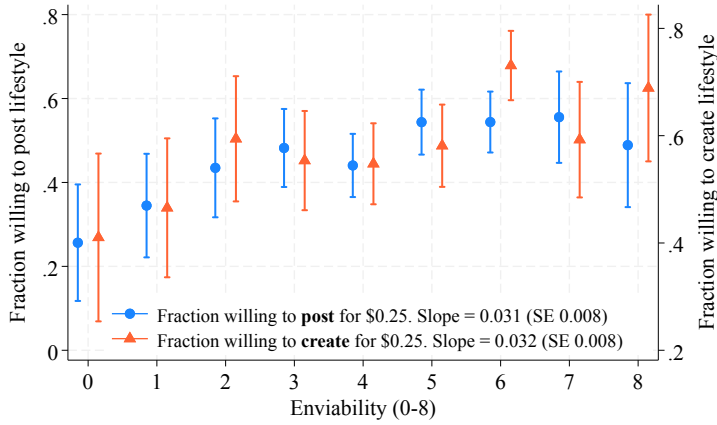
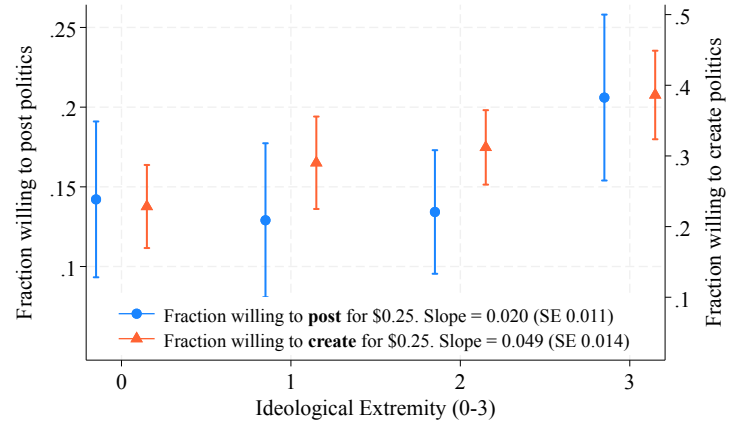
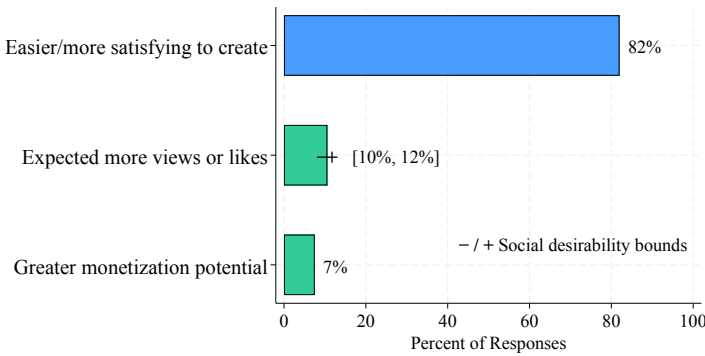
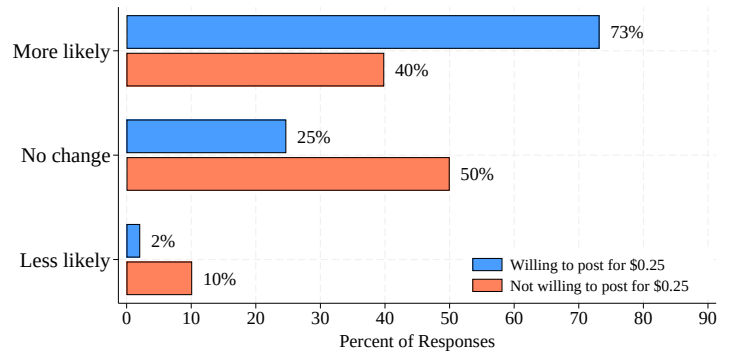
Panel (a) of Figure 10 conducts Oaxaca decompositions on two objects of interest: the gap in equilibrium consumption between entertainment and lifestyle content, and the gap between partisan and nonpartisan political content. The decompositions operate by asking how much consumption would change if we re-assigned one content type to have the other’s supply (or demand) parameters and mechanically recalculated consumption without accounting for factors like diminishing marginal utility from a given content type. As expected given the directions of our demand estimates, the gap in consumption between entertainment and lifestyle content reflects a strong supply-side force favoring lifestyle content colliding with a strong demand-side force favoring entertainment. Meanwhile, the gap between partisan and nonpartisan political content is between half and two-thirds explained by a supply-side force favoring partisan content, with the rest explained by demand favoring partisan content.

**Figure 9: Creator experiment: identification, creator payoff parameters, and drivers of willingness to post****(a) Identification of creator payoff parameters**

| Parameter                     | What pins it down                                                                                                                                                                                    | Data source                        |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| Mean taste $\mu_k$            | Average willingness to accept to post this type in the experiment, partialling out the value of expected views $\beta V^p$ .                                                                         | Creator expmt.                     |
| Deviations $\tilde{\mu}_{kg}$ | Group-specific deviations from overall creator average: creators' willingness to accept to post this type net of the value of expected views $\beta V^p$ , relative to other creators.               | Creator expmt.                     |
| Expected views $V$            | Incentivized expected viewership for a post of this type.                                                                                                                                            | Creator expmt.                     |
| Value of views $\beta$        | WTP for the view-boosting intervention, incentivized expectation of its effect on viewership $\Delta V$ , expected baseline viewership, and curvature: $WTP = \beta[(V + \Delta V)^\rho - V^\rho]$ . | Creator expmt.                     |
| Curvature $\rho$              | Answers to the indifference questions.                                                                                                                                                               | Creator expmt.                     |
| Variance of $\kappa_{jk}$     | Calibrated to match creators' actual posting histories given their experimental answers.                                                                                                             | Creator expmt. & posting histories |
| Group masses $\omega_{kg}$    | Share of creators in each follower bin who do and do not regularly post the type, with a pre-registered weighting procedure that matches the actual distribution of TikTok production.               | Creator expmt. & Research API      |

**(b) Creator payoff parameter estimates**

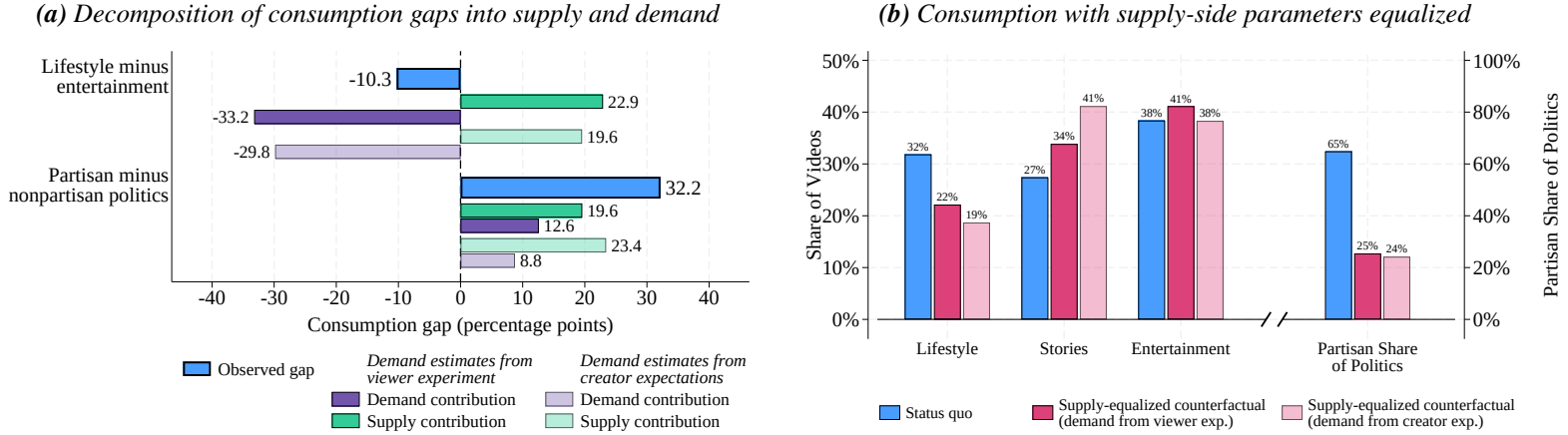
| Parameter                                                                | Estimate | SE      |
|--------------------------------------------------------------------------|----------|---------|
| <i>Mean taste <math>\mu_k</math>, % relative to entertainment</i>        |          |         |
| Lifestyle                                                                | 8.8      | (1.5)   |
| Stories                                                                  | -16.0    | (3.4)   |
| <i>Mean taste <math>\mu_k</math>, % relative to nonpartisan politics</i> |          |         |
| Partisan politics                                                        | 35.3     | (8.5)   |
| Value of views $\beta$                                                   | 1.21     | (0.09)  |
| Curvature $\rho$                                                         | 0.33     | (0.005) |
| <i>Expected views, creators willing to post only some types</i>          |          |         |
| Types willing to post for \$0.25                                         | 813      | (48)    |
| Types not willing to post for \$0.25                                     | 666      | (47)    |
| <i>Expected views, % relative to entertainment</i>                       |          |         |
| Lifestyle                                                                | -32.5    | (2.1)   |
| Stories                                                                  | -25.3    | (2.2)   |
| <i>Expected views, % relative to nonpartisan politics</i>                |          |         |
| Partisan politics                                                        | 53.7     | (15.5)  |
| <i>For the median planned post</i>                                       |          |         |
| Total utility from post                                                  | \$5.00   | (0.69)  |
| Utility from viewership                                                  | \$9.50   | (0.77)  |
| Utility from first 100 views                                             | \$5.57   | (0.40)  |

**(c) Willingness to post lifestyle by envyability****(d) Willingness to post politics by extremity****(e) Why were you willing to post content type A for \$0.25 and not content type B?****(f) If your expected viewership from this type doubled, how would this change your likelihood of posting it?**

**Notes:** Panel (a) explains how creator utility parameters are identified and Panel (b) shows parameter estimates. Panels (c) and (d) show the shares of creators willing to post or privately create a lifestyle or political video for \$0.25, separately by their self-reported envyability in the case of lifestyle or ideological extremity in the case of politics. Panel (e) shows the percent of creators selecting each of three multi-choice options in response to a question about why they were willing to post some content types for \$0.25 and not others (asked to all creators for whom that was true). Panel (f) plots responses to a question we ask separately by content type: hypothetically, if the viewership a creator expected for posts of this content type doubled, how would this change their likelihood of posting this type. We split responses by whether this was a type the creator was versus was not willing to post for \$0.25.



**Figure 10: Oaxaca decomposition of observed consumption gaps, and counterfactual consumption rates after equalizing supply parameters**



Notes: Panel (a) displays a Oaxaca decomposition of observed gaps in consumption rates between lifestyle content and entertainment content, and between partisan and nonpartisan political content, into a supply contribution (how different would consumption rates look if we equalized both types’ demand parameters?) and a demand contribution (how different would they look if we equalized supply parameters?). We plot two versions of the decomposition, one using demand estimates calculated from our viewership experiment and the other using creators’ expected views from our creator experiment. Panel (b) displays, in blue, the “status quo” distribution of consumption simulated from our model using creator-experiment supply estimates and viewership-experiment demand estimates. The red bars show two separate counterfactuals. Left of the axis break, they show the shares of lifestyle, stories, and entertainment in consumption if we give lifestyle and stories the supply parameters of entertainment. Right of the axis break, the partisan share of political consumption if we give nonpartisan political content the supply parameters of partisan political content, holding all other supply parameters fixed.

Panel (b) performs counterfactual simulations that do account for diminishing marginal utility from content types. We perform two separate counterfactuals.

In our lifestyle-focused counterfactual, we assign entertainment’s supply-side parameters to lifestyle and stories, hence eliminating lifestyle’s supply-side advantage over the other two types. In this counterfactual, the equilibrium prevalence of lifestyle content in viewers’ feeds drops by about a third. The magnitude of the drop is attenuated by diminishing marginal utility within content types, which pushes towards equalization of consumption across types.

In our politics-focused counterfactual, we assign partisan political content’s supply-side parameters to nonpartisan content, eliminating partisan content’s supply side advantage. In this counterfactual, the partisanship of consumed political content drops by two-thirds. The supply side hence appears more important for partisan than lifestyle content. This is because—even though lifestyle faces a demand disadvantage and partisan content a demand advantage—the supply advantage of partisan over nonpartisan content is quantitatively much larger than the advantage of lifestyle over entertainment.

Overall, results suggest that the prevalence of both lifestyle and partisan political content is between one- and two-thirds supply-driven.

## 5 Implications: Emotion-Sampling

Section 4 decomposed equilibrium consumption of lifestyle and partisan political content on TikTok into a component driven by supply-side preferences and one driven by demand-side preferences.

A separate question is whether the demand-side preferences we estimate reflect true viewer preferences or successful exploitation of viewers’ self-control problems. This is a secondary question: our model and results are meaningful whether one interprets viewer “utility” as experienced or decision utility. But it may be interesting to collect some evidence on this question.

In particular, our model in Section 3.1 suggested that even if viewers dislike some features of a content type, pieces of this type may end up in their feeds thanks to having high residual appeal on *other* dimensions. Appendix J probes this possibility by using emotion data collected via the iPhone app we asked participants to use for a week before sending us their social media histories.

The app detects when a participant is using TikTok, Instagram, or YouTube and pings her at random intervals to ask how she is feeling. This is “experience-sampling,” a measurement technique with a rich pedigree in economics (Kahneman and Krueger, 2006). We link participants’ responses to the posts they were viewing at the time the pings were randomly delivered. For identification, we restrict to sessions where a participant is scrolling through an infinite vertical feed and therefore not actively choosing which video will appear next. We implement event study specifications that exploit high-frequency video-to-video variation in the content viewers are served. Such local exogenous variation should exist as long as viewers have a taste for unpredictable moment-to-moment variation conditional on the overall composition of their feeds (Eyal, 2014). We evaluate the exogeneity of the variation by examining pre-trends in our event study plots and implementing balance tests checking whether a viewer’s emotions predict the content they are served *next*.

Appendix Figure A28 plots results. The two content types with the largest negative effects on contemporaneous emotional affect are political content and the subset of lifestyle content that is “enviable,” meaning coded as featuring an unusually attractive or wealthy person. This is driven by anger and anxiety in response to political videos and envy- or insecurity-related emotions in response to lifestyle videos.

However, participants seem to like the political and enviable lifestyle videos that appear in their feeds even though they trigger negative emotions on certain dimensions. Our app occasionally asks participants if they regret watching the video they just watched. Panel (c) of Appendix Figure A28 shows that participants only very rarely report that they regret having watched an enviable lifestyle or partisan political video. After participants submit their viewership histories, we conduct a retrospective survey where we show them a sequence of randomly selected creators from their feeds and ask if they would like the opportunity to block those creators from appearing in their feeds in future. Panel (d) shows that participants say “yes” in a substantial minority of cases but are *less* likely to say yes for enviable lifestyle or political creators compared to random other creators. This is consistent with the possibility that the lifestyle and political creators that do appear in viewers’ feeds are sufficiently positively selected on other dimensions to compensate for the

negative emotions they trigger.

These results are severely limited in several respects. They represent only very short-run measures of emotions and could not, for example, capture the possibility that repeated exposure to unfavorable social comparisons over a longer period of time could be psychologically deleterious. They capture only emotions, not other outcomes that might matter more. For example, for political content we may be more interested in attitudes like affective polarization. And participants may be reluctant to give answers that admit self-control problems. We therefore do not draw firm conclusions and simply note that this pattern of results is consistent with our theory.

## 6 Conclusion

In the 20th century, citizens of liberal democracies got their political news from professional journalists and entertained themselves by reading or watching books, television shows, or movies produced by writers, directors, and actors paid to cater to their tastes. In the 2020s, they are likely to get their political news from the tiny slice of the population enthusiastic about diving into political arguments online, and their entertainment from the minority of their peers attractive enough to enjoy showing off in front of a camera.

Citizens of liberal democracies also spend more time on media consumption than ever before. This is because the turn to amateur production was precipitated by the Internet's facilitation of an enormously successful business model based on attracting huge quantities of content to enable ex-post cherry-picking. Media has therefore become more engaging to consumers even as it has de-professionalized.

This paper has characterized this distinctive business model and taken the first step towards tracing out some of its consequences. Both its upsides and potential downsides are potentially enormous. Consumers can access finely personalized content and the low-friction marketplace of social media platforms draws stories and talents from many corners of society. Political discourse has democratized. At the same time, relying on an unrepresentative minority to produce content in pursuit of emotional goals often tied to status and conflict has the potential to intensify patterns of social comparison and political negativity. We leave it to future work to explore these tradeoffs in greater depth.

## References

- Acemoglu, Daron, Asuman Ozdaglar, and James Siderius**, “A model of online misinformation,” *Review of Economic Studies*, 2024, 91 (6), 3117–3150.
- , **Daniel Huttenlocher, Asuman Ozdaglar, and James Siderius**, “Online business models, digital ads, and user welfare,” *National Bureau of Economic Research*, 2024.
- Aguiar, Luis and Joel Waldfogel**, “Quality predictability and the welfare benefits from new products: Evidence from the digitization of recorded music,” *Journal of Political Economy*, 2018, 126 (2), 492–524.
- Allcott, Hunt, Juan Camilo Castillo, Matthew Aaron Gentzkow, Leon Musolf, and Tobias Salz**, “Sources of market power in web search: Evidence from a field experiment,” Technical Report, National Bureau of Economic Research 2025.
- , **Luca Braghieri, Sarah Eichmeyer, and Matthew Gentzkow**, “The Welfare Effects of Social Media,” *American Economic Review*, 2020, 110 (3), 629–676.
- , **Matthew Gentzkow, and Lena Song**, “Digital Addiction,” *American Economic Review*, 2022, 112 (7), 2424–2463.
- , —, **Benjamin Wittenbrink, Juan Carlos Cisneros, Adriana Crespo-Tenorio, Drew Dimmery, Deen Freelon, Sandra González-Bailón, Andrew M Guess, Young Mie Kim et al.**, “The Effect of Deactivating Facebook and Instagram on Users’ Emotional State,” Technical Report, National Bureau of Economic Research 2025.
- Andrews, Travis M.**, “Charli D’Amelio Is TikTok’s Biggest Star. She Has No Idea Why,” *The Washington Post*, May 2020.
- Aridor, Guy, Rafael Jiménez-Durán, Ro’ee Levy, and Lena Song**, “The Economics of Social Media,” *Journal of Economic Literature*, 2024, 62 (4), 1422–1474.
- , **Tevel Dekel, Rafael Jiménez-Durán, Ro’ee Levy, and Lena Song**, “Digital news consumption: Evidence from smartphone content in the 2024 us elections,” 2025.
- Balbuzanov, Ivan, Jared Gars, Mateusz Stalinski, and Emilia Tjernström**, “Incentivizing engagement: experimental evidence on journalist performance pay,” 2025.
- Bao, Renjie**, ““Just One More Clip”: Short Videos, Big Self-Control Problems,” 2026.
- Baumeister, Roy F. and Mark R. Leary**, “The Need to Belong: Desire for Interpersonal Attachments as a Fundamental Human Motivation,” *Psychological Bulletin*, 1995, 117 (3), 497–529.
- Beattie, Graham, Ruben Durante, Brian Knight, and Ananya Sen**, “Advertising spending and media bias: Evidence from news coverage of car safety recalls,” *Management Science*, 2021, 67 (2), 698–719.
- Becker, Gary S and Kevin M Murphy**, “A theory of rational addiction,” *Journal of political Economy*, 1988, 96 (4), 675–700.

- **and** — , “A simple theory of advertising as a good or bad,” *The Quarterly Journal of Economics*, 1993, 108 (4), 941–964.
- Beknazar-Yuzbashev, George, Rafael Jiménez-Durán, and Mateusz Stalinski**, “A model of harmful yet engaging content on social media,” in “AEA Papers and Proceedings,” Vol. 114 American Economic Association 2014 Broadway, Suite 305, Nashville, TN 37203 2024, pp. 678–683.
- , — , **Andrey Simonov, and Mateusz Stalinski**, “Social media advertising loads as prices,” *Columbia Business School Research Paper*, 2025, (5952337).
- , — , **Jesse McCrosky, and Mateusz Stalinski**, “Toxic content and user engagement on social media: Evidence from a field experiment,” Technical Report, CESifo Working Paper 2025.
- Belfi, Amy M, Edward A Vessel, and G Gabrielle Starr**, “Individual ratings of vividness predict aesthetic appeal in poetry.,” *Psychology of Aesthetics, Creativity, and the Arts*, 2018, 12 (3), 341.
- Besley, Timothy and Andrea Prat**, “Handcuffs for the Grabbing Hand? Media Capture and Government Accountability,” *American Economic Review*, 2006, 96 (3), 720–736.
- Boeker, Maximilian and Aleksandra Urman**, “An Empirical Investigation of Personalization Factors on TikTok,” in “Proceedings of the ACM Web Conference 2022” 2022, pp. 2298–2309.
- Bolte, Lukas, John J Conlon, and Jun Hue**, “Social Media and the Limits to Selective Attention,” *Available at SSRN 7448118*, 2026.
- Boxell, Levi and Jacob Conway**, “Journalist ideology and the production of news: Evidence from movers,” *Available at SSRN 4295587*, 2022.
- Braghieri, Luca, Ro’ee Levy, and Alexey Makarin**, “Social Media and Mental Health,” *American Economic Review*, 2022, 112 (11), 3660–3693.
- , — , **and Hannah Trachtman**, “Frictions in news consumption: evidence from social media,” *Available at SSRN 5494670*, 2026.
- , **Sarah Eichmeyer, Ro’ee Levy, Markus M Mobius, Jacob Steinhardt, and Ruiqi Zhong**, “Article-level slant and polarization of news consumption on social media,” *Available at SSRN 4932600*, 2024.
- Brynjolfsson, Erik, Yu Hu, and Michael D Smith**, “Consumer surplus in the digital economy: Estimating the value of increased product variety at online booksellers,” *Management science*, 2003, 49 (11), 1580–1596.
- Bursztyn, Leonardo, Benjamin Handel, Rafael Jiménez-Durán, and Christopher Roth**, “When Product Markets Become Collective Traps: The Case of Social Media,” *American Economic Review*, 2025, 115 (12), 4105–4136.
- , **Benjamin R Handel, Rafael Jiménez-Durán, and Christopher Roth**, “When Product Markets Become Collective Traps: The Case of Social Media,” *American Economic Review*, 2025.

- , **Georgy Egorov, Ruben Enikolopov, and Maria Petrova**, “Social media and xenophobia: evidence from Russia,” Technical Report, National Bureau of Economic Research 2019.
- , **Matthew Gentzkow, Rafael Jiménez-Durán, Aaron Leonard, Filip Milojević, and Christopher Roth**, “Measuring markets for network goods,” *Review of Economic Studies*, 2026, p. rdag099.
- Cagé, Julia, Moritz Hengel, Nicolas Hervé, and Camille Urvoy**, “Political Bias in the Media—Evidence from the Universe of French Broadcasts, 2002-2020,” Technical Report, CESifo Working Paper 2025.
- Carter, Bill**, “TV’s Venturesome Programmers Find It’s Lonely Out Front,” *The New York Times*, November 1990. Section C, Arts.
- Caves, Richard E.**, “Contracts between art and commerce,” *Journal of economic Perspectives*, 2003, 17 (2), 73–83.
- Conover, Michael, Jacob Ratkiewicz, Matthew Francisco, Bruno Gonçalves, Filippo Menczer, and Alessandro Flammini**, “Political polarization on twitter,” in “Proceedings of the international aaai conference on web and social media,” Vol. 5 2011, pp. 89–96.
- Cruz, Christopher**, “Ironmouse Is Twitch’s Top Streamer, But She Doesn’t Know Why You’re Watching,” *Rolling Stone*, October 2024.
- da Silva Pinho, Ana, Violeta Céspedes Izquierdo, Björn Lindström, and Wouter van den Bos**, “Youths’ Sensitivity to Social Media Feedback: A Computational Account,” *Science Advances*, 2024, 10 (43), eadp8775.
- David, Paul A.**, “The dynamo and the computer: an historical perspective on the modern productivity paradox,” *The American economic review*, 1990, 80 (2), 355–361.
- Donati, Dante and Lena Song**, “The Influence of the Vocal Minority: Evidence from Social Media Comments,” April 2026. SSRN Working Paper No. 6529698.
- Durante, Ruben and Brian Knight**, “Partisan Control, Media Bias, and Viewer Responses: Evidence from Berlusconi’s Italy,” *Journal of the European Economic Association*, 2012, 10 (3), 451–481.
- Dutz, Deniz, Ingrid Huitfeldt, Santiago Lacouture, Magne Mogstad, Alexander Torgovitsky, and Winnie Van Dijk**, “Selection in surveys: Using randomized incentives to detect and account for nonresponse bias,” *Review of Economic Studies*, 2026, p. rdag003.
- Euchner, Charles**, “Daniel Menaker on Tragedy, Checking Facts, Writing, and the State of Publishing,” The Elements of Writing May 2019. Interview with Daniel Menaker, former fiction editor of *The New Yorker* and editor in chief of Random House.
- Eyal, Nir**, *Hooked: How to Build Habit-Forming Products*, New York: Portfolio/Penguin, 2014.
- Filippas, Apostolos, John J Horton, Elliot Lipnowski, and Prasanna Parasurama**, “The production and consumption of social media,” *Management Science*, 2025.

- Finke, Nikki**, “Living Legend David Picker’s Movie Memoir,” *Deadline*, September 2013.
- Fishburn, Peter C.**, “Noncompensatory preferences,” *Synthese*, 1976, 33 (1), 393–403.
- Freyaldenhoven, Simon, Christian Hansen, Jorge Pérez Pérez, and Jesse M Shapiro**, “Visualization, identification, and estimation in the linear panel event-study design,” Technical Report, National Bureau of Economic Research 2021.
- Fujiwara, Thomas, Karsten Müller, and Carlo Schwarz**, “The effect of social media on elections: Evidence from the united states,” *Journal of the European Economic Association*, 2024, 22 (3), 1495–1539.
- Gentzkow, Matthew and Jesse M Shapiro**, “Media bias and reputation,” *Journal of political Economy*, 2006, 114 (2), 280–316.
- **and Jesse M. Shapiro**, “What Drives Media Slant? Evidence from U.S. Daily Newspapers,” *Econometrica*, 2010, 78 (1), 35–71.
- **, Nathan Petek, Jesse M Shapiro, and Michael Sinkinson**, “Do newspapers serve the state? Incumbent party influence on the US press, 1869–1928,” *Journal of the European Economic Association*, 2015, 13 (1), 29–61.
- Goldman, William**, *Adventures in the Screen Trade: A Personal View of Hollywood and Screenwriting*, New York: Warner Books, 1983.
- Haidt, Jonathan**, *The Anxious Generation: How the Great Rewiring of Childhood Is Causing an Epidemic of Mental Illness*, New York: Penguin Press, 2024.
- Herman, Edward S. and Noam Chomsky**, *Manufacturing Consent: The Political Economy of the Mass Media*, New York: Pantheon Books, 1988.
- Jeon, Doh-Shin and Patrick Rey**, “Platform competition and app development,” 2024.
- Kahneman, Daniel and Alan B Krueger**, “Developments in the measurement of subjective well-being,” *Journal of Economic perspectives*, 2006, 20 (1), 3–24.
- Kalra, Aarushi**, “Hate in the time of algorithms: Evidence on online behavior from a large-scale experiment,” *arXiv preprint arXiv:2503.06244*, 2025.
- Kleinberg, Jon, Himabindu Lakkaraju, Jure Leskovec, Jens Ludwig, and Sendhil Mullainathan**, “Human decisions and machine predictions,” *The quarterly journal of economics*, 2018, 133 (1), 237–293.
- Kolitz, Daniel**, “Microdramas, TV Shows, and Hollywood Entertainment,” *The New York Times Magazine*, October 2026.
- Levy, Ro’ee**, “Social Media, News Consumption, and Polarization: Evidence from a Field Experiment,” *American Economic Review*, 2021, 111 (3), 831–870.
- Lindström, Björn, Martin Bellander, David T. Schultner, Allen Chang, Philippe N. Tobler, and David M. Amodio**, “A Computational Reward Learning Account of Social Media Engagement,” *Nature Communications*, 2021, 12 (1), 1311.

- Lumet, Sidney**, *Making Movies*, New York: Alfred A. Knopf, 1995.
- Martin, Gregory J and Ali Yurukoglu**, “Bias in cable news: Persuasion and polarization,” *American Economic Review*, 2017, 107 (9), 2565–2599.
- **and Joshua McCrain**, “Local news and national politics,” *American Political Science Review*, 2019, 113 (2), 372–384.
- Mullainathan, Sendhil and Andrei Shleifer**, “The Market for News,” *American Economic Review*, 2005, 95 (4), 1031–1053.
- Noy, Shakked and Aakaash Rao**, “The business of the culture war,” *Available at SSRN 5694622*, 2025.
- O’Keeffe, Gwenn Schurgin, Kathleen Clarke-Pearson et al.**, “The impact of social media on children, adolescents, and families,” *Pediatrics*, 2011, 127 (4), 800–804.
- O’Meara, Victoria**, “The many meanings of measure: Social media metrics and their implications for collective organizing in the influencer economy,” *New Media & Society*, 2026, p. 14614448261445040.
- Postman, Neil**, *Amusing Ourselves to Death: Public Discourse in the Age of Show Business*, New York: Viking Penguin, 1985.
- Pynchon, Thomas**, *Vineland*, Boston: Little, Brown and Company, 1990.
- Qin, Bei, David Strömberg, and Yanhui Wu**, “Media bias in China,” *American Economic Review*, 2018, 108 (9), 2442–2476.
- Rambachan, Ashesh**, “Identifying prediction mistakes in observational data,” *The Quarterly Journal of Economics*, 2024, 139 (3), 1665–1711.
- Ren, Dongning, Olga Stavrova, and Wen Wei Loh**, “Nonlinear Effect of Social Interaction Quantity on Psychological Well-Being: Diminishing Returns or Inverted U?,” *Journal of Personality and Social Psychology*, 2022, 122 (6), 1056–1074.
- Ronson, Jon**, “He Was a Top Lawn Care Influencer on YouTube. Then Everything Went Haywire.,” *The New York Times Magazine*, August 2026.
- Rosen, Sherwin**, “The economics of superstars,” *The American economic review*, 1981, 71 (5), 845–858.
- Rosenbusch, Hannes, Anthony M Evans, and Marcel Zeelenberg**, “The relative importance of joke and audience characteristics in eliciting amusement,” *Psychological Science*, 2022, 33 (9), 1386–1394.
- Scissors, Lauren, Moira Burke, and Steven Wengrovitz**, “What’s in a Like? Attitudes and Behaviors around Receiving Likes on Facebook,” in “Proceedings of the 19th ACM Conference on Computer-Supported Cooperative Work & Social Computing” Association for Computing Machinery 2016, pp. 1501–1510.



- Sibley, Frank**, “Aesthetic concepts,” *The philosophical review*, 1959, 68 (4), 421–450.
- Solheim, Hannah and Nancy Wang**, “Social Media, Body Image, and Discrimination,” 2026. Work in Progress.
- Srinivasan, Karthik**, “Paying attention.” PhD dissertation, The University of Chicago 2024.
- Steel, Benjamin, Miriam Schirmer, Derek Ruths, and Juergen Pfeffer**, “Just Another Hour on TikTok: ID sampling to obtain a complete slice of TikTok,” *arXiv preprint arXiv:2504.13279*, 2025.
- Sweet, Jacob**, “The Instagram Reels Gold Rush,” *The New Yorker*, January 2023. Rabbit Holes column.
- Szeidl, Adam and Ferenc Szucs**, “Media capture through favor exchange,” *Econometrica*, 2021, 89 (1), 281–310.
- Tait, Amelia**, “Unboxing Is So 2012. The Internet Wants Packing Videos,” *Wired*, 2023.
- Tversky, Amos**, “Intransitivity of preferences,” *Psychological Review*, 1969, 76 (1), 31–48.
- Vany, Arthur De and W David Walls**, “Uncertainty in the movie industry: Does star power reduce the terror of the box office?,” *Journal of cultural economics*, 1999, 23 (4), 285–318.
- Vessel, Edward A and Nava Rubin**, “Beauty and the beholder: Highly individual taste for abstract, but not real-world images,” *Journal of vision*, 2010, 10 (2), 18–18.
- Waldfoegel, Joel**, “Cinematic explosion: New products, unpredictabilty and realized quality in the digital era,” *The Journal of Industrial Economics*, 2016, 64 (4), 755–772.
- , “How digitization has created a golden age of music, movies, books, and television,” *Journal of economic perspectives*, 2017, 31 (3), 195–214.
- , *Digital renaissance: What data and economics tell us about the future of popular culture*, Princeton University Press, 2019.
- Wallisch, Pascal and Jake Alden Whritner**, “Strikingly low agreement in the appraisal of motion pictures,” *Projections*, 2017, 11 (1), 102–120.
- Zannettou, Savvas, Olivia Nemes-Nemeth, Oshrat Ayalon, Angelica Goetzen, Krishna P Gummadi, Elissa M Redmiles, and Franziska Roesner**, “Analyzing user engagement with TikTok’s short format video recommendations using data donations,” in “Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems” 2024, pp. 1–16.

## **Online Appendix**

### *“The Supply Side of Social Media”*

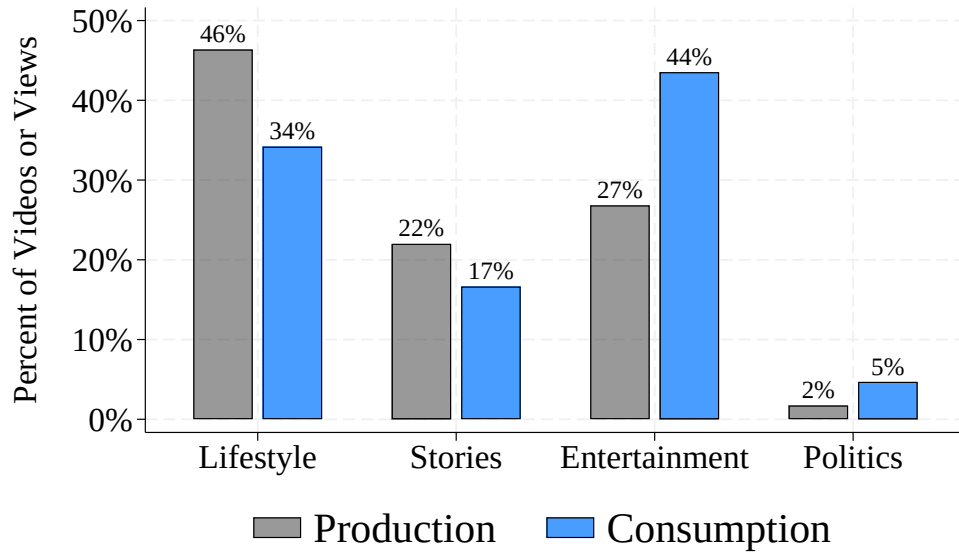
Shakke Noy and Aakaash Rao

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## A Additional Figures

*Appendix Figure A1: Production and consumption of content on TikTok, weighting videos by their view counts*



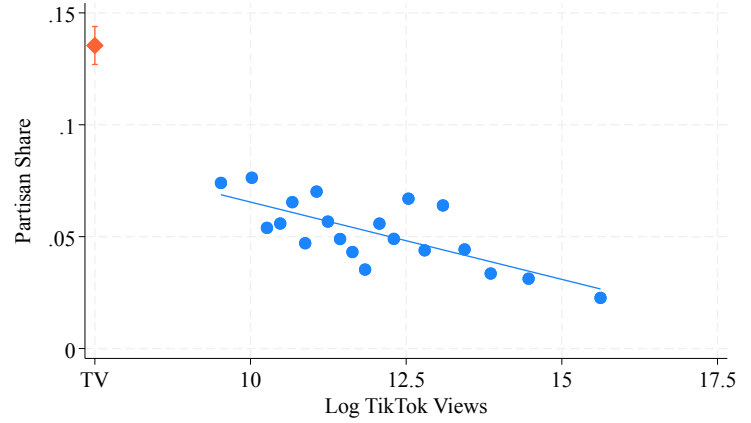
*Notes:* This figure reproduces Panel (d) of Figure 5, but measures consumption by weighting each produced video by its view count instead of by using participants' viewership histories. The grey bars plot production shares, using the same random sample of videos and the same weights as in Figure 5. The blue bars plot consumption shares, multiplying each video's weight by its cumulative view count. Videos with missing or zero view counts count toward production but not consumption.

**Appendix Figure A2: Viewers prefer nonpartisan news outlets on TikTok**

**(a) Top political news outlets in Democrats' and Republicans' feeds**

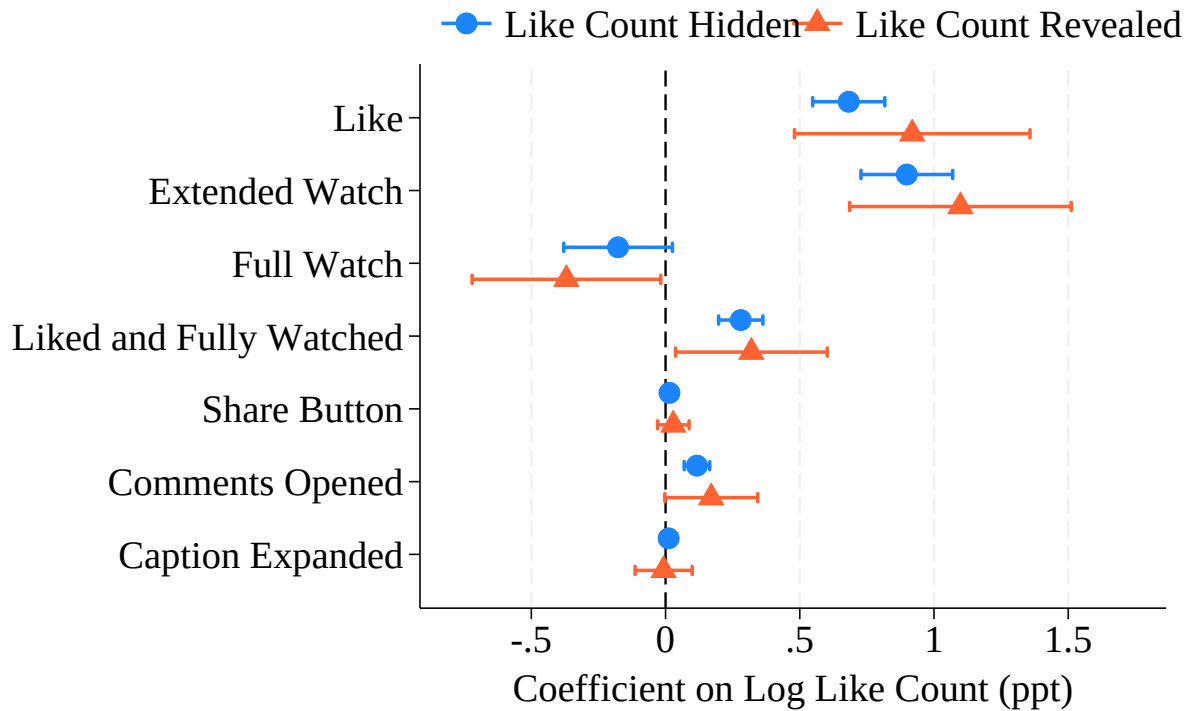
| Democrats         | Republicans      |
|-------------------|------------------|
| 1. Daily Mail     | 1. Daily Mail    |
| 2. CBS News       | 2. CBS News      |
| 3. CNN            | 3. NBC News      |
| 4. NBC News       | 4. Fox News      |
| 5. ABC News       | 5. ABC News      |
| 6. MSNOW          | 6. CNN           |
| 7. The Daily Show | 7. New York Post |
| 8. CBS Mornings   | 8. CBS Mornings  |
| 9. NBC New York   | 9. Metro UK      |
| 10. New York Post | 10. Reuters      |

**(b) News organizations' more partisan videos receive fewer views on TikTok**



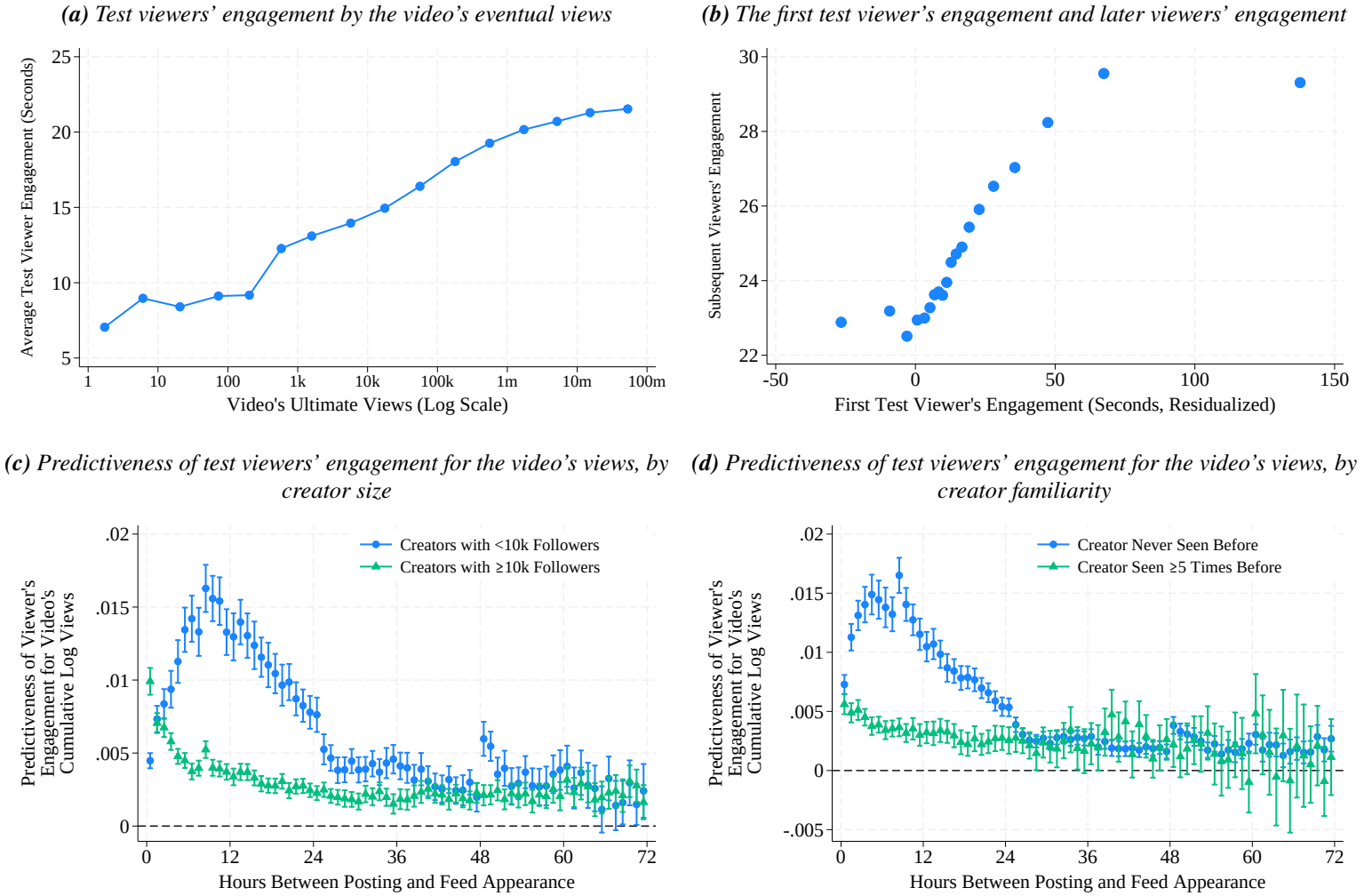
*Notes:* Panel (a) lists the ten news outlets whose TikTok accounts appear most often in the watch histories of Democratic and Republican participants (with party affiliation taken from our screening survey), counting every appearance of one of their political videos in a participant's feed. Panel (b) uses political videos posted by ABC News, CBS News, CNN, Fox News, and NBC News, weighting videos so that each outlet counts equally. The blue dots plot the share of videos expressing a clearly left- or right-partisan perspective within bins of the video's log view count, and the blue line is a linear fit through these points. The orange diamond plots the corresponding partisan share in television news from the same five outlets, with a 95% confidence interval clustered by broadcast. (Its position on the x-axis is arbitrary.)

**Appendix Figure A3:** *Revealing like counts does not increase engagement with highly-liked videos*



*Notes:* In our viewership experiment, we show videos' public like counts to a random 10% of participants and hide them from the rest. This figure plots coefficients from regressions of each engagement outcome on the log of one plus the video's like count, separately for participants who could not see like counts (blue circles) and those who could (orange triangles). Coefficients are in percentage points. The sample is all video encounters of participants who completed the final wave of the experiment, excluding videos from our creator experiment, whose like counts we do not observe. The regressions include fixed effects for participant and, separately by condition, for video length decile, sampling tier, sampling source, and content type. Bars show 95% confidence intervals clustered by participant and video.

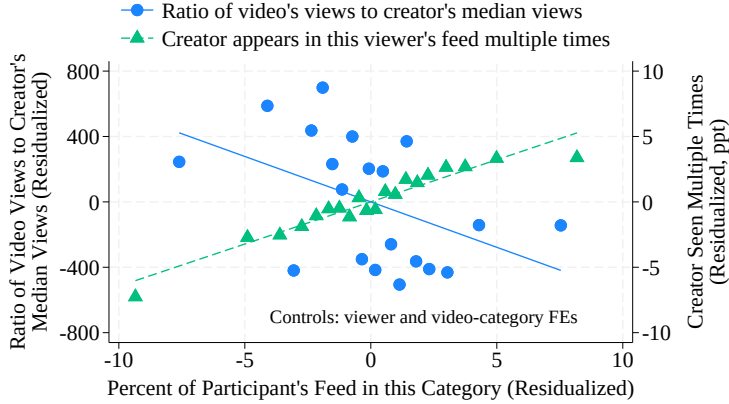
### Appendix Figure A4: Identifying tested videos: additional evidence



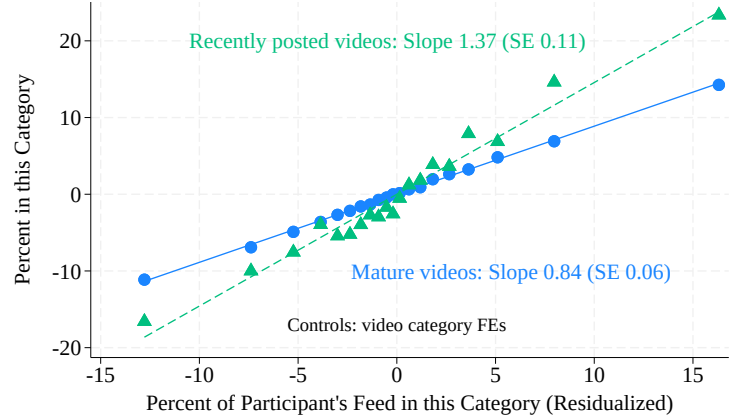
*Notes:* As in Figure 3, a test video is a video appearing in a viewer's feed within five hours of being posted. Panel (a) plots test viewers' average engagement with a video (seconds watched, controlling for viewer by video duration fixed effects, position in the session, time since the beginning of the session, and the video's age in hours) against the video's eventual view count, on a log scale. Panel (b) plots the engagement of a video's first test viewer against the average engagement of the viewers who see it after them, within the following seven days. Panels (c) and (d) repeat the predictiveness series from Figure 3 Panel (b) separately for creators with fewer than versus at least 10,000 followers (Panel (c)) and for viewers who have never seen the creator before versus viewers who have seen them at least five times (Panel (d)). Bars show 95% confidence intervals clustered by viewer and video.

**Appendix Figure A5: Preference dispersion  $\rightarrow$  local content scarcity + pass-through of all content types to consumption**

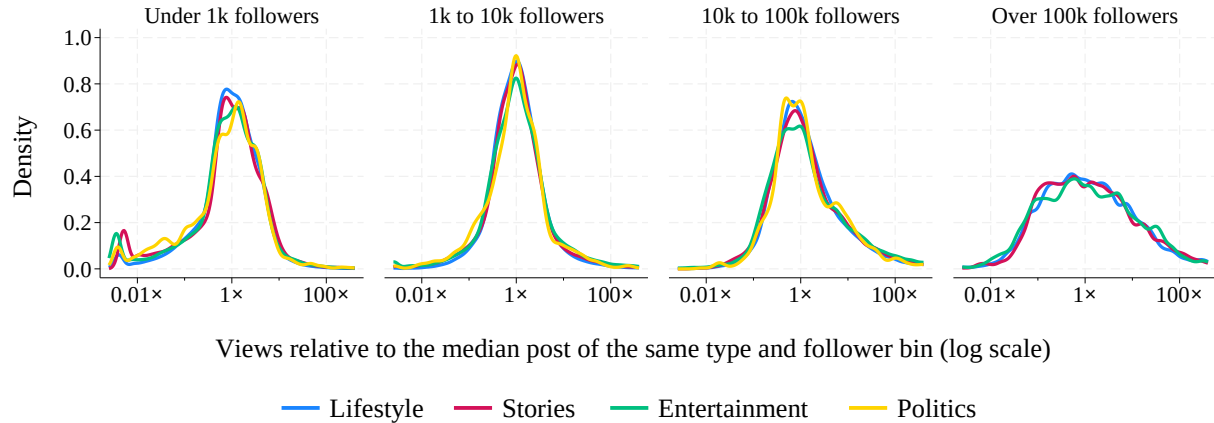
**(a) Pass-through:** Videos from categories a viewer doesn't usually see are larger outliers



**(b) Pass-through:** As platform learns more about a video, its category becomes less predictive of its appearance in viewers' feeds



**Appendix Figure A6: The distribution of views has a similar shape for every content type**



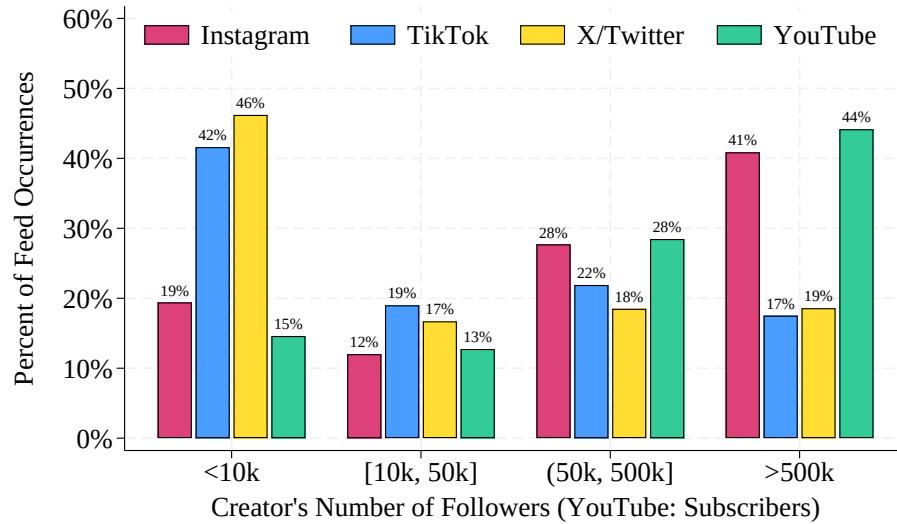
*Notes:* This figure plots the distribution of view counts of the randomly sampled TikTok videos from the Research API, separately for each content type and creator follower bin. Each distribution is centered at the median view count of its content type and follower bin, so the figure compares the shapes of the distributions rather than their levels. (We pool partisan and nonpartisan politics, pool creators with more than 100,000 followers, and drop cells with fewer than 200 videos, which is why politics does not appear in the right-hand panel.) Our model assumes the stochastic component of quality has the same distribution for every content type, and the figure shows the data are consistent with this.

## B Results for Instagram, X/Twitter, and YouTube

This appendix repeats the main descriptives of Section 3.2 for the other platforms in our data. For Instagram and YouTube, we use the viewership histories participants downloaded; for X/Twitter, the feeds captured by our Chrome extension. We scrape the posts in these feeds and the profiles

of their creators from the platforms' public web pages. Instagram's history records only which account a participant saw and when, not which post, so its results are at the level of the creator.

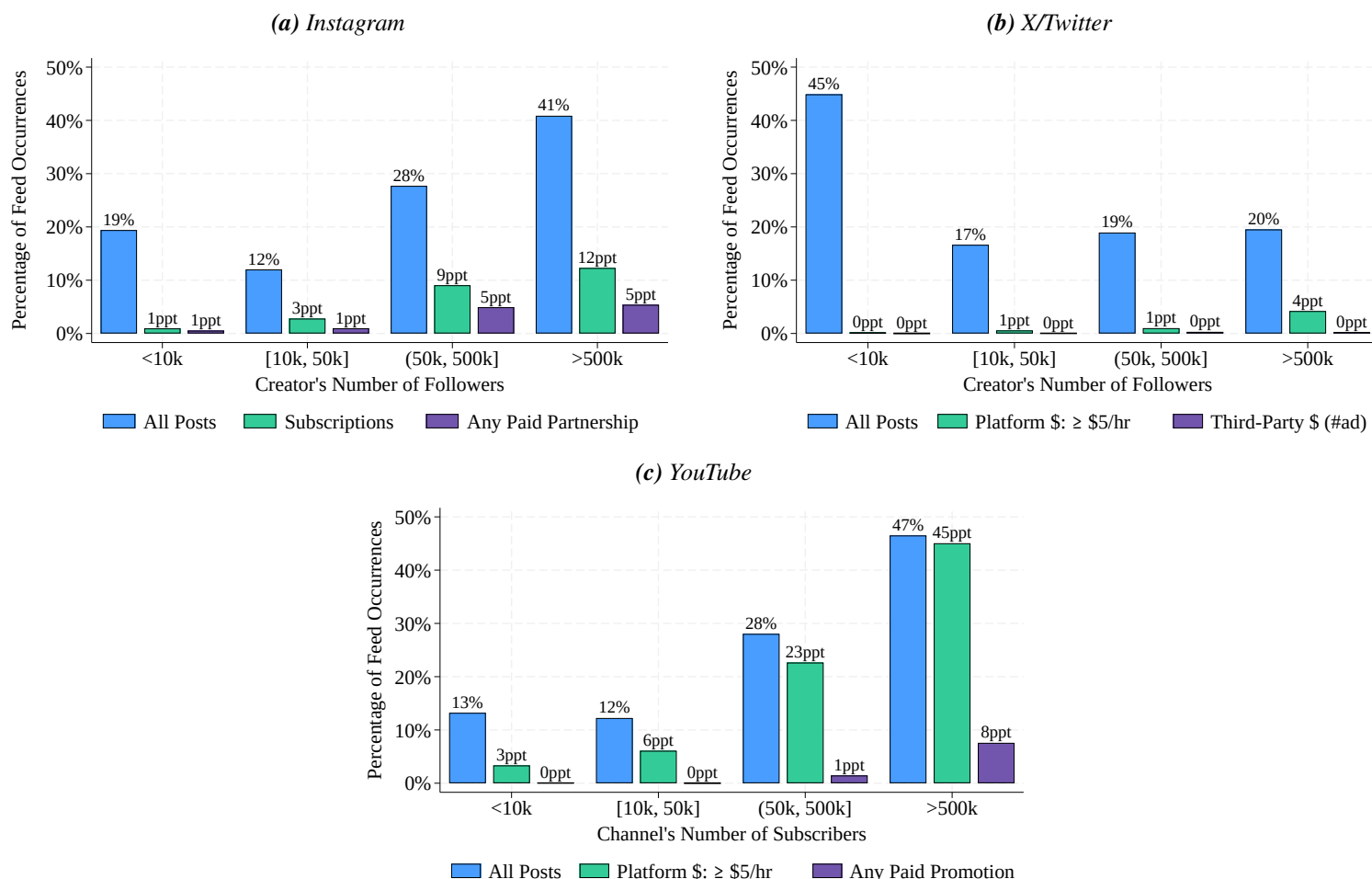
**Appendix Figure A7:** *Distribution of posts in viewers' feeds across account sizes, by platform*



*Notes:* This figure repeats the “All Videos in Feeds” series of Figure 3 Panel (a) for each platform: each bar plots the percent of posts in participants' feeds that come from creators in a given follower bucket (using subscribers for YouTube). We exclude ads and promoted posts, put private Instagram accounts in the smallest bucket, and drop accounts that no longer exist. Instagram feeds include both Reels and feed posts. For YouTube, we use a 20% random sample of views from September 2025 onward.



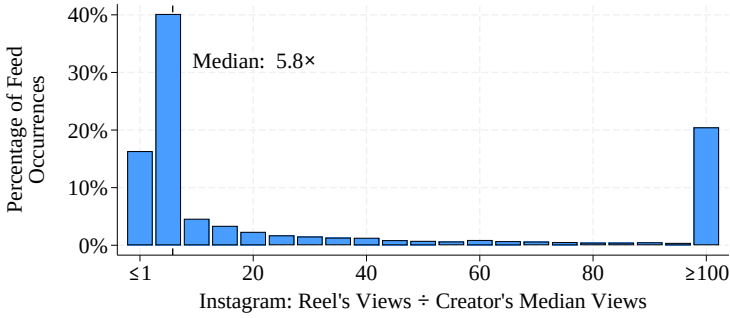
**Appendix Figure A8: What fraction of posts in viewers’ feeds earn money for their creators? Instagram, X/Twitter, and YouTube**



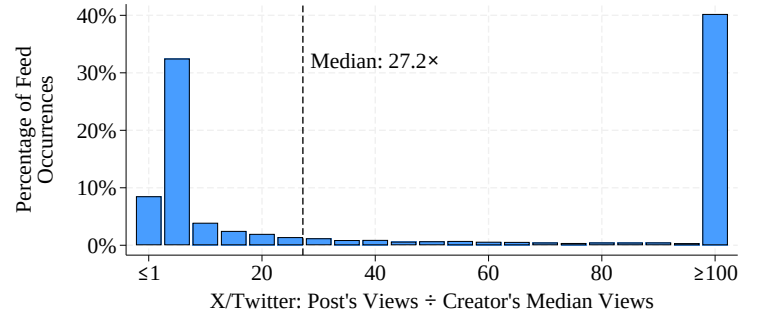
*Notes:* This figure is analogous to Appendix Figure A13. The first bar in each bucket plots the bucket’s share of the posts in participants’ feeds, and the other bars plot how many of these posts (in percentage points of all feed posts) come from creators who earn money in each way. On Instagram, which has no general program paying creators for views, “Subscriptions” refers to creators offering paid subscriptions and “Any Paid Partnership” to creators with at least one Reel carrying Instagram’s paid-partnership label between August 2025 and February 2026. On X/Twitter, “Platform \$” refers to creators with paid verification whose estimated earnings from X’s revenue-sharing program exceed \$5 per hour; we value views at \$8.50 per million (the most commonly reported payout) and assume a text post takes 6 minutes, a photo post 15 minutes, and a video half an hour to an hour. Since X has no label for sponsored posts, “Third-Party \$” refers to creators whose recent posts include #ad, #sponsored, or similar tags. On YouTube, “Platform \$” refers to channels with at least 1,000 subscribers whose estimated earnings from YouTube exceed \$5 per hour, valuing views at industry estimates of \$2 per thousand for long videos and \$0.05 per thousand for Shorts, and assuming a Short takes half an hour and a long video two hours plus an hour per ten minutes of runtime. “Any Paid Promotion” refers to channels with a video in participants’ feeds carrying YouTube’s paid-promotion label.

### Appendix Figure A9: Dispersion on Instagram, X/Twitter, and YouTube

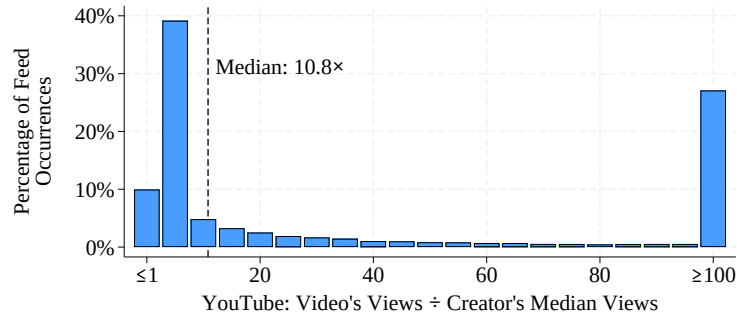
(a) Instagram: Reels' views relative to their creator's median views



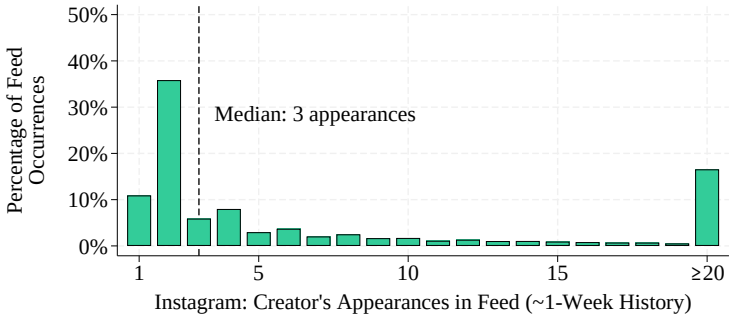
(b) X/Twitter: posts' views relative to their creator's median views



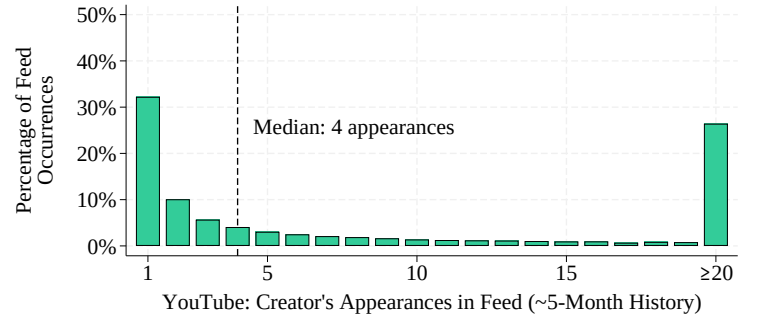
(c) YouTube: videos' views relative to their channel's median views



(d) Instagram: how many times viewers see each creator



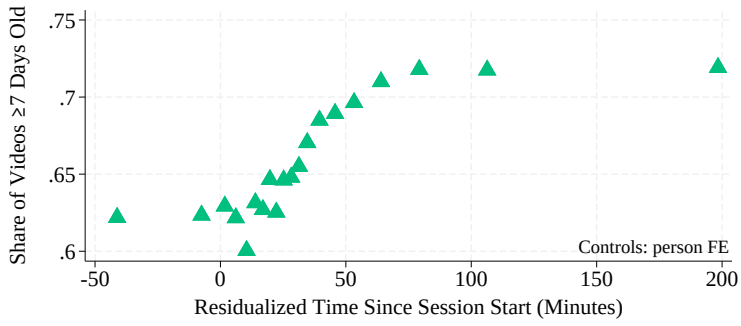
(e) YouTube: how many times viewers see each channel



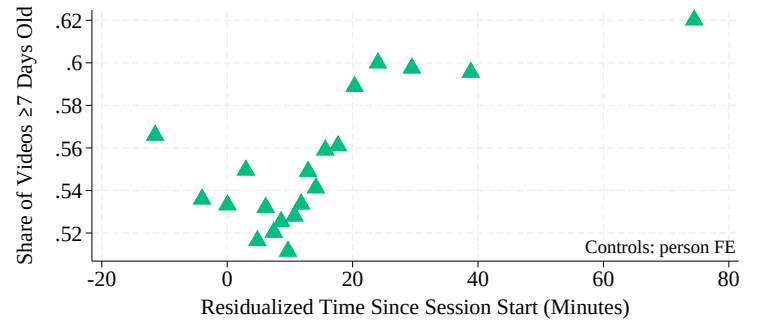
*Notes:* This figure is analogous to Panels (c) and (d) of Figure 4. Panels (a)–(c) plot each post's view count divided by the median view count of the creator's recent posts (for YouTube, the channel's newest uploads in the same format, for a random sample of channels), pooling ratios of 100 or more. Instagram histories do not record which Reel was watched, so Panel (a) spreads each watched Reel over the Reels its creator posted in the 60 days before the view, in proportion to their plays, and divides by the median plays of the creator's Reels posted between August 2025 and February 2026; on TikTok, where the watched video is known, this method reproduces the histogram of Figure 4 Panel (c) closely. Panels (d) and (e) plot how many times each post's creator appears in the participant's viewership history, which covers about a week on Instagram and about five months on YouTube, pooling counts of 20 or more. Dashed lines mark medians.

**Appendix Figure A10: Scarcity on YouTube: stale videos within sessions**

**(a) All YouTube videos**

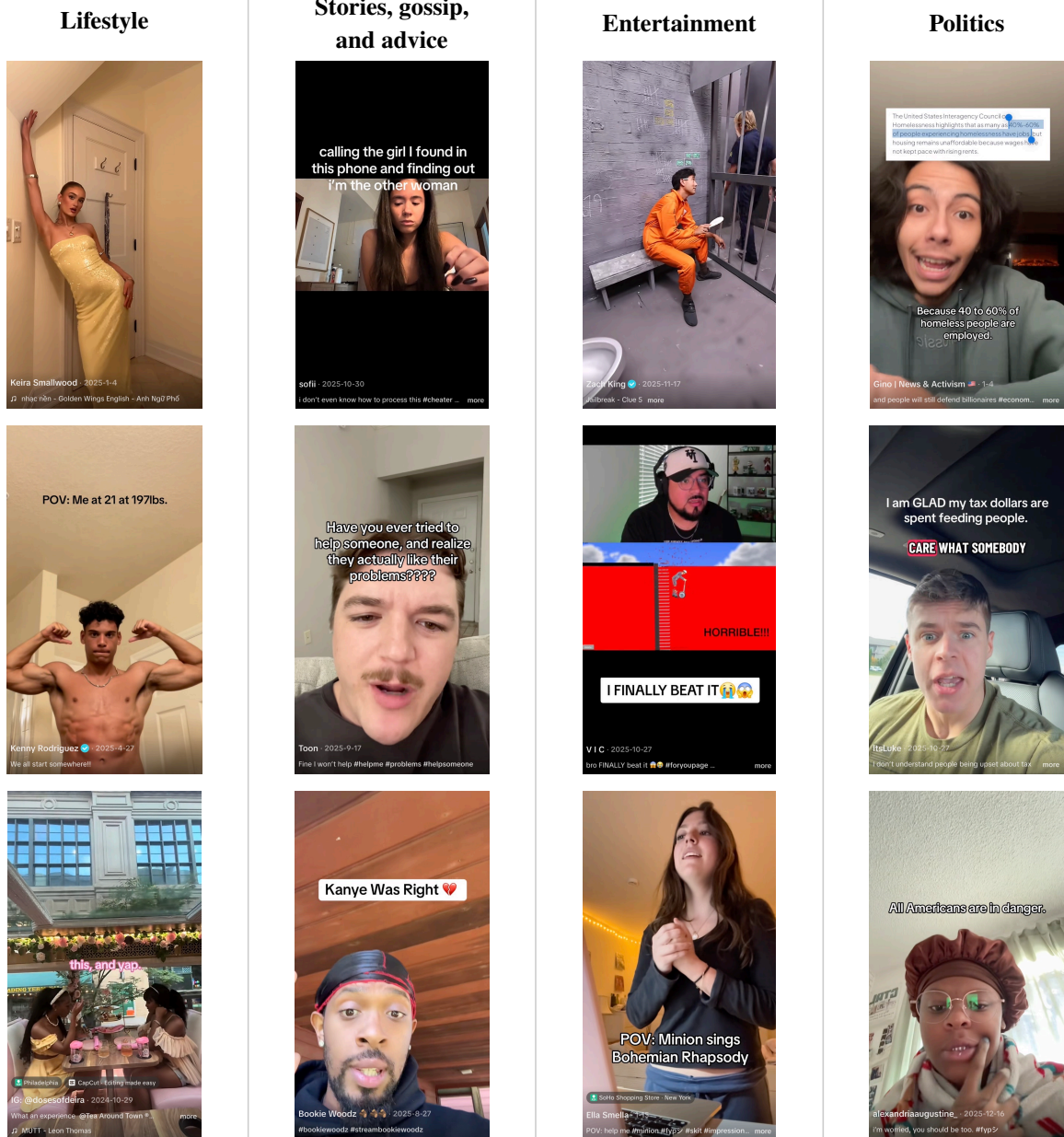


**(b) YouTube Shorts**



## C Observational Data and Classification

Appendix Figure A11: Examples of TikTok content types



**Appendix Table A1: Fit of different distributions to the upper tail of video views**

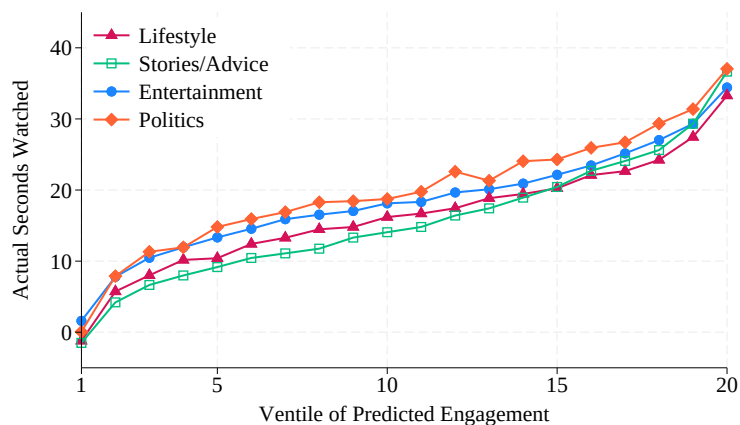
|             | Log-likelihood | $\Delta$ AIC | KS distance | Share of tail videos with $\geq v$ views (%) |      |       |       |
|-------------|----------------|--------------|-------------|----------------------------------------------|------|-------|-------|
|             |                |              |             | $v = 10K$                                    | 100K | 1M    | 10M   |
| Data        |                |              |             | 37.3                                         | 9.00 | 1.19  | 0.054 |
| Lognormal   | -277,183       | 0            | 0.018       | 37.7                                         | 7.94 | 1.45  | 0.230 |
| Pareto      | -277,216       | 63           | 0.012       | 36.6                                         | 8.32 | 1.89  | 0.429 |
| Weibull     | -277,427       | 486          | 0.051       | 41.1                                         | 7.45 | 0.729 | 0.031 |
| Exponential | -317,840       | 81,312       | 0.548       | 89.8                                         | 26.3 | 0.000 | 0.000 |

*Notes:* This table compares how well different distributions fit the upper tail of view counts in our random sample of TikTok videos from the Research API. The sample is the videos with at least 2,102 views (about the top 10% of videos with at least one view); we choose this threshold to make the Pareto fit as good as possible. We fit each distribution to these videos by maximum likelihood. The first three columns report the log-likelihood, the difference in AIC from the best-fitting distribution, and the largest gap between the fitted and empirical distribution functions (the KS distance). The last four columns compare the share of these videos with at least  $v$  views in the data to the share implied by each distribution. The lognormal also beats the Pareto in a Vuong test clustered by creator ( $z = 4.4$ ,  $p < 0.001$ ) and has the lowest AIC in each of the five posting months when we fit them separately, and a bootstrap test rejects an exact power law. The lognormal, Pareto, and Weibull all have decreasing hazard rates over this range, while the exponential has a constant hazard rate.

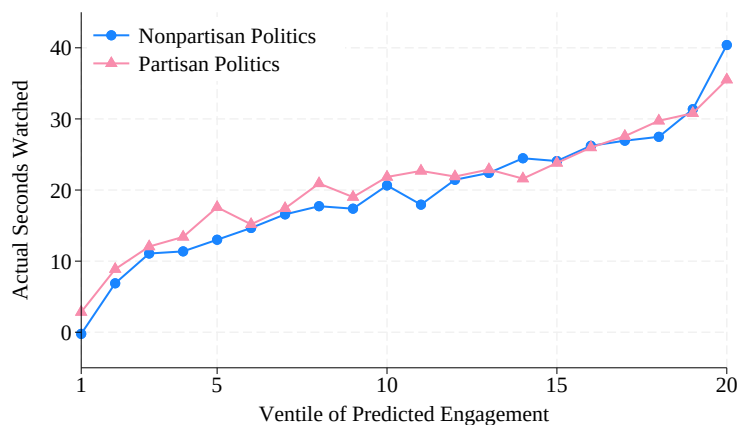
## D Identifying Algorithmic Markups

*Appendix Figure A12: Testing for algorithmic markups: actual versus predicted engagement*

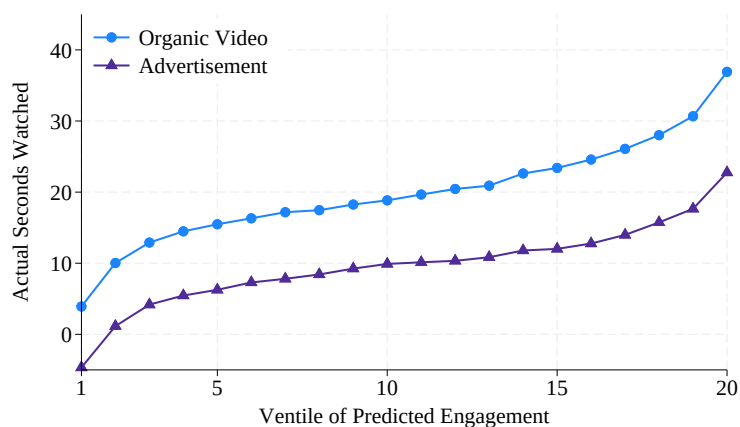
*(a) Content types*



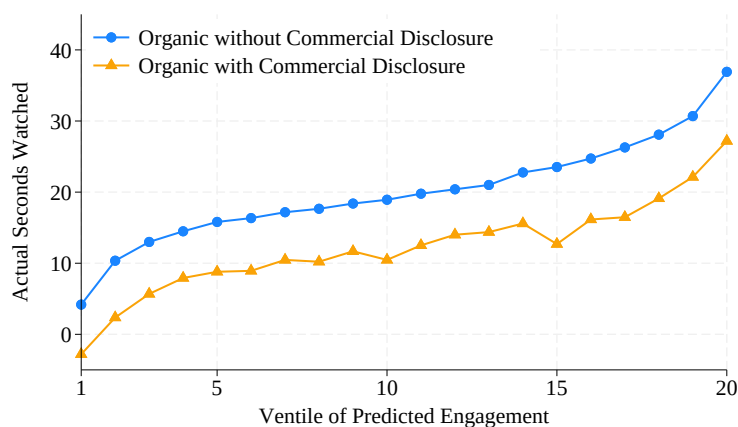
*(b) Partisan versus nonpartisan political content*



*(c) Advertisements versus organic videos*

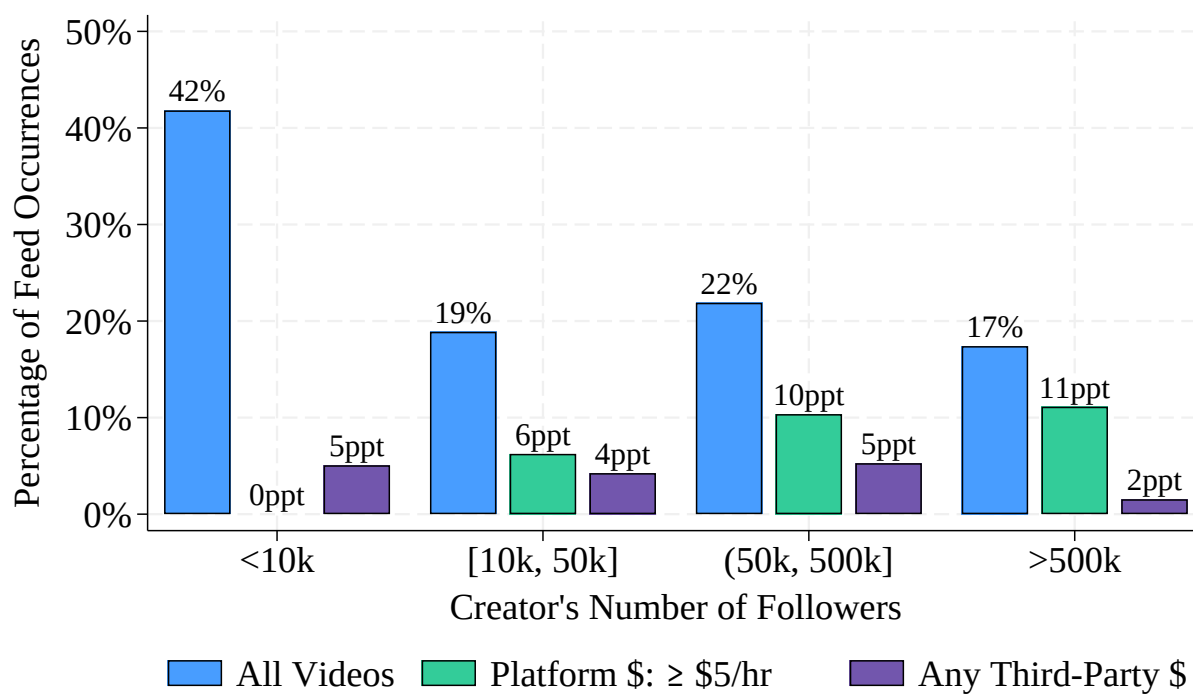


*(d) Organic videos with and without a commercial disclosure*



## E Creator Monetization and Motivations

*Appendix Figure A13: What fraction of the videos viewers see come from creators who ever earn money?*

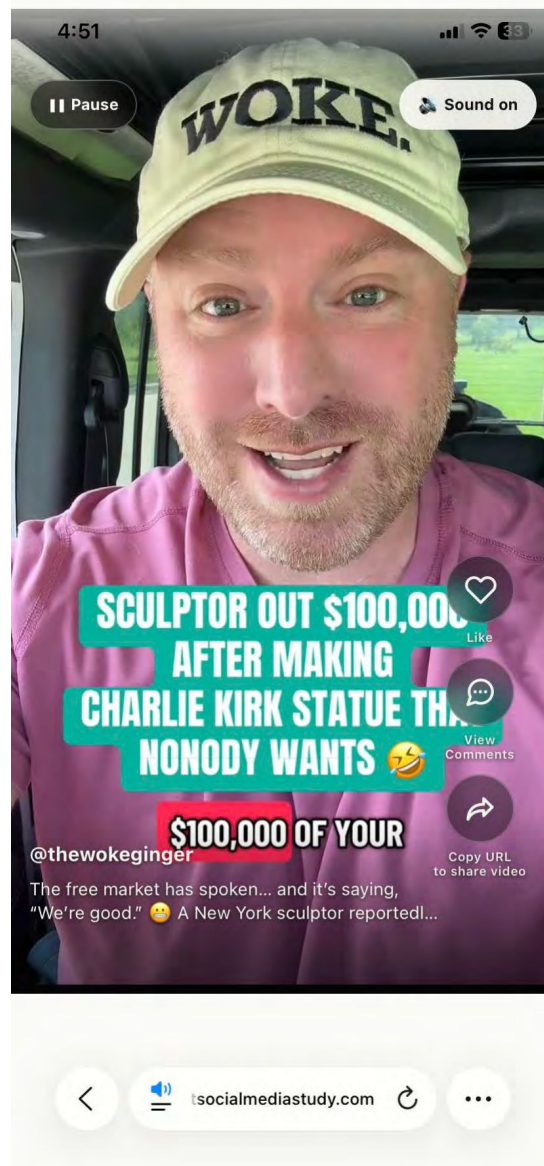


## F Short-Form Content

## G Platform Wages

## H Viewership Experiment

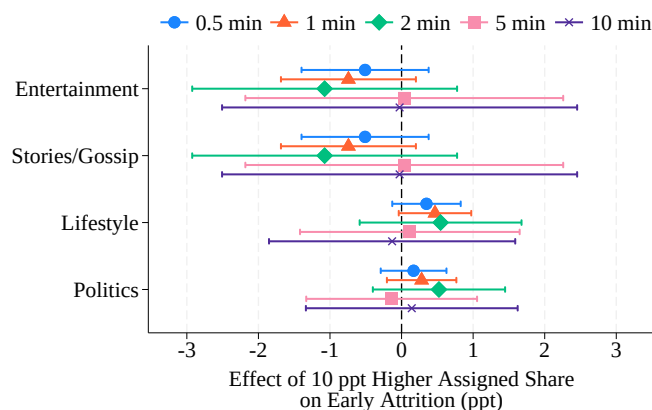
*Appendix Figure A14: Viewership experiment interface*



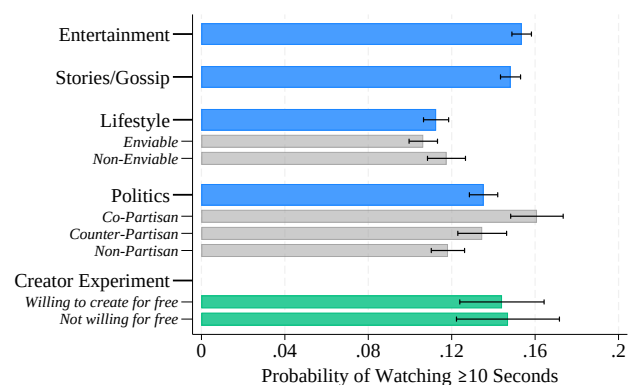


**Appendix Figure A15: Attrition is unrelated to feed composition, and including incomplete sessions does not change results**

**(a) Early attrition by randomized feed composition**

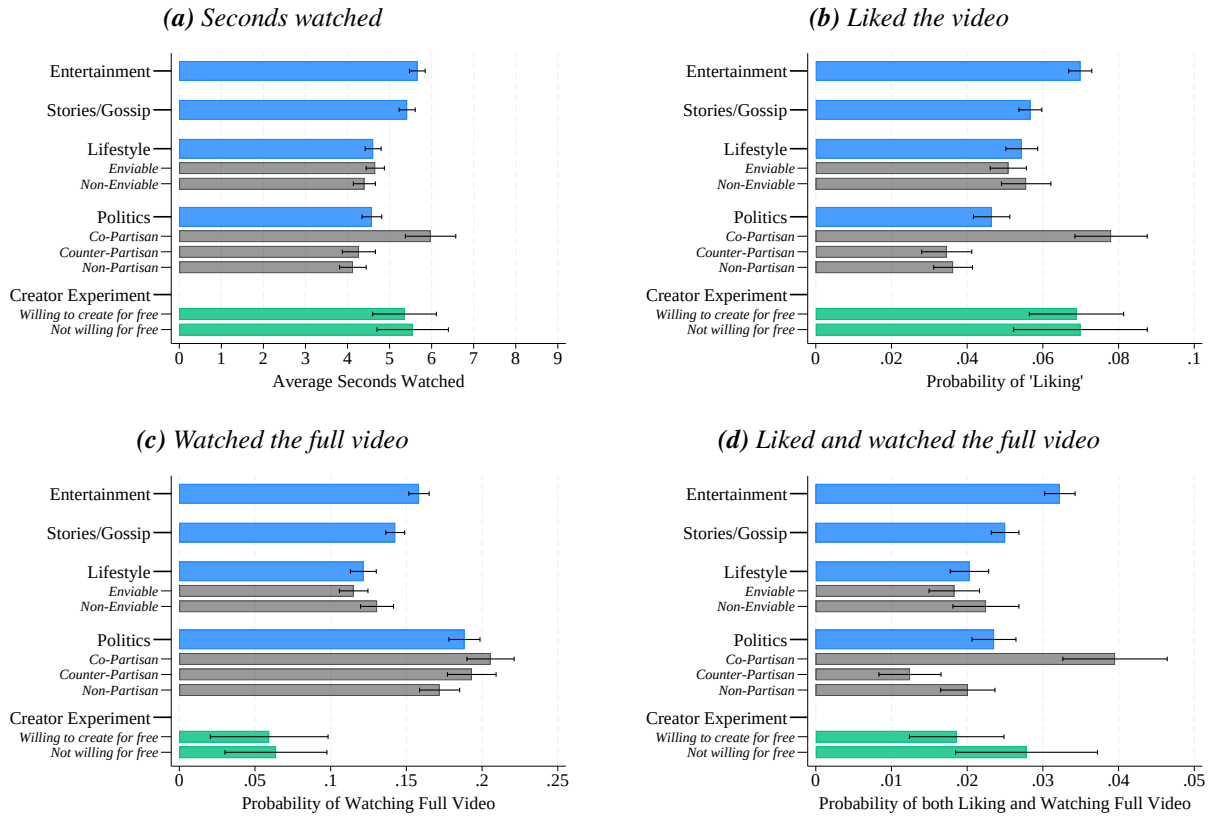


**(b) Probability of watching  $\geq 10$  seconds, including incomplete sessions**



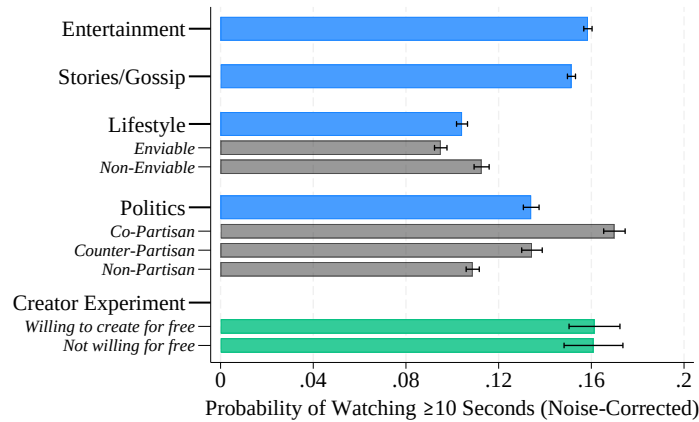
*Notes:* Panel (a) asks whether the randomized composition of a participant's feed predicts whether they drop out of the experiment early. Each coefficient comes from a separate regression of an indicator for dropping out before a given number of minutes of scrolling on the share of the participant's feed assigned to one content type, controlling for which recruitment flow the participant came from. Coefficients are the percentage-point change in the probability of dropping out from a 10 percentage point higher share, with 95% confidence intervals using robust standard errors. The sample is all participants who started the experiment. (Because entertainment and stories split the remaining share of the feed equally, their coefficients are identical.) None of the coefficients is significant at the 5% level, though with so few dropouts the estimates are imprecise. Panel (b) repeats Figure 6 Panel (b) using all video encounters, including those from sessions that were not completed.

**Appendix Figure A16: Engagement by content type: other pre-registered measures**



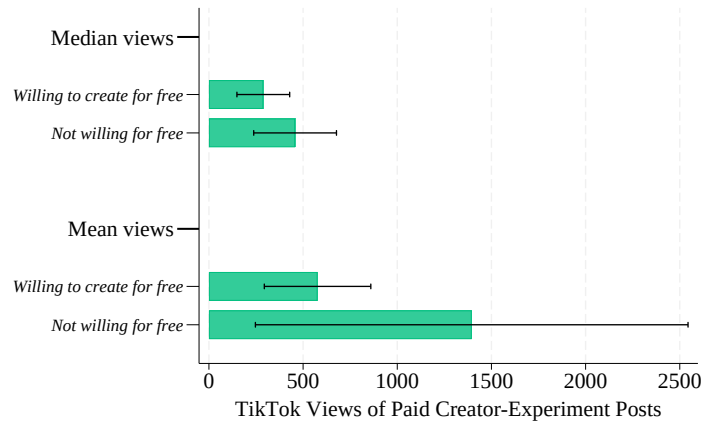
*Notes:* Each panel repeats Figure 6 Panel (b) for one of our other pre-registered engagement measures: (a) seconds watched, (b) whether the participant liked the video, (c) whether they watched the full video, and (d) whether they both liked it and watched it in full. The bars plot average engagement with each type of video, controlling for fixed effects for participant, video length decile, sampling tier, and sampling source, with 95% confidence intervals clustered by video. The creator experiment bars use only the videos created in our creator experiment, controlling instead for the content type the creator was asked to make.

**Appendix Figure A17: Probability of watching  $\geq 10$  seconds by content type, correcting for behavioral noise**

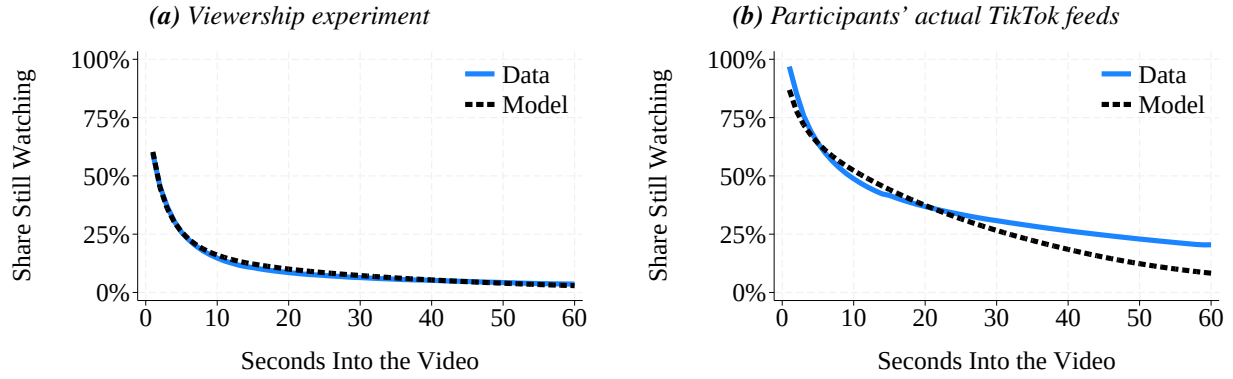


*Notes:* This figure repeats Figure 6 Panel (b) after correcting for noise in participants' engagement behavior. We use the re-pick module at the end of the experiment, where we show participants two videos they saw earlier, one they watched for at least 5 seconds and one they did not, and ask which they preferred. In comparisons where participants were not told which video they had engaged with more, they pick the video they engaged with more 74% of the time. If we treat these choices as accurate, this implies that 29% of the variation in participants' propensity to watch a video for at least 10 seconds reflects stable preferences rather than noise. We then replace each indicator for watching a video for at least 10 seconds with the probability that the participant's stable preference alone would lead them to do so (an empirical Bayes correction), keeping the overall rate of watching at least 10 seconds unchanged at 14.4%. The correction widens the gap between entertainment and lifestyle from 4.1 to 5.4 percentage points (5.6 percentage points, with a 95% confidence interval of 4.8 to 6.7, when we bootstrap the whole procedure). Since participants' choices in the re-pick module are probably themselves somewhat noisy, this is an upper bound on the role of noise.

**Appendix Figure A18: TikTok views of paid creator-experiment posts, by creators' willingness to post for free**

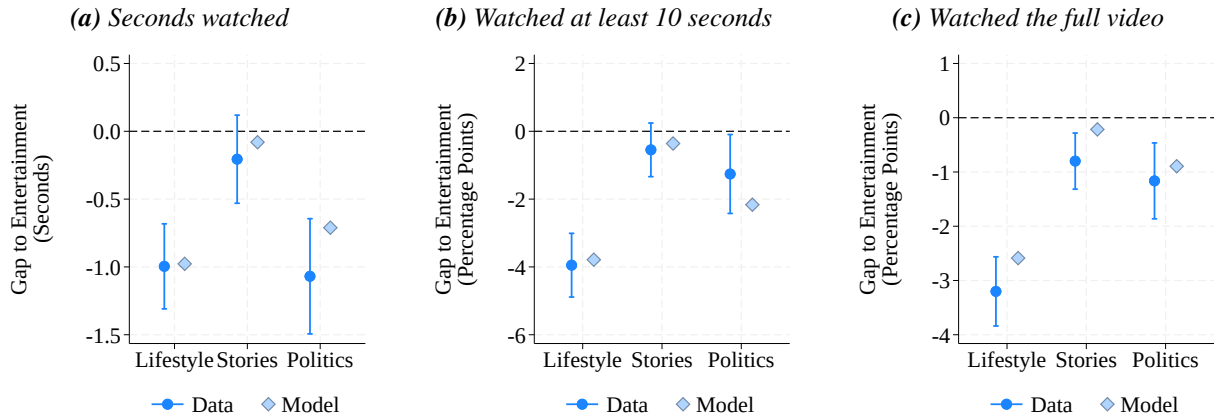


**Appendix Figure A19: Watch times in the data and in our model**



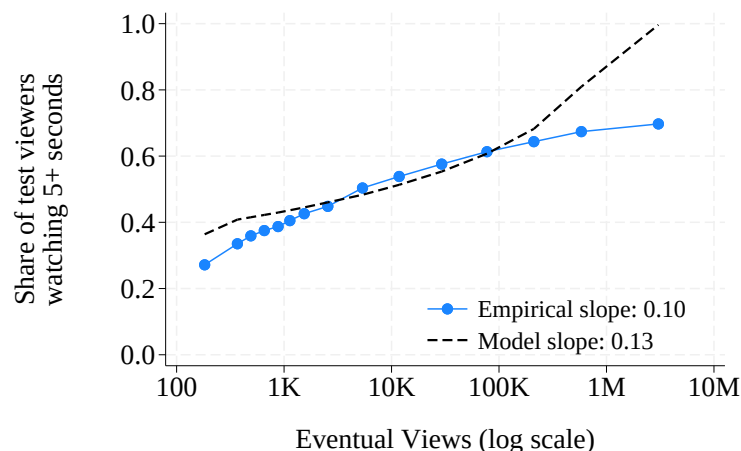
*Notes:* The blue line in Panel (a) plots the share of views in which the viewer is still watching the video  $t$  seconds in (restricting to videos at least  $t$  seconds long) across all occurrences in our viewership experiment. The black dashed line shows the distribution simulated by our model. Panel (b) reproduces this in our observational viewership-history data, re-calibrating the reservation value of swiping to the next post to match the share of viewers watching videos for at least five seconds and assuming that each video in a person's actual feed is drawn from the model's utility distribution truncated above this reservation value.

**Appendix Figure A20: Engagement by content type in the viewership experiment: data and model**



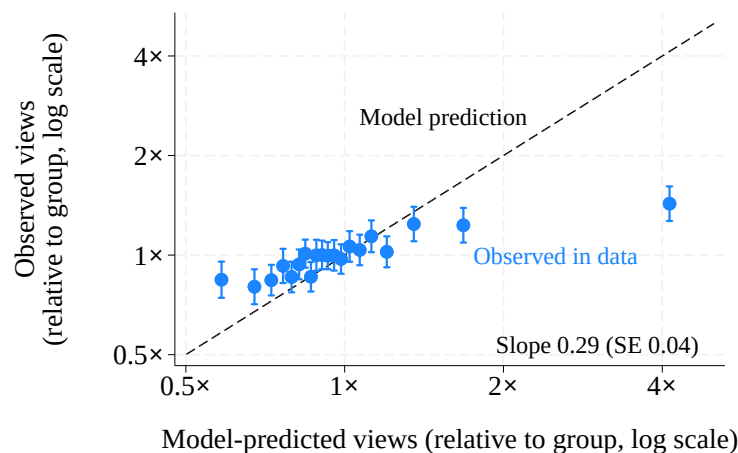
*Notes:* These figures plot coefficients from pre-registered regressions of engagement outcomes on dummies for content types, with fixed effects for participant, video length decile, and sampling source. Next to these coefficients, we display the analogous coefficients using viewership simulated from our model estimates.

**Appendix Figure A21: Test viewers' engagement and videos' eventual views: data and model**



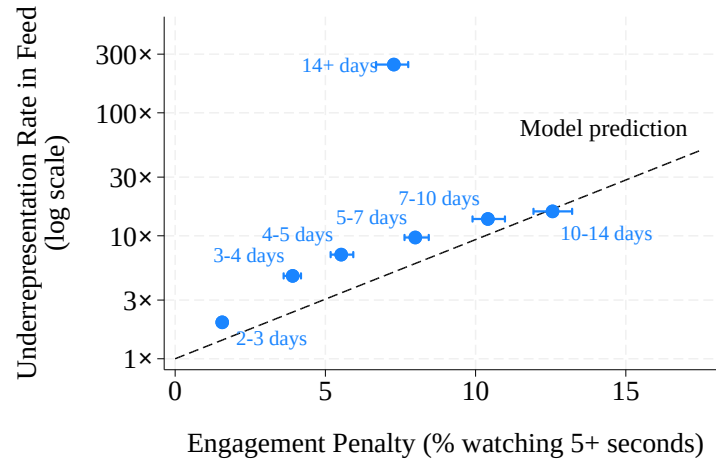
*Notes:* This figure studies “test videos” in our observational viewership-history data—videos that appear in a viewer’s feed within 5 hours of being posted. The x-axis plots the eventual view count of the video on a log scale while the y-axis plots the share of viewers served this video as a test who watch it for at least five seconds. The dashed black line plots our model’s prediction of the share of viewers watching each of these videos for at least five seconds. As in Figure A19, the model predictions are generated by re-calibrating the reservation value of swiping to the next post to match the share of viewers watching videos for at least five seconds in their actual feeds, and assuming that each video in a person’s actual feed is drawn from the model’s utility distribution truncated above this reservation value.

**Appendix Figure A22: Predicted and actual TikTok views of the videos in our viewership experiment**



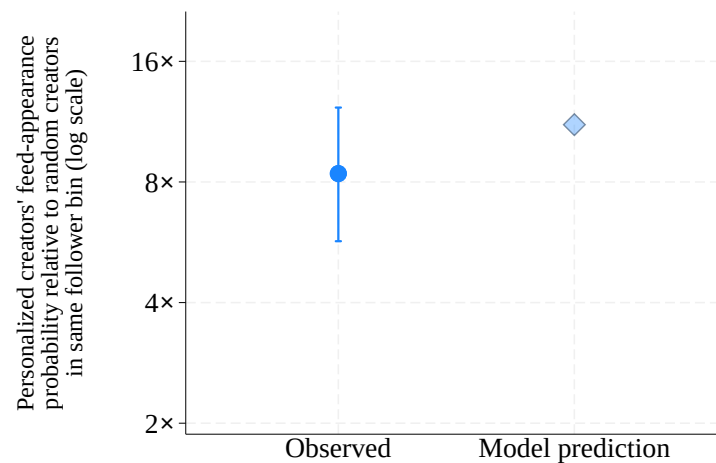
*Notes:* This figure plots the realized view counts on TikTok of the videos used in our viewership experiment against the view counts predicted by our model based on utility estimates from the engagement behavior of experimental participants. It plots this slope within groups (follower count  $\times$  content type  $\times$  sampling source). The black dashed line is the 45 degree line, reflecting perfect concordance between our model estimates and actual viewership.

**Appendix Figure A23:** *Stale videos' engagement and feed-appearance penalties: data and model*



*Notes:* This figure buckets posts in our viewership history by their “age” (the time between their posting and their appearance in a viewer’s feed). On the x-axis it plots the average engagement penalty associated with a bucket relative to videos 1-2 days old, measured as the percent of viewers watching them for at least five seconds. On the y-axis it plots the feed appearance penalty of this bucket, meaning how much less common videos from this bucket are relative to videos 1-2 days old. The dashed line shows our model’s prediction about the effects of a given decrease in the engagement measure on a video’s probability of appearing in a viewer’s feed.

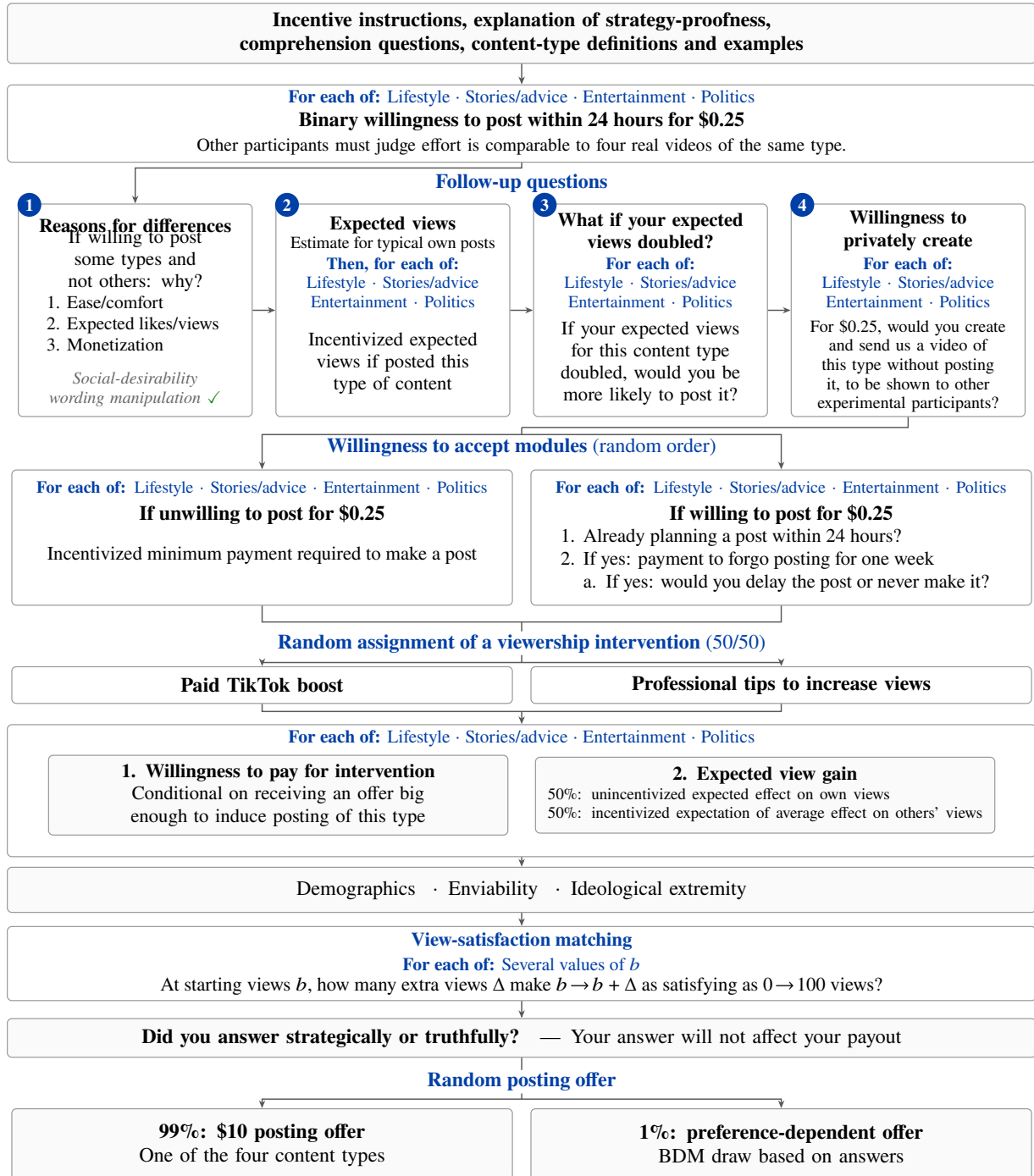
**Appendix Figure A24:** *Feed appearances of videos by creators viewers regularly see: data and model*



*Notes:* This figure studies the subset of participants in our viewership experiment for whom we pipe in “personalized” videos recently posted by a creator who appeared at least 3 times in their TikTok viewership history. It plots how much more likely a random video from such a creator is to appear in their feed than a random video from another creator from our experiment in the same number-of-followers bin, based on comparing the historical overlap between their viewership histories, the posting rates of the personalized creators, and the production and consumption shares of creators in this follower bin. It also plots our model’s prediction for this ratio.

# I Creator Experiment Details

*Appendix Figure A25: Creator experiment design*



**Appendix Figure A26:** *Creator payoff parameter estimates: creators with at least 1,000 followers*

| Parameter                                                                | Estimate | SE      |
|--------------------------------------------------------------------------|----------|---------|
| <i>Mean taste <math>\mu_k</math>, % relative to entertainment</i>        |          |         |
| Lifestyle                                                                | 13.4     | (3.2)   |
| Stories                                                                  | -6.5     | (4.3)   |
| <i>Mean taste <math>\mu_k</math>, % relative to nonpartisan politics</i> |          |         |
| Partisan politics                                                        | 6.0      | (25.8)  |
| Value of views $\beta$                                                   | 1.45     | (0.25)  |
| Curvature $\rho$                                                         | 0.33     | (0.005) |
| <i>Expected views, creators willing to post only some types</i>          |          |         |
| Types willing to post for \$0.25                                         | 1111     | (117)   |
| Types not willing to post for \$0.25                                     | 822      | (115)   |
| <i>Expected views, % relative to entertainment</i>                       |          |         |
| Lifestyle                                                                | -37.1    | (4.7)   |
| Stories                                                                  | -33.0    | (5.2)   |
| <i>Expected views, % relative to nonpartisan politics</i>                |          |         |
| Partisan politics                                                        | 106.8    | (55.7)  |
| <i>For the median planned post</i>                                       |          |         |
| Total utility from post                                                  | \$7.14   | (1.85)  |
| Utility from viewership                                                  | \$12.77  | (2.26)  |
| Utility from first 100 views                                             | \$6.70   | (1.14)  |

*Notes:* This figure repeats Panel (b) of Figure 9, restricting to creators with at least 1,000 followers. We re-estimate all of the creator parameters on this sample except the curvature  $\rho$ , which we hold at its full-sample value. Standard errors come from 200 bootstrap draws resampling creators. Very few partisan or nonpartisan creators in this sample post about politics, so the comparison between partisan and nonpartisan political content is imprecise.



**Appendix Table A2:** *Creator results are robust to the valuations of creators already planning to post*

| Specification                              | Creators | Mean taste, % cheaper |                          | Oaxaca supply contribution (pp) |                        | Supply-equalized counterfactual |                       |
|--------------------------------------------|----------|-----------------------|--------------------------|---------------------------------|------------------------|---------------------------------|-----------------------|
|                                            |          | Lifestyle<br>vs. ent. | Partisan<br>vs. nonpart. | Lifestyle<br>– ent.             | Partisan<br>– nonpart. | Lifestyle<br>share (%)          | Partisan<br>share (%) |
| Main estimates                             | 770      | 8.8<br>(1.5)          | 35.3<br>(8.5)            | 22.9                            | 19.6                   | 22.1                            | 25.4                  |
| Creators who would forgo rather than delay | 486      | 9.3<br>(2.5)          | 37.7<br>(11.1)           | 23.7                            | 23.0                   | 21.1                            | 34.4                  |
| Planning-branch payments $\times 1.1$      | 770      | 8.9<br>(1.5)          | 34.8<br>(8.3)            | 22.6                            | 19.4                   | 22.2                            | 25.7                  |
| Planning-branch payments $\times 1.2$      | 770      | 8.9<br>(1.5)          | 34.4<br>(8.2)            | 22.3                            | 19.2                   | 22.2                            | 25.9                  |
| Planning-branch payments $\times 1.5$      | 770      | 8.9<br>(1.4)          | 33.1<br>(7.7)            | 21.6                            | 18.8                   | 22.2                            | 26.3                  |
| Planning-branch payments $\times 2$        | 770      | 8.9<br>(1.4)          | 31.3<br>(7.0)            | 20.7                            | 18.3                   | 22.2                            | 26.6                  |
| Planning-branch payments $\times 3$        | 770      | 9.0<br>(1.4)          | 28.6<br>(6.0)            | 19.6                            | 17.6                   | 22.3                            | 26.7                  |

*Notes:* Creators who were already planning to post a given type of content tell us how much we would need to pay them not to post it for a week. Most of them say they would postpone the post rather than not make it at all, so these answers may understate how much they value posting. The first row repeats our main estimates. The second row drops creators who say they would postpone, and the remaining rows multiply all of these creators' valuations by the factor listed. In each row, we re-estimate all of the creator parameters and repeat the decomposition and counterfactual in Figure 10, holding the viewer-side estimates fixed. Mean tastes are percent differences in the net cost of posting relative to the reference type, with standard errors from 200 bootstrap draws resampling creators in parentheses. The Oaxaca columns report the supply contributions to the consumption gaps in Panel (a) of Figure 10, using demand estimates from the viewership experiment. The counterfactual columns report lifestyle's share of consumption when lifestyle and stories get entertainment's supply parameters, and the partisan share of political consumption when nonpartisan political content gets partisan content's supply parameters (31.8% and 64.8% in the status quo).

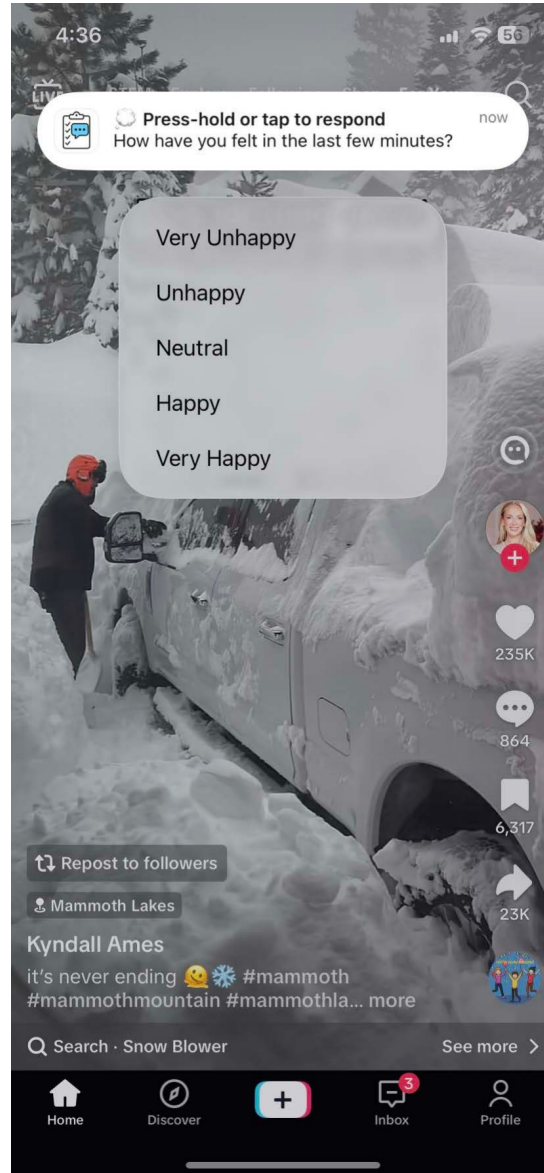
**Appendix Table A3:** *Creator results are robust to how we elicit the expected effects of the boost*

| Boost question               | $\beta$        | Mean taste, % cheaper |                          | Oaxaca supply contribution (pp) |                        | Supply-equalized counterfactual |                       |
|------------------------------|----------------|-----------------------|--------------------------|---------------------------------|------------------------|---------------------------------|-----------------------|
|                              |                | Lifestyle<br>vs. ent. | Partisan<br>vs. nonpart. | Lifestyle<br>– ent.             | Partisan<br>– nonpart. | Lifestyle<br>share (%)          | Partisan<br>share (%) |
| Both groups (main)           | 1.21<br>(0.09) | 8.8<br>(1.5)          | 35.3<br>(8.5)            | 22.9                            | 19.6                   | 22.1                            | 25.4                  |
| Own views (unincentivized)   | 1.25<br>(0.14) | 8.8<br>(1.5)          | 35.2<br>(8.5)            | 23.1                            | 19.7                   | 22.1                            | 25.3                  |
| Others' views (incentivized) | 1.16<br>(0.11) | 8.8<br>(1.5)          | 35.4<br>(8.5)            | 22.7                            | 19.5                   | 22.2                            | 25.7                  |

*Notes:* We identify the value of views  $\beta$  from participants' willingness to pay for an intervention that boosts their viewership and their expectation of how much it will increase their views. Half of participants answer an unincentivized question about the intervention's effect on their own views, and the other half an incentivized question about its average effect on other participants' views. The first row repeats our main estimates, which pool both groups. The second and third rows estimate  $\beta$  using only one of the two groups, and then re-estimate the remaining creator parameters, the decomposition, and the counterfactual as in Appendix Table A2. Standard errors from 200 bootstrap draws resampling creators are in parentheses, and the Oaxaca and counterfactual columns are defined as in Appendix Table A2.

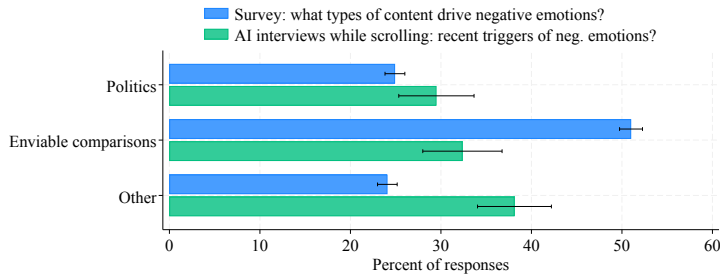
## J Emotion-Sampling: Details and Results

*Appendix Figure A27: Emotion-sampling app interface*

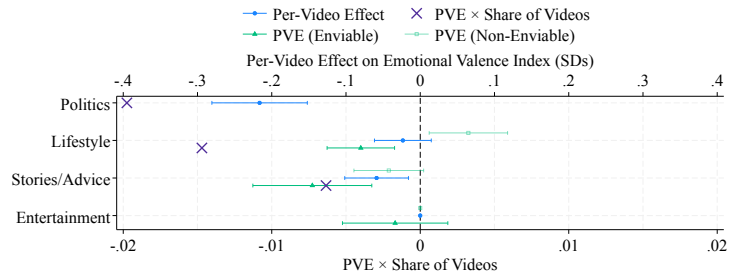


**Appendix Figure A28: Political and enviable lifestyle videos trigger negative emotions, but viewers do not regret consuming the ones appearing in their feeds**

**(a) Open-ended surveys and interviews: what content types drive negative emotions on social media?**



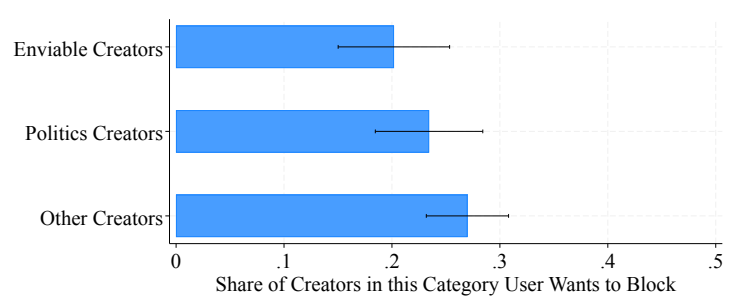
**(b) Emotion-sampling: what content types reduce contemporaneous emotional valence?**



**(c) App questions: what did you get out of the video you just watched? Did you regret watching it?**

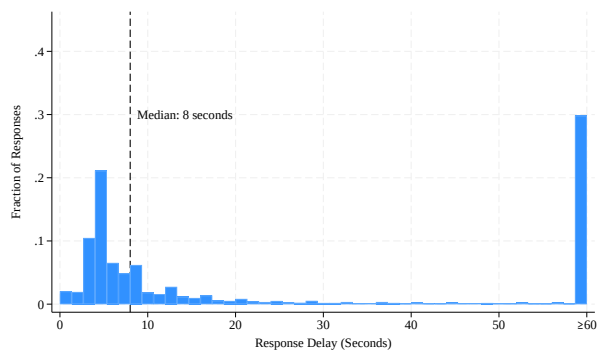
| Reason                              | All | Enviably | Politics |
|-------------------------------------|-----|----------|----------|
| Entertaining                        | 55% | 55%      | 34%      |
| Funny                               | 32% | 22%      | 24%      |
| Staying Informed                    | 22% | 18%      | 58%      |
| Visually Appealing                  | 17% | 23%      | 7%       |
| Self-Improvement Advice             | 8%  | 13%      | 4%       |
| Product Recommendation              | 8%  | 11%      | 3%       |
| Regret Watching (Negative Emotions) | 4%  | 4%       | 12%      |
| Regret Watching (Waste of Time)     | 4%  | 3%       | 2%       |

**(d) Retrospective survey: which of these specific creators from your feed would you want to block?**

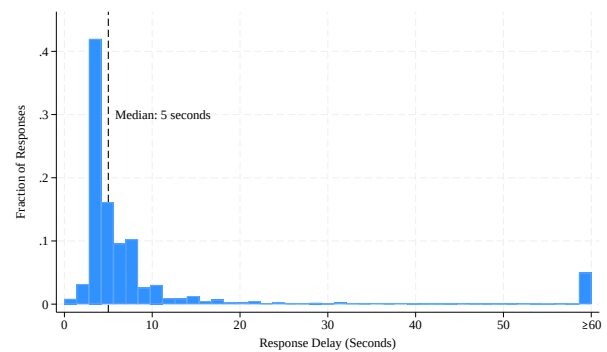


**Appendix Figure A29: Response delays to wellbeing notifications**

**(a) Notifications delivered while off social media**

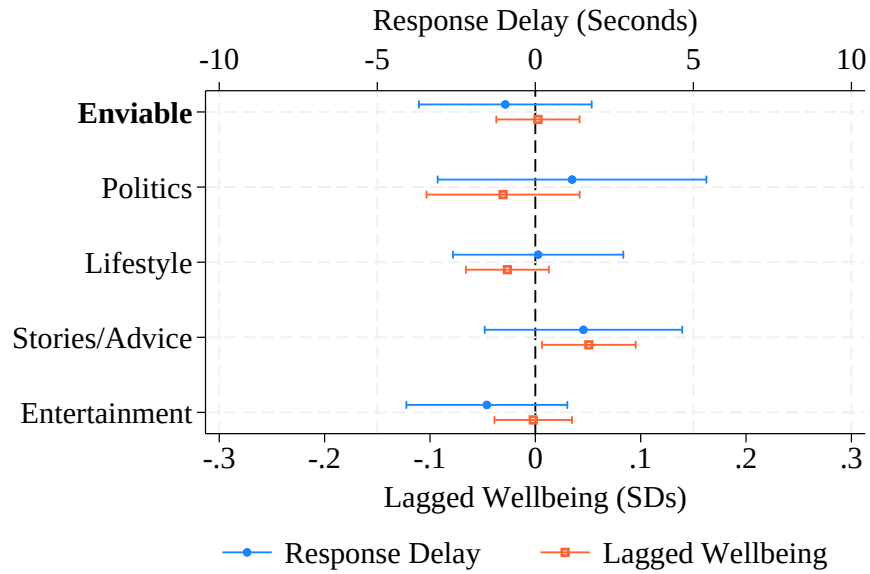


**(b) Notifications delivered while on social media**



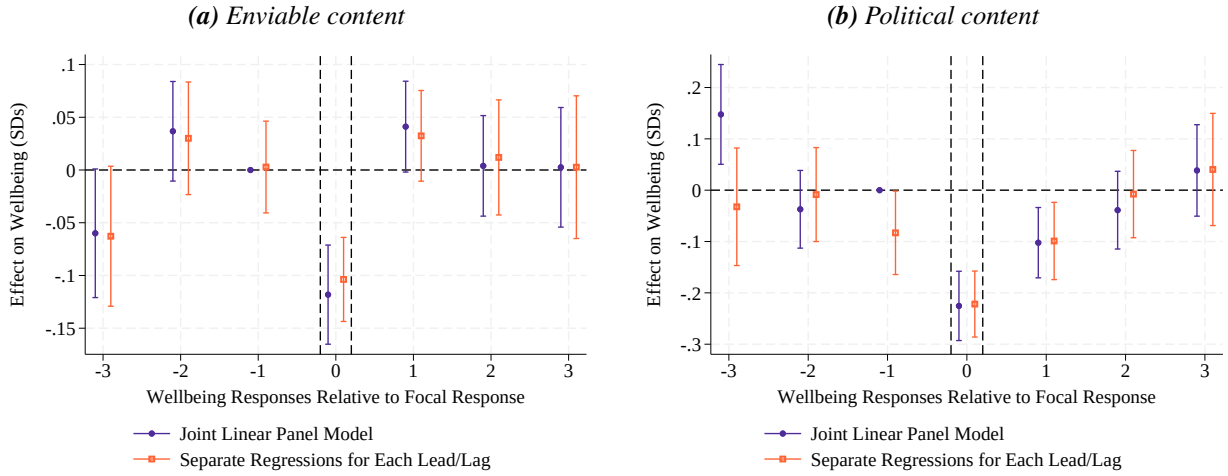
**Notes:** This figure plots distributions of notification response delays, defined as the number of seconds between notification delivery and notification response for notifications that are responded to. Panel (a) plots the distribution for notifications delivered while the recipient is not on social media, while Panel (b) plots notifications delivered while the recipient is on social media. (Social media usage is measured through viewership histories.)

**Appendix Figure A30: Balance tests: content on screen, response delays, and lagged wellbeing**



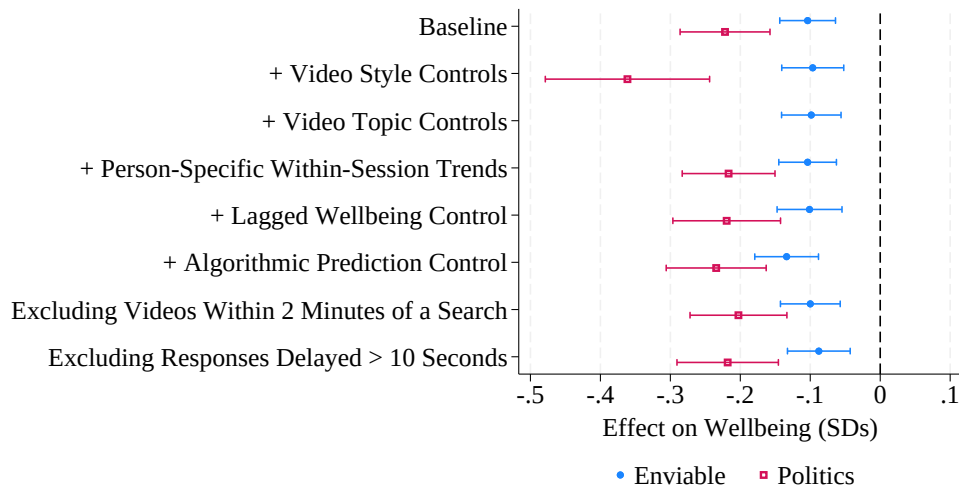
*Notes:* This figure restricts to videos that were being viewed at the time that a user responded to a notification from the app containing a wellbeing question; the blue coefficients show the correlation between the amount of time it took the user to respond to the question (top-coded at 10 minutes) with the characteristics of the content being viewed at the time of the response; the orange coefficients show the correlation of the *most recent* wellbeing response submitted by the respondent (within the last 30 minutes) with the characteristics of the content being viewed at the time of the response. (We use the content being viewed at the time of the response rather than at notification delivery because our classifications cover the former; the latter is classified only when the user is still watching it when they respond, which would select on short delays.)

**Appendix Figure A31: Wellbeing effects of enviable and political content: linear panel models**



*Notes:* Panels (a) and (b) plot estimates of the effects of enviable and political content exposure, respectively, on contemporaneous wellbeing using two linear panel model specifications. The purple dots use, as a panel unit, a social media usage session, and as a time unit the  $n$ th wellbeing response in that session; we estimate coefficients using the package developed by Freyaldenhoven et al. (2021) and decumulify the  $t > 0$  coefficients. The orange dots regress contemporaneous wellbeing on the characteristics of the video being viewed while the wellbeing response was received ( $t = 0$ ) and, one at a time, the characteristics of the videos being viewed at the next 3 wellbeing responses in the session ( $t < 0$ ) or the previous 3 ( $t > 0$ ), controlling for session fixed effects. Both specifications control for the other content-type indicators (and, in Panel (b), for enviable content).

**Appendix Figure A32: Robustness of the wellbeing effects of enviable and political content**



*Notes:* This figure plots the coefficients on enviable and political content from a regression of contemporaneous wellbeing on indicators for enviable content and for the political, lifestyle, and stories/advice content types (entertainment omitted), with session fixed effects, in each case with the modification listed on the y-axis. The video topic controls are shown for enviable content only, because the topics include news and politics and so absorb the political classification itself.